

**THE DOMAIN FACTORS INFLUENCING
ON LEARNING GEOMETRICAL CONCEPTS
OF SENIOR SECONDARY STUDENTS**

BY

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**FOR THE DEGREE OF
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By

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(B.Sc, B.Ed, M.Sc in Science Education)

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of the

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DECLARATION

I do hereby declare that the work reported in this thesis was exclusively carried out by me under the supervision of Mr.P.K.J.E.Nonis. It describes the results of my own independent research except where due reference has been made in the text. No part of this thesis has been submitted earlier or concurrently for the same or any other degree.

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Abstract

The domain factors influencing on learning geometrical concepts of senior secondary students

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The importance of geometry is vital and evident in every step of human life. Geometry plays an important role in studying the branches of other mathematics. Therefore, it is increasingly important that students should gain knowledge and develop the concepts of geometry. The performance of students in geometry is influenced by several factors including teacher-related factors, school-related factors and student-related factors. Although a number of factors fall under these categories, due to the practical convenient domain factors are focused in the study. The purpose of this study was to analyze the impact of domain factors on the development of geometrical knowledge and skill in secondary schools in the Jaffna district. This study further attempts to identify the approaches that the teachers adapted to teach geometrical concepts and the relevant strategies used by the students in learning geometrical concepts. A mixed methods design was used to collect and examine data. A sample of 379 students and 50 teachers was drawn from different types of 22 schools situated in Jaffna district Educational zones using the purposive sampling technique. The sample also consisted of in-service advisors of mathematics, additional directors of mathematics and subject coordinators of mathematics. Data for the study was collected through questionnaires, interviews, documents and a geometry achievement test. To identify whether the students achieve the development of geometric knowledge and skills, the geometry achievement test which includes all the levels of Van hiele geometric thinking was conducted as Van hiele theory describes how the students learn geometry by achieving each level of geometrical thinking step by step. The analysis was done to show the impact of domain factors on the development of geometric knowledge and skills. Here the impact of each domain factor was analysed through regression analysis technique and thematic analysing technique was used to analyse qualitative data. By utilizing both analysing techniques and from literature, the current status of geometry in secondary schools was also

revealed. Furthermore, results indicated that the status of students' performance in geometry is poor among the senior secondary students and the domain factors teaching methodology, ability of students and instructional materials and tools affect the performance of the students in geometry. Among three domain factors teaching methodology impact a lot than the ability of students and instructional materials and tools. Moreover, it was also used to establish the influence of the components of each domain factor on students' development of geometric knowledge and skills.

Keywords: concepts, achievement, geometry, factors, impact, performance, Van Hiele levels, knowledge.

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LIST OF ABBREVIATIONS

G.C.E (O/L)	General Certificate of Education (Ordinary Level)
G.C.E (A/L)	General Certificate of Education (Advanced Level)
TIM	Teacher Instruction Manual
NIE	National Institution of Education
NEC	National Education Commission
ICMI	International Commission on Mathematics Instruction
QCA	Qualifications and Curriculum Authority
NCTM	National Council of Teachers of Mathematics
NVQ	National vocational qualification
SBTD	School Based Teacher Development
NSBTD	Non- School Based Teacher Development
ADE	Additional direct of education
ISA	In service advises
JSME	Japan Society of Mathematics Education

CHAPTER 1

INTRODUCTION

1.1 Introduction

Mathematics is the prime source to create an active community to reach its success. Each step of life is depending on mathematics. It aids to organize patterns and regularities in the world. Albert Einstein stated that "as far as the laws of mathematics refer to reality, they are not certain; and as far as they are certain, they do not refer to reality."

Mathematics education enhances several skills of the students like problems solving, logical argument and reasoning. Since Mathematics creates challenges in the thought of students and promotes creativity, intelligence and curiosity it brings out a feeling of society.

‘Geometry’ concepts are used very often in mathematics contexts and real-life situations. They are very essential to the study of Mathematics as well as many other related fields and their uses are infinite. Geometry is one of the most important branches of mathematics education because it aims to provide students with the ability of critical thinking, problem solving and a better understanding of the other subjects in mathematics by making the students have a high level of geometric thinking skills (Sahin. O, 2008). As geometry is a major component in mathematics, geometrical thinking is a fundamental way to engage with mathematics. Although geometry is an important area from the mathematical point of view it is a difficult subject for our students; a subject scarcely considered till now by researchers in mathematics education (NIE, 2011). The latest international studies and some cognitive research studies suggest that there is space for improvement in the geometric knowledge of the students.

Students’ performance in geometry is not adequate to reach its standard. International Commission on Mathematics Instruction (ICMI, 2016), a world recognized mathematics organization says that in recent years many countries have been failed to perform geometrical concepts in a better way. Qualifications and Curriculum Authority (QCA, 2014) for England also confirmed that poor performance in

geometry is obvious in European countries and many other countries. Poor performance in geometry is observed not only in other countries but also in Sri Lanka too. An analysis of the GCE O/L mathematics papers carried out by the Research and Development branch of the Department of Examination reveals that in Sri Lanka in the years 2014 and 2015 the performance of students in geometry was extremely poor. The poor performance of geometry is caused by a number of factors. To identify the factors influencing learning geometrical concepts and how young people learn geometry Van Hiele theory helps a lot. An understanding and application of the Van Hiele level of geometrical thinking would influence teacher practice, teaching strategies, students' ability, attitude and teaching and learning geometry (Gates, 2011).

Scholars and researchers have the responsibility to make our youngsters realize the actual usage of geometry and make use of them in their day to day life. Most of the studies on geometry focus on weaknesses of the methodology employed for teaching geometrical concepts. Only a few studies on geometry were focused on students' conceptions of geometry, rather than reporting on misconceptions. It seems that a new study on geometrical concepts is necessary to find out the reasons for the less achievement in geometry.

Since students from grades 10 and 11 show low performance in mathematics especially in geometry (NEC, 2011) steps have to be taken to find the causes for the poor performance of geometry. Scholars point out so many factors as the reasons for the poor performance of geometry. As per the literature, less interest in geometry, inappropriate study habits of the students, students are not much clear about the basic concepts of geometry, less ability of students, inappropriate teaching methods, inadequate qualified teachers, inadequate and inappropriate resources and tools in schools, less importance given in the curriculum and unsuitable textbooks influence a lot in the poor performance of students in geometry. As students' performance is affected by several factors including school-related variables, student-related variables and teacher-related variables, this study analyses the impact of factors and reveals the causes for the poor performance of students in geometry and explore some remedies to the knowledge development of the students in geometry. Based on the purpose of the study, the sample was selected from the schools located in the Jaffna district and targeted to the senior secondary students and teachers from the Jaffna district schools.

1.2 Statement of problem

Geometry captures a major role in mathematics education. But in a real situation, our student community does not pay attention to geometry and show poor performance (NEC, 2011). Since the study was targeted to analyze the domain factors influencing learning geometrical concepts of senior secondary students, the ability of students, teaching methodology and instructional materials and tools is considered to be the major problems to analyze the impact of domain factors on the development of geometrical knowledge and concepts.

Srilanka ranks poorly in terms of mathematics education in schools (World Bank, 2003). Sri Lankan students rarely realize the actual usages of geometry and do not show much interest in learning geometrical concepts. This leads them to perform low in geometry and there are some drawbacks in learning geometrical concepts (Kalamani, 2014). As the causes were analyzed in students' view, ability of students is the major cause that affects the performance of the students in geometry. If students are much clear about the basic concepts of geometry and they are eager to learn geometrical concepts, they can develop their geometrical knowledge by utilizing proper study habits. But, though they spend nearly eleven years in learning mathematics from grade one they struggle a lot in the subject, especially in the portion of geometrical concepts. As the students don't follow the appropriate method to learn the concepts in geometry and are with inadequate prior knowledge, their interest in learning geometry become poor. Lack of ability of the students forces them to neglect the geometry questions in G.C.E O/L mathematics exam paper. Therefore the poor performance of geometry affects the secondary education of the students and their future performance too.

G. C.E O/L and G.C.E A/L are considered the most important exams in the Sri Lankan education system. The effect of 6 years of education from grade 6-11 is reflected in G.C.E O/L exam results. To continue their studies in A/L, students must get through the O/L exam. Hence, the G.C.E O/L examination functions like a hurdle for the students to move on to G.C.E A/L. As the G.C.E O/L exam paper contains 22% of questions in geometry, it's obvious that without attempting the questions in the geometry portion no one can score good results.

Table 1.1: Content of mathematics portion in G.C.E O/L exam paper

Content	percentage
Number	23
Measurement	15
Algebra	20
Set and Probability	10
Geometry	22

Source: NIE Report (2010)

According to the above information, it is clear that 22 percentage of the content consists of geometrical concepts on the paper. Students are eligible enough to choose the stream in A/L as per their results in O/L. Those who wish to choose the science stream should obtain a credit pass and others who wish to study in commerce and arts stream should get at least an ordinary pass in mathematics. The pass in mathematics would be easy if they achieve geometrical knowledge. Therefore, knowledge in mathematics as well as in geometry will help a lot to choose any stream in Advance Level.

Table 1.2: Island-wide Students' Achievement level in G.C.E O/L

Description	2015	2016	2017	2018
No. sat	273224	286251	296812	296029
Not Qualified for A/L	83796	86043	79997	73742
Not Qualified for A/L (Percentage)	30.57%	30.06%	26.95%	24.91%

Source: DOE, 2019

The table shows the achievement level of the students in the G.C.E O/L exam. As per this table, the percentage of failures comes down in the recent past. As a result, the number of dropouts who fail to complete secondary education decreases in our society. Failure in mathematics is the main cause of dropouts. In 2018 the percentage of dropouts is 24.9%. Among this 24.9% of dropouts, most of the students failed in mathematics. According to DOE (2018) report, at the national level, half of them become failures as they fail in mathematics.

Table 1.3: Students' Achievement level in Mathematics in G.C.E O/L

Description	2017	2018
No. sat	295406	296205
No. of failures in maths	93642	97032
Percentage	31.7%	32.76%

Source DOE(2019)

It is not possible to achieve the target in mathematics without focusing our attention on geometry. To succeed in mathematics, students should succeed in geometry. In the Sri Lankan education system, if a student fails to succeed in mathematics in G.C.E ordinary exam he will be disqualified in the whole exam. Therefore, failure in mathematics will force the students to fail in their entire education progress.

The “13 years” Guaranteed system was introduced in 2017 to minimize the dropouts of the country. Though this system creates a student community with vocational talents, some drawbacks were identified in this system. This stream was introduced to provide 13-years of continuous education for all students, disregarding their performances at the GCE O/L exam. By this system, students will get the attitude that they can continue their education without passing the G.C.E O/L exam. This thought will lead them to neglect secondary education too. As most of them consider mathematics as a difficult subject 13years guaranteed system lead our students to neglect the subject from their secondary education. This allows students to ignore geometry and perform low. Less focus on mathematics, especially on geometry will increase in the percentage of failures. Therefore the ability of students including prior knowledge, study habits and interest in learning geometry affects the performance of students in geometry and their entire education progress.

Mathematics is very important as well as a very interesting subject and it is an easy subject if the students are given opportunities to learn the concepts of geometry logically, this subject will be interesting forever (Jaffer, 2004). Teachers are the major facilitators to the students but they fail to pay attention on student misconceptions, and they don't take remedial actions for them (Clements, 2011). Moreover, they don't understand the ability of the students and plan the lessons

accordingly. Since the teachers don't update themselves educationally, they fail to use proper learning theories and modern teaching tools too.

Teachers with good mathematics knowledge and effective teaching practices are needed for imparting excellent mathematics education. However, in Sri Lanka teachers who are entitled to teach mathematics do not have subject knowledge up to the standard (NEC, 2011). So, the quality of the mathematics teachers in the school system influences the students' performance in geometry. At present, geometry is taught in the classroom through a series of activities. Even though all activities are described in step-by-step detail, most of them follow the same format and prescribe the same teaching strategy, teachers are not capable enough to choose the more appropriate teaching strategies and tools.

In 2008, a study was conducted by NEC to assess the mathematics knowledge of teachers in schools. A question paper contained 9 questions on the themes of measurement and algebra was distributed among the teachers. An analysis of the marks obtained by the teachers revealed that 36 % of teachers from the Western province and 51 % of the teachers from the other provinces scored less than or equal to 5 marks out of 10 (NIE, 2008). DOE conducted another study in 2010 with 207 teachers who had applied for paper marking. They were given a question paper similar to the GCE O/L mathematics paper (DOE, 2010). Although most of the teachers performed well on paper- I, they fail to score marks on paper- II. Paper-II tested higher order skills and 50 % of the paper contained problems on algebra and geometry. Only 42 % of the teachers were able to score over 80 % on paper- II. Question by question analysis revealed that teachers' performance was poor on geometry questions. The same weakness was reflected in students' performance at the GCE O/L examination.

As teachers are with inadequate geometrical knowledge, they are not capable enough to impart the concepts through appropriate teaching methods. As a result, the student community faces many challenges in gaining geometrical knowledge from teachers and show poor performance in geometry.

Along with students and teachers, instructional materials and tools also affect the performance of students in geometry. Inadequate and improper instructional materials and tools in schools influence a lot in learning geometry. Less importance given to geometry in the curriculum, content and arrangements of the textbooks, inappropriate

usage of mathematics activity rooms in schools shows an impact on the performance of students in geometry.

The Sri Lankan government implemented several reforms from time to time. According to the World Bank report inapplicability of the Sri Lankan curriculum influence a lot in the performance of mathematics.

In the curriculum before 1972 students had been given chance to learn mathematics consciously. During that time mathematics was categorized under three divisions as geometry, algebra and arithmetic. Geometry was taught as a main subject and students were divided in grade 9 based on the stream they had chosen. They were taught only that particular subject throughout the following years. Students who selected science stream in A/L were taught a high level of mathematics concepts and they were given higher order thinking of geometry. For science students, the chance was given to learn all the levels of geometry. This benefited them to capture the advanced level themes easily and promoted them to get good results. This helped to fill the gap between G.C.E (O/L) and G.C.E (A/L).

But, in the reforms which came after 1977, the weight given to the geometry was minimized step by step and at present, the G.C.E O/L exam paper doesn't contain the real application of proof. Although the proof is given in the structure method, students often ignore this particular section (Ekanayake, 2002). From 1990, a student could pass the mathematics subject without learning geometry. Less importance of geometry in the curriculum affects the performance of the students and they fail to grasp knowledge in geometry and face many challenges when they join in advanced level.

With the implementation of new reforms and curriculum, the importance given to mathematics as well as geometry has become poor and the majority of students from secondary schools try to avoid the subject. As the importance given to geometry was decreased step by step in the Sri Lankan curriculum, grade 10 and 11 students are not given adequate geometrical conceptual knowledge relevant for the stream they wish to study in G.C.E A/L. This increase the gap between G.C.E (O/L) and G.C.E (A/L).

Moreover, the Sri Lankan government provides textbooks to facilitate the teaching and learning process. Despite its good arrangement and content inappropriate usage of textbooks in learning and teaching geometry affects the performance of the

students. The mathematics activity room is another recent concept introduced in the curriculum to promote activity based learning. But this concept is not utilized rightly in geometry teaching and learning. Therefore, to analyze the impact of domain factors on learning geometrical concepts of senior secondary students, the ability of the students, teaching methodology and instructional materials and tools are identified as problems of the study. Identifying the factors for the lack of student achievement in geometry will help to make an immense change at present. Hence the change will occur and it will reduce the percentage of failures in mathematics and develop geometrical knowledge and concepts.

1.3 Background of the study

Geometry is one of the most established branches of mathematics. It has been placed in a central place in mathematical education all over the world. Geometry is an important subject, with wide applications and a long history. It deals with matters we find attractive and for which we have a strong visual capacity (Battista, 1992). On the surface, then, it would appear that geometry should be one of the easiest branches of mathematics to teach. But this is not the case in much of the developed world (ICMI, 2010).

The importance of geometry to the study of mathematics was recognized by the Russians in the 1960s and then by the Americans in the 1970s accepted the importance of geometry in the study of all mathematics, increased its prominence in all levels of mathematics study in schools (NCTM, 1989).

In many countries during a period of 1950s and 1960s new syllabi were developed where the emphasis was on formal structures which were predominantly algebraic. At the same time, the range of approaches to geometry was broadened from its traditional Euclidean base to include the use of transformations, vectors, matrices and some topology.

In recent years the countries have been reviewing the aims, content and approach of their geometry curricula. NCTM focuses on developing students with the reasoning abilities and communication skills required to solve everyday problems, by integrating the traditional content of mathematics to make meaning of the real world. Although Sri Lankan educators realized its validity our student community has not understood its necessity. Especially in the Jaffna district, secondary school students

show low performance in geometry (DOE, 2017). There are so many factors that contribute to the poor performance of geometry. Factors like the ability of students, teaching methodology and instructional materials and tools are considered as major causes for the lack of performance in geometry.

The government of Srilanka accepts the necessity of mathematics education and has taken several steps from time to time, from primary education to advanced level education (NEC, 2003). At the same time, the Srilankan government tries a lot to eradicate the drawbacks in the mathematics education process to develop student ability by introducing proper teaching methods and instructional materials and tools. But still, both the teachers and students struggle a lot to reach the target in teaching and learning mathematics education.

In Sri Lanka, mathematics is a compulsory subject, and it is the basic requirement to continue their higher studies. In the Sri Lankan curriculum, geometry is introduced in grade 6 and taught step by step until grade 11. Sri Lankan syllabus contains all the common topics in geometry: Plane geometry, Solid geometry covering the surface area and volumes of solids such as cubes, pyramids, cones and cylinders, Transformation geometry and Analytical geometry. However, the depth given to the topics is not sufficient for the improvement.

Analysis of the GCE O/L mathematics papers carried out by the Research and Development branch of the Department of Examination reveals that in the years 2014 and 2015 since the new curriculum was introduced, the performance of students in geometry has been extremely poor. In the year 2014, questions on geometry were selected by less than 45 percentage of the students and over 75 percentage of those who attempted these questions scored less than 25 out of a total of 100. Similar trends were seen in 2015 too.

Sri Lankan government introduced a reform in 1997 to overcome the problems faced in the past. In the 1997 reforms, the following problems were identified in mathematics education. This reform specifies the problems related to the shortcomings in the curriculum, lack of professional and content knowledge of the teachers and the other related factors which affect the performance of the students in mathematics.

According to this reform:

1. The education system does not produce young people with basic mathematics skills and competencies required by the world of work.
2. The curriculum has not been modernized to suit the needs of the modern world.
3. There is a dearth of mathematics qualified personnel in the job market.
4. Modern techniques are not used in the learning teaching process.
5. Teacher guides are neither comprehensive nor adequate.
6. Teachers are neither adequately trained nor motivated.
7. Education software and other essential teaching materials are not available.
8. Textbooks are scarce or not available.
9. Monitoring mechanisms are not effective.
10. Mathematics education is too theoretical with little emphasis on applications.

By identifying the above-mentioned problems, the NEC had recommended the following proposals to improve the quality of mathematics education as per 1997 reform. With 1997 reforms some changes were made in the mathematics curriculum and teaching learning process. In spite of all these, students were not able to update their standard of mathematics to its maximum. The poor performance of geometry was also followed by its implementation.

1. The extensive revision or re-formulation of the curricula.
2. Modern learning-teaching methodologies such as an activity-based student-centred approach. Use of IT as a tool in the learning teaching process.
3. Preparation of textbooks and teacher guides based on the curricula.
4. Both pre-service and in-service teacher training in mathematics strengthened.
5. Promotion and popularization of mathematics.

Source: NEC (2003)

In 2007 competency-based teaching and learning method was introduced and the major aim of the mathematics curriculum under these reforms is to create individuals who are able to think mathematically and apply mathematical knowledge effectively and responsibly in problem solving and decision making(NIE, 2009). The main target of the 2007 reform was to develop the basic competencies of each subject. Activity-based instructional lessons were introduced to achieve the targeted competencies by creating interest in the subject. Students were given chance to show active performance and choose their preferred study habits. In this system 5E

method of teaching was utilized to maintain student-centred learning. A continuous assessment method was introduced to check the prior knowledge and current knowledge of the students. However, the steps introduced in the 2007 reforms also did not help much in learning geometrical concepts.

To enhance the mathematical knowledge along with technological skills NEC proposed a number of acts recently. In 2015 technology stream was introduced. In order to minimize this circumstance and to create a society with the students of technological knowledge and skills, the introduction of a technological stream helps a lot.

Students in the technological stream must achieve the competencies in mathematics to develop their technical knowledge and skills. Especially they must improve their knowledge in geometrical concepts to enhance their technological skill. So they have to be given geometrical knowledge at least on the basic level. In this case, it is almost enough to give geometrical thinking levels 1 & 2 to promote them to compete in the vocational unit. So, by considering all these facts our curriculum must be remodified.

Sri Lanka implemented compulsory education as per the concept of UNESCO's education for all. Students have the opportunity to get degrees like NVQ through the institutions like technical colleges and NITA. These institutions offer degrees in students' preferred fields. But the dropout students failed to get a chance to continue their studies. To make a solution to this problem Sri Lankan government introduced 13 years of guaranteed education in 2017. The objectives of this system are providing opportunities for all the children to enrol on vocational stream after O/L and provide flexible learning opportunities to meet the needs of all children after O/L. In this system, students who fail in G.C.E O/L also get a chance to continue their school education & develop their skills in vocational training. In 13 years guaranteed education system some schools were selected in area wise and the students from that area are inculcated and they are given the NVQ level 4 at the end of this course.

This stream is referred to as the Professional Curriculum, which consists of two parts as General Curriculum and Applied Curriculum. The general curriculum covers nine main subjects and it is compulsory to complete these subjects within the first six months of the academic year. This Common Curriculum intends to assure the

personality and other basic skills necessary for the practical workings of the students.

However, in 13 years of guaranteed education syllabus mathematics is not included. But for considerable achievement, students should be taught the basic level of geometrical thinking without focusing on other high levels of geometrical thinking. For better understanding, these basic concepts should be given at least in their secondary education. It is obvious that no one can learn any course by neglecting the mathematics subject. Unless the students develop their geometrical knowledge in mathematics every single student will face difficulties to meet challenges in G.C.E A/L class.

Students who face difficulties in learning the mathematical concepts in secondary education will struggle a lot to grasp the concepts when they are engaged in 13 years guaranteed system. Therefore, to learn these geometrical concepts prior knowledge in mathematics is essential for them to understand the concepts. So, the application and usage of geometry should be developed at the secondary level.

The geometrical content of the new National Curriculum, with a few adjustments, form an appropriate basis for good geometry education. To achieve this, considerable changes are needed in the syllabus and the way geometry is taught. In the year 2019, the Northern Provincial education department conducted a special exam for grades 9, 10 and 11 on geometry in addition to the year-end exam. The purposes of this exam were to develop the standard of geometry, to make the students realize the importance of geometry and to create interest to achieve the high level of mathematics in Northern Province.

The project was implemented in the Jaffna district in Northern Province. Since our zonal director and ADEs of mathematics observed a decreasing trend in mathematics pass rates during these years they realized the importance of geometry in passing mathematics. So, they conducted this exam to promote knowledge in geometry.

Mathematics teachers who have been involved in teaching mathematics at the secondary school level for several years have seen how learners struggle with the solving of geometry problems. Bringing about improvements in geometry teaching will require a significant commitment to continue professional development alongside the development of appropriate supporting materials.

Teacher quality is the most important factor affecting students' performance. According to the school census data in 2012, there were 223,333 teachers in

government schools. Of them, 43,061 (19.28%) were not professionally qualified. It has also been revealed that 38.8 percent of teachers are graduates, but it is not mentioned how many of them are professionally qualified.

The Ministry of Education in Sri Lanka has recognized that good teachers are keys to enhance the quality of education (MOE, 2004). NIE designed and developed the curriculum for general and teacher education, providing professional development of the educational community. MOE (2005) also states that Sri Lanka has already considered professional development for teachers as an essential part of the education system because professional development enhances the quality of good teaching (Sethunga, 2007). 'No untrained teacher enters the classroom' is one of the policies the government has attempted to implement during the past decade.

Sri Lanka has a variety of teacher training programmes in operation. Professional development programmes for teachers in Sri Lanka are divided into two groups. The first is the School-Based Teacher Development (SBTD) programmes, these programmes are considered an effective strategy for teacher professional development because they are designed, organized, implemented, owned and controlled by principals and teachers of the schools (Bandara, 2010). The activities under SBTD typically include School-based mentoring, peer learning, peer coaching, individual consultation, and visits to other classrooms. The second is the Non-School-Based Teacher Development (NSBTD) programme, these programmes are organized and implemented by external organizations or institutes of the schools, such as Zonal Education Offices, the Provincial Ministry of Education, the Ministry of Education, the National Institute of Education as well as various Non-Government Organization. As a result, these teacher development programmes may create ineffectiveness in teacher professional development.

But, Sri Lanka has been lagging behind the developed countries as regards teachers' professional qualifications. Though they spend much interest and willingness for teacher development still there is a room for improvement.

So it is important to make sure students are properly prepared by obtaining quality education through qualified teachers using quality geometry resources and it is very necessary to analyze the impact of factors that affect the performance of geometry and help the students to succeed in the lesson. This study was an attempt to identify the Domain factors influencing learning geometrical concepts of senior secondary students in Jaffna government schools.

1.4 Purpose of the Study

The purpose of this study was to analyze the impact of domain factors on learning geometrical concepts in senior secondary students of the Jaffna district. Due to many causes students from secondary schools in the Jaffna district show poor performance in geometry. By analyzing several factors three factors were identified as domain factors that affect the performance of geometry. Teaching methodology, ability of student and Instructional materials and tools are treated as domain factors that affect the performance of geometry. How the domain factors influence the development of geometrical concepts was analyzed. By analysing the impact of domain factors it was expected to overcome the challenges faced by both teachers and students in teaching and learning geometry and improve the performance in geometry. Moreover, it was also expected to identify the approaches that the teachers adapted to teach geometrical concepts in a more effective way along with the relevant strategies used by the students in learning geometrical concepts.

1.5 Significance of the study

Mastering knowledge in geometrical concepts tends to be one of the most difficult skills to acquire. This study intended to make several points regarding teaching and learning geometrical concepts. The findings of this study help the teachers to identify the factors which affect the Students' achievement in geometry and to choose relevant strategies. It also helps them to identify the students' attitudes and teachers' role in understanding geometry concepts. It guides them to identify the challenges faced by the students in understanding geometry concepts and how they influence their performance in geometry. In addition, this study may provide a number of teaching techniques that will enhance the development of geometric knowledge and thinking and produce materials and tools that will help to improve geometric thinking. The findings of this study will also provide information about geometry and assist the student community. It may give information to the Ministry of Education and NIE about the needful changes in the mathematics curriculum. Moreover, it can be used by educators to design a syllabus for the academic curriculum. Further, it can be useful to construct written activities and to prepare exam papers for the G.C.E O/L examination.

1.6 Sample of the study

This study was conducted in the Jaffna district in the northern province of Sri Lanka. The target population was comprised of 379 students, 50 teachers, a provincial ADE and two zonal ADEs of mathematics from senior secondary schools. The sample consisted of the students and teachers from senior secondary schools in the Jaffna district. As Jaffna district has 5 educational zones, this study was targeted to the teachers and students from senior secondary schools from each educational zone. Moreover, the participants were chosen from all types of schools like 1AB, 1C and type 2 schools.

1.7 Research Questions

To fulfil the purpose of the study following research questions were targeted.

- Research question: What is the impact of domain factors on the development of geometric conceptual knowledge and skills?

As domain factors are categorized into teaching methodology, instructional materials and tools and ability of students the following subsidiary research questions are formed to have a depth analysis on the factors.

Subsidiary research questions

1. Research question 1: What is the impact of ability of students on the development of geometric conceptual knowledge and skills?
2. Research question 2: What is the impact of teaching methodology on the development of geometric conceptual knowledge and skills?
3. Research question 3: What is the impact of instructional materials and tools on the development of geometric conceptual knowledge and skills?

1.8 Research Objective

At the end of the study we will get to know:

- To identify the current status of students' performance in geometry.
- To analyse the impact of the domain factors on the development of geometric conceptual knowledge and skills?
- To analyse the impact of the ability of students on the development of geometric conceptual knowledge and skills?
- To analyse the impact of teaching methodology on the development of geometric conceptual knowledge and skills?
- To analyse the impact of the instructional materials and tools on the development of geometric conceptual knowledge and skills?

1.9 Hypotheses

The following hypotheses were chosen in order to meet the research objectives.

- $H_{O,1}$: There is no impact of the domain factors on the development of geometric conceptual knowledge and skills.
- $H_{O,4}$: There is no impact of ability of students on the development of geometric conceptual knowledge and skills.
- $H_{O,2}$: There is no impact of teaching methodology on the development of geometric conceptual knowledge and skills.
- $H_{O,3}$: There is no impact of instructional materials and tools on the development of geometric conceptual knowledge and skills.

1.10 CONCEPTUAL FRAME WORK

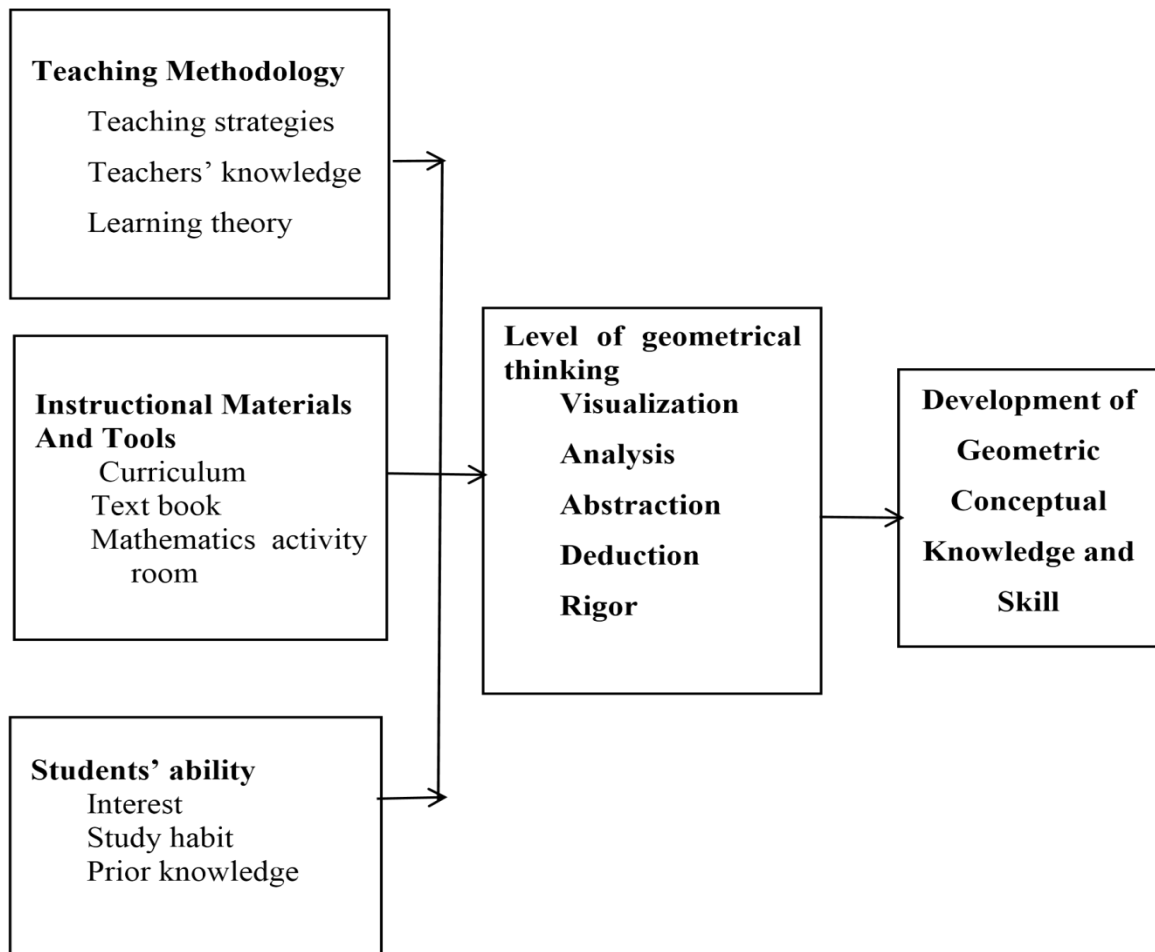


Fig1: Conceptual framework

From the scholarly articles the factors influencing learning geometrical concepts by senior secondary students are categorized under three main factors. They are the teaching methodology, Instructional materials and tools and the ability of students. At the same time, geometrical thinking is levelled in five steps according to Van hiele theory. Such as Visualization, Analysis, Abstraction, Deduction and Rigor. If the students succeed to pass the prescribed levels step by step through their learning, definitely they will develop geometrical knowledge and skills. How the development of geometrical knowledge and skills is influenced by the above mentioned factors is analyzed in this study.

1.11 Definition of Terms

It is best to define any key terms in the research upfront. So that everyone has a shared understanding. The following terms are defined in relation to the study.

Achievement in mathematics could be all possible tasks accomplished by students in and out of school

Analytical geometry is the geometry that deals with the relation between algebra and geometry, using graphs and equations of lines, curves and surfaces to develop and prove the relationship.

Euclidean geometry is geometry based on postulates.

Geometry is the branch of Mathematics that deals with space, figures in space and with properties of those figures such as size and shape

Plane geometry is the geometry that deals with figures in a two dimensional plane.

Postulates or axioms are unproven but universally accepted assumptions such as ‘there is one and only one line that passes through two distinct points.

Solid geometry is the geometry that deals with figures with three-dimensional space.

Spherical geometry is the geometry that deals with figures on the surface of the sphere.

Strategy is the overall way in which the process of instruction is organised and executed.

A syllabus is a document that shows the content to be covered in a given subject and time at a particular level.

Theorems are logically deduced postulates.

Concepts

A general idea usually expressed by a word, which represents a class, or a group of things or actions having certain characteristics in common (Quillen & Hanna, 2006).

Mathematics Achievement

Successful accomplishment of mathematical content, especially by means of skills and practice (Olga, 2008).

Misconceptions

Conceptual knowledge that differs from commonly accepted scientific consensus (Sanger, 1996).

Motivation

A display of live curiosity, concern and interest towards the subject matter which is being taught

1.12 Limitation of the study

The limitations the researcher faced while conducting this study are stated briefly as follows: Though the factors influencing learning geometrical concepts are common to all parts of Sri Lanka, the sample was limited only to the Jaffna district level rather than the national level due to the limitation of time, cost, accessing problem of the researcher and as per the limitation of the target population. Although there are many schools in the Jaffna district only 22 schools were selected as sample schools. As the target group was small it may not reflect the actual outcome of the study. The researcher has faced many practical challenges to get the approvals and to meet the interviewees such as Zonal addition directors of mathematics, mathematics teachers and the subject coordinators. As teachers were keen on completing the syllabus they were reluctant to give the periods to the researcher. As a result, the researcher couldn't spend adequate time with the students and grasp their actual potential. Though the students were very keen to learn from the researcher, the classroom atmosphere did not support much in some schools.

1.12 Ethical consideration

Participation in this study was voluntary and participants were told that they can withdraw anytime if they no longer want to participate. All the participants were treated with respect. Before starting the research, the participants would be given information outlining the purpose of the research, the types of data and how the data would be used and reported.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter deals with important reviews on domain factors influencing learning geometrical concepts of senior secondary students. The purpose of this chapter is to review literature related to three domain factors that influence the Students' performance of geometry: students' ability, instructional materials and tools, and teaching methodology. This chapter also focuses literature on the impact of domain factors on Van Heile level of geometrical thinking. Moreover, this chapter covers an in-depth discussion and critical analysis of the van Hiele theory and describes the five van Hiele teaching phases. Chapter two begins by giving a brief overview of the importance and development of geometric conceptualization worldwide. This is followed by an analysis of student performance and achievement which highlights the current performance and problems issues in geometry. Also, the literature related to the factors influencing learning geometrical concepts is deeply analysed as per Sri Lankan education system to see the impact of domain factors in detail.

2.2 Geometric conceptualization

Geometric conceptualization is fundamental for many other fields. This section focuses on the literature that concerns the importance of geometrical concepts in general and in mathematics and the necessity of why students have to be given geometrical concepts to buildup an active society to lead a convenient life.

2.2.1 Importance of geometrical concepts

Many researchers try to say the importance of geometry in their research. Geometry is not only an academic subject but it is mingled with each and every step of our real life. Geometry has many applications in everyday life. It can be found in every part of

natural and artificial creations of the world and the elements of geometric form can be seen everywhere.

Other subjects in the curriculum make use of geometrical ideas and techniques. Geometry is also be used to build foundations for further studies in mathematics. For example, fraction concepts are related to geometric part-to-whole constructs. Ratio and proportion are directly related to the geometric concept of similarity. Measurement and geometry are clearly related.

Geometry has always been associated with practical situations. As such geometry is absorbed and combined with every individual's everyday activities. And geometry is a logical system and it uses deductive reasoning in developing new truths from accepted facts. For example, understanding the geometrical properties of the universe continues today in the work of Weeks (1998) and others. Kepler (1571-1630) stated that "where there is matter there is geometry" (World Book, 1999). Using geometry and astronomical observations it was shown that the planets travelled in elliptical and not in circular orbits around the sun and by doing so a 2000 years old belief was destroyed.

Since Geometry is used daily by many people, Van de Walle (2001) also describes the importance of geometry. Scientists of all sorts, architects and artists, engineers, and land developers are just a few of the professions that use geometry regularly. Euclidean geometry forms the basis of engineering today and is used, for example, in architecture, carpentry, navigation, metalwork, art and sport. Those who work in construction will need to ensure angles are measured very precisely and that all lines are straight for a building to be safe; architects and interior designers need to use their geometry knowledge to guide their designs, and even those who work in computer animation will use geometry while putting together an animated film.

Hollerands (2003) reveals there are three important reasons to learn geometry, which gives an opportunity to the students to think about the important concepts in mathematics, provide a context in which students can see mathematics as a discipline are interconnected and provide the opportunity for students to engage in high-level reasoning activities using a variety of representations.

As many literatures emphasize the importance of geometrical concepts to all the fields of study, even in day to day activities, this study tries to bring out the causes that affect learning geometrical concepts in mathematics.

2.2.2 Importance of geometrical concepts in Mathematics

Mathematics plays a key role in raising a country's knowledge economy. The general aim of mathematics is stated as making an individual acquire the mathematical knowledge needed on daily basis, teaching how to solve problems, making him/her have a method of solving problems and acquiring reasoning methods (Altun, 2008). Geometry plays an enormous role in mathematics. It is not just dealt with some sort of theorem; it is the most important content for our day to day life. Geometry covers a whole range of aspects which includes spatial thinking, geometric measurement, and visualization related to geometry, as well as encompassing theoretical developments and research.

As Geometry is one of the key branches in mathematics education, it functions as part and partial of mathematics. The Greek called Geometry as 'earth measure', it is the science of measuring land. The meaning of Geometry is not as straightforward as one may think (Royal Society, 2001). Geometry encompasses a whole range of concepts including triangles and trigonometry, area and perimeters, lines and angles, and all other shape-related problems which will be encountered in everyday life. Studying geometry is important to develop skills such as logic, problem solving and spatial understanding (Anderson, K, 2008). Geometric concepts start from organising spatial experiences that lead to the formulation of definition and theorems and they do not start from formulating definitions and theorems,

Every unit whether it is built around a particular kind of geometric property will aim at the understanding and mastery of several kinds of things (Charles,1990). This includes the mastery of an associated mathematical vocabulary, clarification of certain geometrical concepts, understanding of relationships within and among particular geometric figures under given conditions, the ability to make acceptable and helpful drawings and to use a suitable notation, the ability to understand certain geometric facts (Charles,1990).

Geometry may be Euclidean or non-Euclidean depending on the postulates used. Generally, in mathematics curricula all over the world Euclidean geometry is included. Euclidean geometry contains the studying of the shape, size and position of both plane shapes, such as triangles, and solids, such as cubes, and includes congruence, similarity and parallel lines. Moreover, Euclidean geometry fosters creative and imaginative thinking through the solution of problems and the writing of proofs hence the Sri Lankan mathematics curriculum deals with Euclidean geometry.

According to David Joyce(1994), Geometry, Euclidean or otherwise, offers visual structure to abstract concepts of mathematics, and in certain cases offers insight into the logic of various parts of mathematics such as calculus, group theory, or abstract algebra. In the essence of engineering, geometry typically is thought of as a visual “measurability,” providing simulation and modelling of real-world applications to offer a sort of predictability to concepts applied to structures we find in our everyday lives.

In Anderson’s and Kosslyn’s research on “Performance on middle school geometry problems” (2007), they mentioned several geometrical concepts and the importance of geometry.

In National Council of Teachers of Mathematics (NCTM, 2000) revised work ‘Principles and Standards for School Mathematics’, the NCTM reinforced the importance of geometry. They stated that “Geometry has long been regarded as the place in the school mathematics curriculum where students learn to reason and to see the axiomatic structure of mathematics” and that “Geometry offers a means of describing, analysing, and understanding the world and seeing beauty in its structures”. Supporting their 1989 statement that geometry was useful for the learning of other branches of mathematics and problem solving, the NCTM further stated in 2000 “Geometric ideas can be useful in other areas of mathematics and in applied settings”.

Spatial reasoning is an important form of problem solving, and problem solving is one of the major reasons for studying mathematics. Geometry can be used to develop students’ spatial awareness, intuition and visualization. It can also be used to solve practical problems. Geometry is linked to many other areas of mathematics. When children begin learning about shapes they will study 2D shapes and their properties,

and this knowledge is consistently built upon as they progress through their education. Geometry problems might also become number problems or algebra problems (Kosslyn, S. M., 2008).

The rapid development in a range of technologies means that citizens now and in the future will interact with a variety of forms of displayed images. Geometry has a role to play within the development of citizenship in enabling students to interpret, manipulate, control and create such images. If geometry increases students' interest in mathematics that will make the effort worthwhile.

Therefore it is noted that geometrical concepts and mathematics are combined with each other. If a student fails to understand the geometrical concepts he/she will fail in mathematics too. So to raise the mathematical knowledge and geometrical concepts it should be given through clear explanation and application. As there is a need to understand the current status of the performance of geometry in and around the world to make a significant development, this study analyses the students' performance of geometry in some countries to give a world view. Then the status of student performance of geometry in Sri Lanka is explored to identify the factors that influence the student performance of geometry.

2.3 Students' Performance in geometry

Many studies reveal the poor performance of geometry all over the world. According to Higgins (2005), many people today believe geometry has failed learners because it is difficult. Most of the learners consider that geometry is not an easy subject to learn. To master it, students have to realize the actual value of it and do some hard work.

It also needs careful and sustained attention, perseverance and a measure of ingenuity. To get real achievement in the subject, the government of each country implements some changes in their curriculum from time to time. In this study, how the Countries like England, USA, Russia, Malaysia, Singapore, China and Japan treat geometry is taken into brief discussion and have a look on the status of geometry worldwide and then move on to Sri Lanka.

2.3.1 Worldwide Performance of students in geometry

The importance of geometry to the study of mathematics was recognized by the Russians in the 1960s and then by the Americans in the 1970s (Hoffer, 1988; Teppo, 1991), who, having accepted the importance of geometry in the study of all mathematics, increased its prominence in all levels of mathematics study in schools (NCTM, 1989).

According to Clements and Battista (1992), “primary and secondary school learners in many countries, especially in US are failing to learn basic geometric concepts and become proficient in geometric problems solving”.

A. M. Pyshkalo (1981) commented on the Soviet educators' extensive research and experimentation on van Hiele's theory. It was reported that the Soviets have substantially revised their geometry curriculum based on the van Hiele levels of thinking in geometry.

Developed countries such as England and Germany teach Geometry in their own method. In order to improve the learners' performance in Geometry, these countries focus on designing appropriate and relevant procedures for the assessment and recording of learners' progress and attainment. Teachers use these new concepts strategies and methodology in a dynamic creative learning environment in the teaching of Geometry (Jayaprakash, 2005).

In Malaysia, geometry is taught at primary and secondary school levels. Geometry was defined as a basic skill by the National Council of Teachers of Mathematics (NCTM, 2000). Learning geometry may not be easy and a large number of students fail to develop an adequate understanding of geometry concepts, geometry reasoning and geometry problem-solving skills (Battista, 1999; Idris, 2006).

Until recently, the geometry curriculum in the United States has been very poorly defined. Children in the early grades learned to identify a few basic shapes. Beyond that primitive bit of knowledge, teachers have had little guidance on what is important. However, the work of two Dutch educators, Pierre van Hiele and Dina van Hiele-Geldorf is beginning to have an impact on the design of geometry instruction and curriculum (Van de Walle, 1998).

National Council of Teachers of Mathematics (NCTM ,2000) accepted the importance of Geometry in USA and constructed the objectives. According to them, learners should be geometrical literate to cope with everyday life, Geometry should empower learners to solve problems at the workplace, Mathematics should prepare the most able learners to be mathematicians, scientists, statisticians and engineers and Geometry should be viewed and taught as part of the culture.

In South Africa's mathematics curriculum, Euclidean geometry is persisted as a major part of it. A possible rationale for this situation was given by De Villiers (1986), who claimed that Euclidean geometry allows pupils to learn about the formal axiomatic systems with the aim of developing their deductive thinking and reasoning. Recent studies, however, have shown that poor achievement in Euclidean geometry is mainly due to the tendency by pupils to struggle with content that requires the use of logical deductive reasoning (De Villiers, 1987; Mudaly, 1997)

In South Africa, the performance of learners in Geometry is associated with lack of learning materials, poverty, lack of poor learners motivation, teachers' relationships, lack of teachers' subject knowledge, lack of parental support, lack of clearly defined learner support systems, and serious lack of discipline in many schools (Van der Westhuizen, Mosoge, Nieuwoudt, & Steyn, 2002).

Japan and China have National Curriculum for Mathematics that covers Geometry, among other mathematical topics (JSME, 2000). Countries such as U.K and Singapore address this by providing different mathematics syllabi to cater to the varying needs and abilities of students.

As most of the countries show poor performance in geometry and it is not adequate to reach its standard nowadays these countries have taken different activities and necessary remedies to improve the performance of geometry Therefore as Sri Lankans essential move has to be taken to give remedial measures to improve the standard of geometry. As such this study reviews the current status of geometry and analyses the impact of domain factors on learning geometrical concepts through Van hiele level of thinking.

2.3.2 Students' Performance of geometry in Sri Lanka

It is obvious that geometry is a very important as well as very interesting subject. But unfortunately, the people of Sri Lanka rarely realized the actual usages of geometry. As our students are not exposed to the subject much they tend to ignore the subject. (Kalamani, 2014).

In Sri Lanka, mathematics education is compulsory up to grade 11 and is commonly termed “mathematics for all”. In 1972, a common general curriculum was introduced and mathematics was taught to all students in grades 1 to 9. But by analyzing the curriculum of Singapore and America, The Education Ministry of Srilanka have thought to adopt a similar practice and have two syllabi, an advanced mathematics syllabus for students intending to pursue mathematics-intensive subjects in high school, and a second practical mathematics syllabus for the others. But several problems associated with mathematics education at the secondary level were identified and new proposals were put forward in 2003 to improve the quality of mathematics education at the junior secondary level (NEC, 2003).

Mathematics education up to grade 9 focuses on developing the basic skills required to function effectively in daily life, as well as on enhancing the analytical abilities of students. Students in the secondary level are expected to be led away from primary level methodologies where mathematical concepts are introduced with concrete examples and pictures to a logical understanding of abstract mathematical concepts. More emphasis is given to the basics of higher mathematics including algebra and geometry in grades 10 and 11; all students in this age group are compelled to learn mathematics as a ‘subject’ distorting the concept of ‘mathematics for all’ to a concept of ‘same mathematics for all’ (NIE, 2006). This is a major contributing factor for the high failure rate in mathematics at the GCE O/L examination (Upali Mampitiya, 2014).

In the GCE O/L examination, more emphasis is given to algebra and geometry in Paper II of mathematics. This paper mainly tests medium and higher order knowledge and skills. Out of 12 questions, 6 are on algebra and geometry. Students have to answer 5 questions from each of the parts A and B. During the past three years, the 3 algebra questions have appeared in Part A and the three geometry questions in Part B. Thus students are compelled to answer at least 2 algebra questions and 2 geometry questions. (NIE, 2009). Hence, a student would have to perform well in these two

areas to obtain an 'A' grade in mathematics which tests students' knowledge in higher mathematics.

In 2007 reforms, a competency-based curriculum was introduced. The main aim of the mathematics curriculum under this reform is to create individuals who can think mathematically and apply mathematical knowledge effectively and responsibly in problem solving and decision making (NIE, 2009). The NIE report (2009) revealed that the activities in the TIMs focus mainly on basic concepts and mathematical skills and do not provide opportunities for open-ended and creative problem solving.

World Bank report of strengthening mathematics education in Sri Lanka (2011) reveals that Sri Lankan teachers felt that the training they had received on implementing the curriculum was inadequate, that there was insufficient time allocated for all the activities proposed in the Teacher's Instructional Manuals and that the 5E method is not the most appropriate teaching strategy in all situations.

The government of Sri Lanka, recognizing the need for a high quality mathematics education has implemented several reforms in the past (NEC, 1997). Although the progress achieved is commendable, there is still room for improvement in certain important aspects of the learning and teaching geometry process. So it is very necessary to make the students understand the importance of geometry and investigate the problems in enhancing the performance on geometry and find out the solutions to solve these problems. In this study, several factors are identified as the cause for the poor performance of geometry among Jaffna district students. These factors are analysed further and three domain factors were chosen and analysed the relationship between the development of the geometrical conceptual knowledge and skills and the domain factors.

2.4 Geometrical thinking

Geometrical thinking is metacognitive thinking. Geometrical thinking should be developed to understand geometric concepts meaningfully. Van Hiele's level of thinking helps both students and teachers to develop geometrical knowledge and skills. It creates a much simpler environment in the teaching and learning of geometry.

2.4.1 Van Hiele Theory

Pierre Van Hiele and Dina van Hiele-Geldof as teachers in Montessori secondary schools were concerned about the difficulties their students were having with their studies of geometry. It became apparent to them that secondary school geometry involved a high level of thinking and primary school geometry involved lower levels of thinking. After observation and discussion of their students' progress, the van Hieles concluded that in learning geometry, the students seemed to progress through a sequence of five reasoning levels, from wholistic thinking to analytical thinking to rigorous mathematical deduction. They also concluded that to progress from one level to the next, students seemed to pass through five phases from an inquiry phase through to an integration phase. The van Hieles supported this composite approach of levels and phases to teaching by defining the subject matter to be learned but at the same time defining the role of the teacher as a helper who guided the student through levels of understanding of the subject matter. Fuys (1988) stated that The van Hieles' research work focused on levels of thinking in geometry and the role of instruction in helping students move from one level to the next and van Hiele (1986) quoted from his 1955 work and stated: "The attainment of the new level cannot be effected by teaching, but still, by a suitable choice of exercises the teacher can create a situation for the pupil favourable to the attainment of the higher level of thinking."

American study tested the ability of the van Hiele theory to describe and predict the performance of students in secondary school geometry (Usiskin, 1982). Hiele theory helps a lot to identify the factors influencing learning geometrical concepts and how young people learn geometry. The Van Hiele level theory offers an explanation and a remedy for student difficulty with higher-order cognitive processes required for success in secondary school geometry.

This study also focuses on how levels of Van Hiele theory helps the students to achieve the geometrical concepts step by step and further reveals how the domain

factors impact the development of geometrical conceptual knowledge and skills through Van Hiele's levels of theory.

2.4.2 Level of geometrical thinking (Van Hiele levels)

Van Hiele (1986) established an enabling framework to interrogate the development of cognition in geometry. He established a five-level framework which has been the subject of much research around the world. The van Hiele theory does not offer a deterministic view of a fixed progression, but rather an empirical description of relatively stable stages that can provide educators with guidance on structuring learners' experiences appropriately.

According to Van Hiele theory, there are five major steps to understand geometry topics, namely: Recognition (L0), Analysis (L1), Order (L2), Deduction (L3) and Rigor (L4). Van Hiele (1959) states that the levels are "characterized by differences in objects of thought" (p. 14).

Level 0: The student identifies, names, compares and operates on geometric figures (e.g., triangles, angles, intersecting or parallel lines) according to their appearance.

Level 1: The student analyzes figures in terms of their components and relationships among components and discovers properties/rules of a class of shapes empirically (e.g., by folding, measuring, using a grid or diagram).

Level 2: The student logically interrelates previously discovered properties/ rules by giving or following informal arguments.

Level 3: The student proves theorems deductively and establishes interrelationships among networks of theorems.

Level 4: The student establishes theorems in different postulational systems and analyzes/compares these systems.

According to van Hiele, the learner, assisted by appropriate instructional experiences, passes through the above five levels, where the learner cannot achieve one level of thinking without having passed through the previous levels.

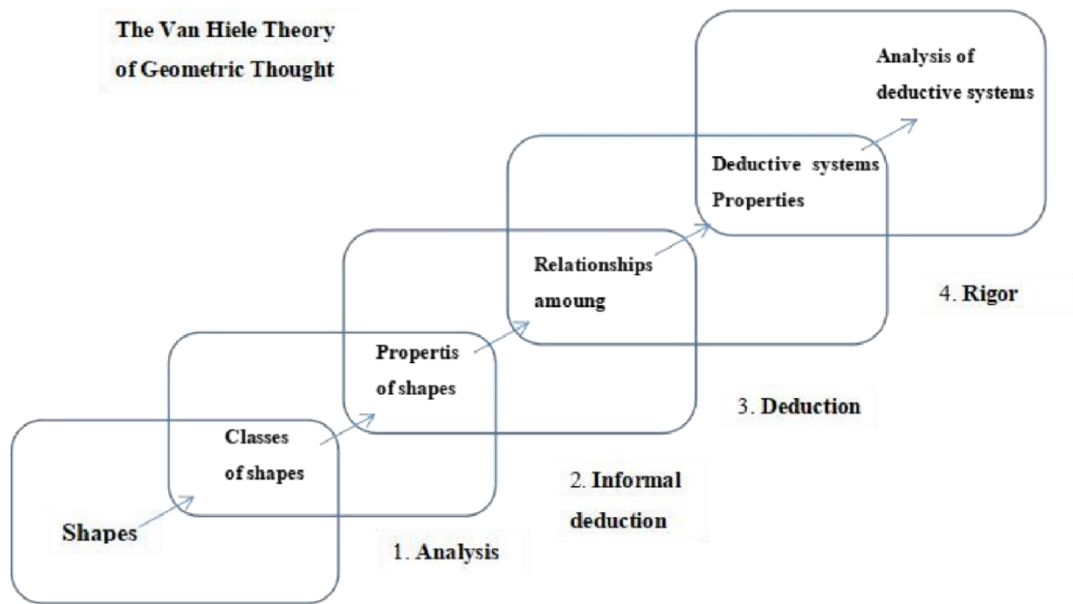


Fig 2.1: The Van Hiele Theory of geometric Thought

Van Hiele (1958) noted that learning is a discontinuous process and that there are jumps in the learning curve which reveal the presence of "levels." They observed that at certain points in the instruction

According to van Hiele, a student progresses through each level of thought as a result of instruction that is organized into five phases of learning. The phases are described below.

(a) Phase 1: Information. At this stage, a two-way teacher-student interaction is essential in understanding certain geometrical shapes such as making observations, asking questions and understanding the vocabulary for that particular geometrical shape.

(b) Phase 2: Orientation. Students explore the topic about geometry as arranged by the teacher. The activities involved should enable students to identify the geometrical shape that is to be learned. Therefore, for students to master the level, they have to be assigned simpler tasks.

(c) Phase 3: Explanation. Based on previous experiences, students are to express their opinions and discuss about the geometrical shapes that had been observed. At this stage, teachers act merely as facilitators.

(d) Phase 4: Free Orientation. Students are able to solve more complicated problems such as open-ended problems. They gain experience by finding their own solutions or by completing tasks. Much of the relationships among objects are clarified through the interaction among students when making investigations.

(e) Phase 5: Integration. Students survey and summarize what they have learned by

A student may need to cycle through some of the five phases more than once with a particular topic.

So the major characteristics of the van Hiele "levels" are that the levels are sequential; each level has its own language, set of symbols, and network of relations; what is implicit at one level becomes explicit at the next level; material taught to students above their level is subject to reduction of level, and one goes through various "phases" in proceeding from one level to the next.

Many studies say that for the development of geometrical concepts, students have to achieve the levels of Van heile step by step. They further say that number of factors have influenced in achieving the levels. This study attempts to explore the effect of some of the factors which influence teaching and learning geometry.

2.5 Factors influencing learning Geometrical concepts

The development of geometrical concepts is influenced by many non-defined reasons. It's not an easy task to specify the factors and put them together to analyse further. Scholars put forward their view on factors influencing mathematical achievement rather than factors influencing geometrical concepts and only a few studies were done to explore the factors which influence the development of geometrical concepts.

Ercikan (2002) expressed that various independent variables that include both student personal and environmental variables such as students' attitudes toward mathematics, parents' highest level of education attained, self-expectations and the expectations of parents, teachers, and friends, students' confidence in mathematics, home support for learning can affect the dependent variables (students' mathematics achievement and participation in advanced mathematics courses).

Reyes and Stanic (1996) believed that the society, mathematics curriculum, the activity in class, the attitude of students, and the behaviour related to academic achievement can affect mathematics achievement. Moreover, they stated that the comparison between students also affects their expectation which succeeds in achievement, thus having a higher performance in mathematics achievement.

Okebukola (1992) found that factors such as home environment, school environment and quality and quantity of instruction are responsible for students' poor performance in mathematics. The above-mentioned literatures revealed that various factors affect the performance of students in mathematics.

Particularly, when investigating factors that facilitate performance in Geometry, a more extensive investigation should consider learner, teacher and school variables (Malcolm, 2000). In Deppe, Sonderegger, Stice, Clark & Streuling (1991) research, they found some of the problems encountered in the development of geometrical concepts in developed countries. They listed out some problems. They expressed that the learners did not explore the conceptual foundation and real-world relevance of geometry information and failed to relate to the child's environment and thus the child cannot see the importance and immediate application of Mathematics, particular geometry in his day-to-day living. According to them a lack of programme continuity was seen in the curriculum. Geometry was treated as a subject with no integration with other learning areas. Learners were not given chance to participate in extensive group work in most classes as most of the teachers encourage writing skills in Geometry. Since teachers had never been involved in the development of the new

curriculum they were unaware of each new curriculum. They also stated that non-availability of instructional material such as textbooks, workbooks, slides and other media to meet the pedagogical demand of new curriculum imposed on teachers is another reason for the lack of performance in geometry.

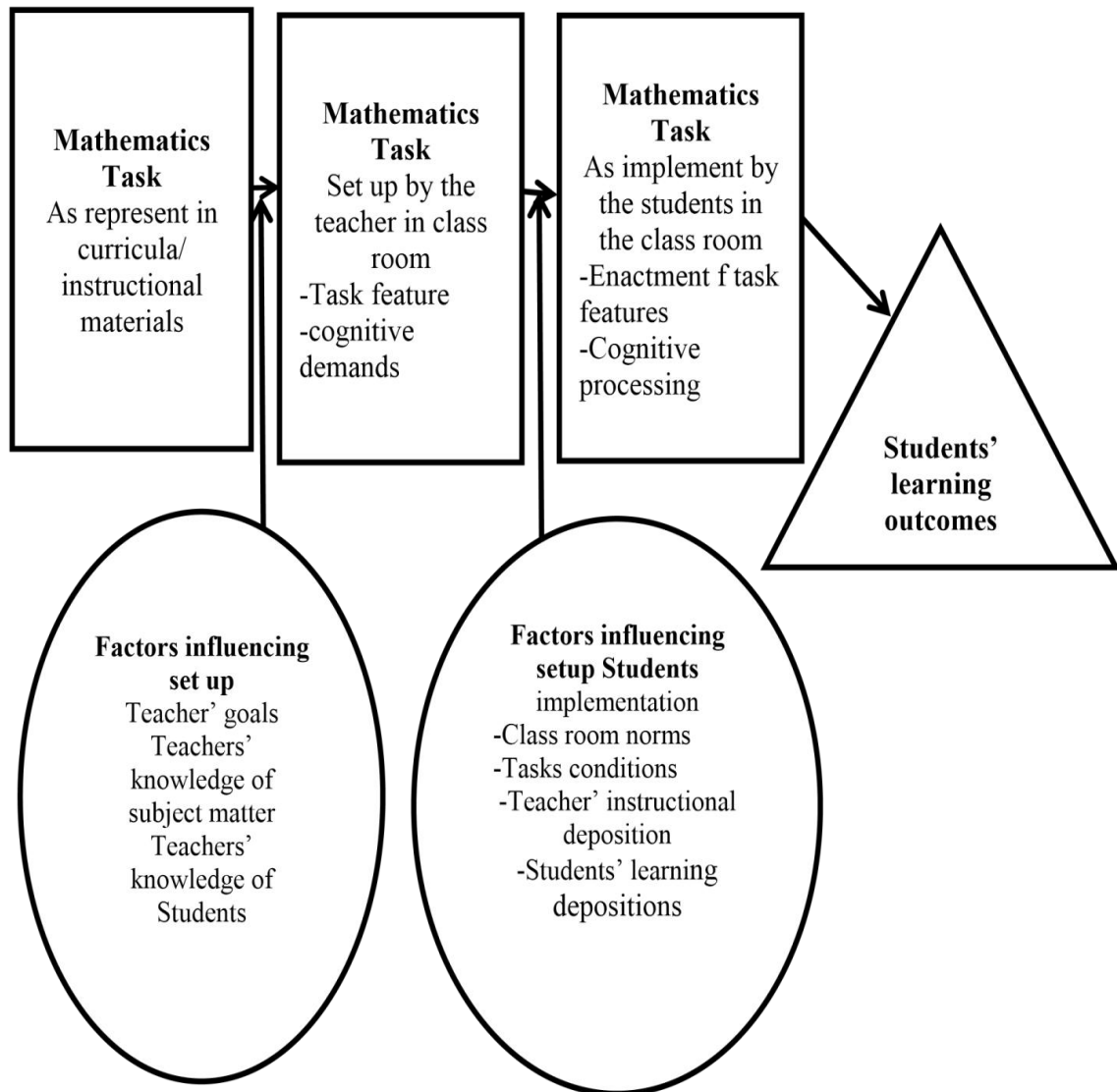


Fig 2.2: Influencing factors

Source: NCTM Report (2011)

Therefore it is true that infinite factors contribute to the development of geometrical concepts and affect the performance in geometry. But due to the practical convenience and the reliability of the study, this study focuses only on some of the specific factors which affect the performance of geometry. To fulfil the purpose of the

study three domain factors were analysed in deep to analyse their effects on teaching and learning geometry and the development of geometrical conceptual knowledge. Domain factors were selected from teacher-related factors, student-related factors and factors related to the materials used for instruction.

2.5.1 Teacher related factors

Teachers are the lifting ladders who guide the students to uplift their talents. Then teaching of geometry needs proper assistance and support from the teachers and the understanding of geometry depends on the teachers' subject knowledge and appropriate teaching strategies. Therefore teachers with excellent mathematical knowledge and effective teaching methods are necessary for teaching geometry.

Knowledge development for mathematics students is varied. Lesh and Harel (2003) argue that examining students' performances over a variety of activities will result in different 'developmental stages for the tasks. This has significant implications for organizing abstractions. Lesh and Harel continue stating that mature levels of thinking are needed to organize those abstractions. These levels of thinking will only be developed through challenging students, allowing them to express their constructs, and encouraging them to go beyond thinking with those constructs (Lesh&Harel).

Therefore, as teachers are responsible persons for imparting knowledge to the students they must understand that development is varied. Rather than focusing on a single developmental skill at a time, teachers must understand that skills can be developed simultaneously. As such teachers should adopt themselves as well as teaching strategies to give proper knowledge and skills to the students.

2.5.1.1 Teachers' knowledge

The quality of the educational opportunities that a country provides depends on the quality of its teachers. Nowadays, teaching is becoming more and more complex and the demands placed upon teachers are increasing day by day. Teachers have to update their knowledge in order to meet the challenges in the modern mathematical world. Especially in geometry teaching, teachers should pay attention to the changes in geometry.

A number of studies report that elementary and middle school teachers' geometry content knowledge is reliably insufficient, leaving them underprepared to teach in ways that help students internalize salient concepts (Ball & Bass, 2000; Murphy, 2011).). Notably, pre-service teachers report that geometry is a difficult subject, to learn and to teach (Barrantes& Blanco, 2006; Clements &Sarama, 2011).

In Sri Lanka, the NEC (2011) report says that one of the chief causes for the poor performance in geometry is the subject knowledge of teachers. And the quality of the mathematics teachers in the school system is another major factor. NEC report further says that teachers who are appointed to teach mathematics for senior secondary students fail to update their knowledge and they fail to perform well in geometry.

In 2008, a study was conducted by NIE to assess the mathematics knowledge of teachers in schools. Teachers were selected from the schools where students score less than 30 percent in the GCE O/L mathematics paper. 172 teachers participated in the study. For them, a question paper was given containing 9 questions on the themes measurement and algebra. An analysis was done to see the subject knowledge of the teachers. The marks obtained by the teachers revealed that 36 percent of teachers from the Western province and 51 percent of the teachers from the other provinces scored less than or equal to 5 marks out of 10 (NIE, 2008)

In another study conducted by the DOE in 2010 to determine the causes for the poor performance in mathematics at the GCE O/L examination, 207 teachers who had applied for paper marking were selected and given a question paper similar to the GCE O/L mathematics paper (DOE, 2010). An analysis of the marks obtained is represented in the graph given below. The data shows that although most teachers performed well in paper I, this is not the case for paper II. Paper II tests higher order

skills and 50 percent of the paper contains problems on algebra and geometry. Only 42 percent of the teachers were able to score over 80 percent on paper II. Question by question analysis revealed that teachers' performance was poor on geometry questions. The same weakness is reflected in students' performance at the GCE O/L examination.

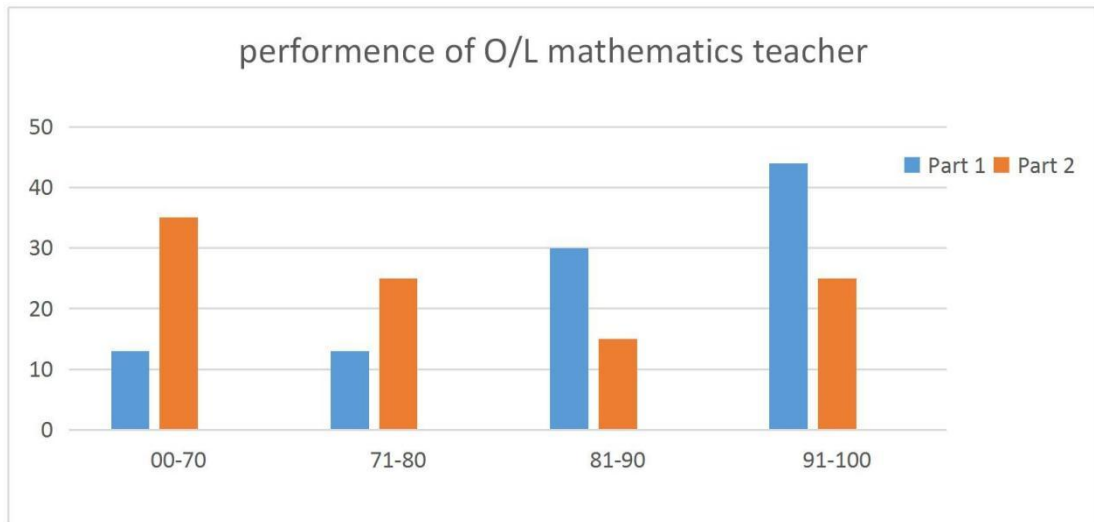


Fig 2.3:performance of O/L mathematics teachers

source:DOE(2010)

This graph portrays that the teachers are not with adequate knowledge in geometry and lack of knowledge of teachers affects a lot in students' performance in geometry. Therefore, it is vital that subject content knowledge is essential for successful teaching.

2.5.1.2 Teaching strategies

There are many strategies and techniques adopted by teachers to impart geometric knowledge to their students. By analyzing the factors for the poor performance in geometry, teachers have been changing their approaches and finding new strategies to increase the performance rate of the students in geometrical learning.

The Royal Society and Joint Mathematical Council (2001) report argues that there is a problem: "We believe that there are many teachers who have been taught geometry through styles of teaching which we would not advocate as appropriate".

The present teachers have difficulties in the way of their own style of teaching and adopting strategies to understand students' strengths and weaknesses with regard to geometry. To meaningful learning, teachers require an understanding of how students

construct their knowledge of various geometric topics (Battista, 1999). So, it is necessary that mathematics teachers should understand how students construct geometrical knowledge as a result of their learning experiences in school.

An understanding and application of the Van Hiele's level of thinking would influence teacher practice and teaching strategies in teaching geometry (Andrew, 2001; Gates, 2001; Kilpatrick, 2001). Teacher practice and teaching strategies are instruments that can activate learners' pre-knowledge in the development of geometric understanding.

Many teachers have observed that many young children have numerous misconceptions about geometry when a teacher discusses a geometry proof problem in class, it generally involves oral presentation of a formal proof and body movements pointing at different parts of the figure of the problem. Students must watch, listen, jot notes, and think as a lecture proceeds. They have to refer to many elements of the instruction and incorporate them into their memory (Sweller, 1988). This often causes cognitive overload and poses a negative effect on students' learning. Teachers may not uncover students' mathematical misconceptions that proliferate throughout the elementary and middle school years (Lin, Luo, Lo, & Yang, 2011; Ma, 2010). In many cases, elementary and middle school teachers rely on no more than their own secondary school coursework for geometry content knowledge in their teaching practice (Jones, & Thornton, 1997).

Numerous researchers have experimented different ways of teaching and found serious problems in geometry learners: incomplete comprehension of the problem and mathematical symbols, producing proofs based on direct visual elements (Healy & Hoyles, 2000), lacking strategic knowledge in producing proofs, etc.

Moreover, solving geometrical problems often requires the interactions of different apprehensions, and "what is called a 'geometrical figure' always associates both discursive and visual representations, even if only one of them can be explicitly highlighted according to the mathematical activity that is required" (Duval, 2006). The recent report on the teaching and learning of geometry by the Royal Society and Joint Mathematical Council (2001) argues that "the most significant contribution to improvements in geometry teaching will be made by the development of good models of pedagogy, supported by carefully designed activities and resources". The report suggests that "in many respects, we need to develop a completely new pedagogy in geometry".

For successful teaching, teachers should understand how students' misconceptions of mathematical geometry influence their learning (Indiana Professional Standards Board, 2001). Teachers should seek out problems that students are interested in. Teachers can take an interest survey of their students and build those interests into word problems for students to solve. Hartman and Kaiser (1998) argue that these problems can be challenging and that students will achieve at very high levels. Focusing on students' interests and their prior knowledge can be a departure point.

Campbell (2014) said that student achievement is positively affected by the interaction of teacher attitudes and mathematics content knowledge. Teachers who practise an effective method according to the needs of the children in teaching geometry will motivate the students more positively. The teaching of geometry provides not only a key vehicle for developing learners' spatial thinking and visualization skills but also a major opportunity to develop their ability in deductive reasoning and proving (Battista, 2007; Royal Society, 2001). Ineffective instruction will lead to the poor performance of the students.

In the research on geometry problem solving strategies used by elementary mathematics teacher candidates, (CenkKesan, 2014) focuses on some useful teaching strategies to teach geometrical concepts. As per their view, geometry instructions incorporate with new and tested approaches such as using visual and multimedia tools in the classroom is very useful for the students to capture the picture in mind.

In Srilanka, there are approximately 1,600 Sinhala medium and 800 Tamil medium teachers in the system with appointments for other subjects who teach mathematics in the secondary education cycle. These 2400 teachers are without the required training to teach mathematics. Studies by the NIE and the DOE revealed the need for a comprehensive teacher development programme to enhance the knowledge and skills of these types of mathematics teachers.

Accepting the necessity in education progress Sri Lankan ministry of education along with the National Institution of Education has put forward many teacher development programmes to provide teachers with required subject knowledge and training in good teaching practices. The programmes are designed to fulfil the expectations of the curriculum. Therefore, from the literature, it is observed that it is essential to ensure that all mathematics teachers are subject-competent and have good pedagogical

knowledge to teach mathematics since they play a vital role in providing students with a high quality mathematics education.

2.5.2 School-related factors

Every student spends nearly six hours in school per day. So school plays a prime role in the development of each child. Parents almost believe that imparting education is in the hands of the teachers. So in geometry knowledge development, school-related factors like materials used for instruction which include curriculum, textbooks, availability and usage of mathematical activity room or math lab are given attention in this study.

2.5.2.1 Materials used for instruction

Instructional materials constitute a wide range of materials that can be used to extend the range of varied experiences of learners in a teaching-learning situation. Teachers use instructional materials to convey their ideas more vividly to learners. This allows teachers to teach the same lessons in multiple ways. Instructional materials facilitate both teachers and students in the teaching and learning process in and outside the classroom to function effectively.

Many studies have explored the relationship between teaching materials and other related inputs and students' learning and achievement in developing countries (Chepkurui 2004). Curriculum, prescribed textbooks and the availability of mathematics activity room are selected as the important factors related to instructional material which affects the knowledge development of the senior secondary students in geometry in this study.

2.5.2.2 Curriculum

The curriculum is the major component of the education system of a country. A curriculum can be defined as a collection of lessons, assessments, and other academic content that is taught in a school, programme, or class by a teacher. Promoting elite citizens is depend on the perfect curriculum. Especially, the mathematics curriculum aims to create learners with basic skills in problem solving and reasoning. (Lehren&Chazan, 1998). They also state that curricula in which learners' knowledge

and skills grow, is significantly connected to their learning, and therefore their achievement.

The geometry curriculum should aim at the development of geometric reasoning and spatial sense. Many researchers find the short comes of the curriculum in their country. For example, Usiskin (1987) argues that because there is not an adequate geometry curriculum at the primary school level, learners entering secondary school are not performing well in geometry. Usiskin (1987) found in his research that the secondary school geometry curriculum was studied by only a few learners. He also found that only one third of these learners could perform well in geometry.

Sri Lanka has high education achievements and learning in the South Asia region (De Mel, 2007) especially achievements in literacy, educational enrolment and equal opportunity and access to education (Little, 2010). A well designed curriculum is one of the reasons for its success. Sri Lankan mathematic curriculum is designed to cope with international standards. Sri Lankan content standards and process standards are aligned with international trends

Table 2.1: Comparison of the NCTM standards and Sri Lankan standards

NCTM content standards	Sri Lankan content standards
Number and Operations	Numbers
Measurement	Measurement
Geometry	Geometry
Algebra	Algebra
Data Analysis and Probability	Statistics
	Sets and Probability
NCTM process standards	Sri Lankan process standards
Communication	Communication
Representation	(Knowledge and) Skills
Connections	Relationships
Reasoning and Proof	Reasoning
Problem Solving	Problem Solving

Source: National Council of Teachers of Mathematics (www.nctm.org) & National Institute of Education, Sri Lanka.

In Sri Lanka, the authority for the national mathematics curriculum is the National Institute of Education (NIE). The present education system has been influenced by

historical factors and the pressures from the global and local environment (NEC, 2003). While determining the mathematics curriculum, it must be kept in mind that the majority of students would leave formal education at the end of grade 11. They would need to apply math skills and competencies in their work situation. Only a small number would go on to higher education. The curriculum needs to distinctly accommodate the requirements of both groups (U.Mampatiya, 2014).

At present, the mathematics curriculum is developed by taking into account student characteristics, teacher characteristics, resources available, individual needs, social needs, research done on the current curriculum and the new trends. (NIE,2009). National Institute of Education implemented new curriculum reforms in 2007 to face the 21st-century challenges and make the students cope with the modern world namely the Modernized Competency-based Curriculum. But, the less weight given to geometry in the present curriculum causes negative results in students' performance in geometry. The importance given to geometry is decreasing year by year. Then the mathematics curriculum allows the students to give less attention to geometry and leads to poor performance in geometry.

According to NEC (2009), the most important factor for poor performance in mathematics is that all students follow the same mathematics syllabus and sit the same mathematics examination paper at the GCE O/L. As is the practice in many countries, Sri Lanka should also consider having at least two different syllabuses for GCE (O/L) classes. One syllabus would cater to the needs of those who wish to pursue GCE (A/L) subjects that require a high level of mathematical knowledge and skills and the other one for those who only need practical knowledge of mathematics. The second syllabus should consist of Essential Learning Concepts to function effectively in a Knowledge Society. The first will consist of Essential Learning Concepts delivered at a higher level than the above and additional topics that will be beneficial to those who pursue higher studies, especially in the GCE (A/L) Science Stream (U.Mampatiya, 2014).

Burger and Shaughnessy (1986) used student interviews to study students' van Hiele levels. According to them "The results of the interviews and the van Hiele theory have implications for the way geometry is taught in the school and for the way students learn geometric concepts". Burger and Shaughnessy recommended activities that encourage development through the van Hiele levels that need to be incorporated into the geometry curriculum. Burger and Shaughnessy believe that geometry course

material should have materials that promote development from one level to the next. Geometry concepts need to be introduced in mathematics classes in the elementary and middle grades as well. “Many students have only had brief encounters with geometric concepts during their elementary school years. We must allow students to explore geometric concepts and shapes informally for many years prior to a high school course in geometry” (Burger & Shaughnessy, 1985).

Many geometry curricula overemphasize memorization of facts and under emphasize understanding and application of these facts to discover, make connections, and test geometry concepts. Memorization must be raised to conceptualization, application and problem solving for learners’ to successfully apply what they learn(NCTM, 2000).

In the Sri Lankan education system geometry is introduced in grade 6. But the way they develop the concepts and the coordination among the levels seems to be confused (NEC, 2003). Sri Lankan government designed many mathematics curricula to develop mathematical knowledge. Although we make changes in the curriculum from time to time, still our students show negative feedback in geometry. In order to meet the needs of the students in learning geometry, when designing the curriculum, it should be considered that adequate importance should be given to geometry in the curriculum. Moreover, geometrical concepts have to be developed through a series of activities and learning by doing should be encouraged to follow the Van Hiele’s level of thinking. Therefore geometry curriculum should be modified to develop the geometrical conceptual knowledge of senior secondary students of Sri Lanka.

2.5.2.3 Textbooks

Textbooks are considered to be among the most powerful influences on school mathematics (Mullis, Martin & Foy, 2008). In the process of teaching and learning, the textbook is considered to be a very important source of learning. . A textbook contains all kinds of materials used to help teachers or instructors to carry out the teaching and learning activities. Textbooks are equally important resources for students as well as teachers. Textbooks facilitate learners to learn coherently and systematically. Teachers use textbooks to plan their lessons and choose suitable strategies to teach their mathematics lessons

Although the mathematics curriculum specifies skills, communication, relationships, reasoning and problem solving as process standards, in textbooks little emphasis has been placed on developing the full spectrum of process skills that are specified, in particular, relationships, reasoning and problem solving.

Mathematics classroom instruction is, in many cases, generally organized around and delivered through the mathematical tasks and activities found in textbooks. Therefore, textbooks are probably among the most immediate determinants of educational practice. (Weiss & Ziebarth, 2012).

Textbooks are the most powerful instructional resource used to deliver knowledge to learners. They have a significant influence on the teaching-learning process. Moreover, the quality and the content of the textbooks have been affecting the learners' attitudes towards learning. Their availability and use of such materials have a positive effect on students' achievement in Mathematics and in particular in geometry since the students can use it as a guide in geometric skill development and offers exercise for further practice of learning concepts. (Chepkurui, 2004).

Sri Lankan government provides free textbooks from C.W.W.Kannangara's period. The Department of publication is responsible for issuing the textbooks. Though Sri Lanka tries to provide high quality books with adequate relevant activities, our students do not pay much interest in textbooks. Rather than go through the lessons and activities in the textbooks they tempt to search from some other instructional resources. In the current trend students prefer to use electronic devices and to meet the demand of students, our government put effort to give the books in digital print.

In Sri Lanka, good quality mathematics textbooks are given to senior secondary students. Though the design and the quality of the textbooks facilitate our students and motivate them to be engaged in the teaching and learning process, students fail to realize the usage of textbooks. In, Grade 10 and 11 mathematics textbooks geometry units are compiled per Van Haile Levels of thinking. Also, they are formed with appropriate activities based on the relevant competencies.

The following table shows the relationship between geometrical concepts and Van Haile Levels of thinking in Grade 10 and 11 mathematics textbooks.

Table 2.2: Content of mathematics textbook

Grade 10,11 Mathematics

TOPIC	TITLE
Van Haile Level 1	
2-D shapes	Spatial properties of quadrilaterals
Geometry-quadrilaterals	Quadrilaterals
Geometry-quadrilaterals	Classifying Quadrilaterals
Geometry-quadrilaterals	Using the Properties of a Parallelogram
Geometry-quadrilaterals	Proving a Shape is a Parallelogram
Geometry-quadrilaterals	Properties of the Rectangle, Square and Rhombus
Geometry-quadrilaterals	Properties of the Trapezium and Kite
Geometry-quadrilaterals	The quadrilateral family and coordinate methods in geometry
Van Haile Level 2	
Functions and graphs	Quadratic polynomials of the form $y = ax^2 + bx + c$.
Functions and graphs	Graphing perfect squares: $y = (a-x)^2$
Coordinate geometry	Solve by graphing
Graphing-polynomials	Graphing complex polynomials: quadratics with no real roots
Graphing-polynomials	General equation of a circle: determine and graph the equation
Graphing-cubic curves	Graphing cubic curves
Graphs, polynomials	Graphs of polynomials
Volume	Finding the volume of prisms
Volume	Volume of a cylinder and sphere.
Volume	Volume of pyramids and cones.
Volume	Composite solids.
Geometry-constructions	Geometric constructions
Geometry	To identify collinear points, coplanar lines and points in 2 and 3 dimensions

Geometry-constructions	Angle bisector construction and its properties (Stage 2)
Geometry-constructions	Circumcentre and incentre (Stage 2)
Geometry-locus	Constructions and loci – single condition
Geometry-locus	Constructions and loci – multiple conditions
Area	Area of a trapezium.
Area	Area of a rhombus.
Area	Area of a circle.
Area	Area of regular polygons and composite figures.
Trigonometry-ratios	Trigonometric ratios.
Trigonometry-ratios	Using the calculator.
Trigonometry-ratios	Using the trigonometric ratios to find unknown length. [Case 1 Sine].
Trigonometry-ratios	Using the trigonometric ratios to find unknown length. [Case 2 Cosine].
Trigonometry-ratios	Using the trigonometric ratios to find unknown length. [Case 3 Tangent Ratio].
Trigonometry-ratios	Unknown in the denominator. [Case 4].

Van Haile Level 3

Pythagoras	Find the hypotenuse
Pythagoras	Pythagorean triples

Circle Geometry	Theorem – Equal arcs on circles of equal radii subtend angles at the centre. Theorem – Equal angles at the centre subtend equal arcs on equal circles.
Circle Geometry	Theorem – Equal chords in equal circles are equidistant from the centres. Theorem – Chords in a circle which are equidistant from the centre are equal.
Circle Geometry	Theorem – Angles in the same segment of a circle are equal.

	circle are equal.
Circle Geometry	Theorem – The angle of a semi-circle is a right angle.
Circle Geometry	Theorem – The opposite angles of a cyclic quadrilateral are supplementary.
Circle Geometry	Theorem – The exterior angle at a vertex of a cyclic quadrilateral equals the interior opposite angle.
Circle Geometry	Theorem – The tangent to a circle is perpendicular to the radius drawn to it at the point of contact.
Circle Geometry	Theorem – Tangents to a circle from an external point equal.
Circle Geometry	Theorem – The angle between a tangent and a chord through the point of contact is equal to the angle in the alternate segment.
Circle Geometry-chords	Theorem – The products of the intercepts of two intersecting chords are equal.
Circle Geometry-cyclic quads	Theorem – If the opposite angles in a quadrilateral are supplementary then the quadrilateral is cyclic.
Van Haile Level 4	
Circle Geometry-tangents	Theorem – The square of the length of the tangent from an external point is equal to the product of the intercepts of the secant passing through this point. [Including Alternate Proof]
Pythagoras	Proofs of Pythagoras theorem
Circle Geometry subtending	Theorem – If an interval subtends equal angles at two points same side of it, then the end points of the interval and the two are concyclic.
Circle Geometry	
Circle Geometry-nonlinear	Theorem – Any three non-collinear points lie on a unique circle whose centre is the point of concurrency of the perpendicular bisectors of the intervals joining these points.

2.5.2.4 Mathematics Activity room

The availability of mathematics activity rooms also plays an important role in learning geometry. Learning by doing motivate the students to be engaged in the learning process actively. At present modern technology is used to encourage the students. Modern equipment like computers, overhead projectors, multimedia and even smart boards are utilized by mathematics teachers to teach geometry. The availability of these resources helps the students in understanding geometrical concepts more efficiently. Mathematics lab or activity room is the new concept introduced by the ministry of education to promote teaching and learning. Many schools have allocated a separate room with needed equipment and resources for mathematics learning. As geometry has to be learned with meaningful explanation and visual images learning by doing in the mathematics activity room promotes teaching and learning geometry.

Greg (1992) asserts that computers have had an enormous impact on the teaching of geometry. According to him, teaching geometry has become more open-ended and exploratory than the traditional emphasis on memorizing theorems and proofs with the help of good graphing programs and visualization software. This new approach is consistent with research findings that students pass through different levels of geometric thinking, visualization, analysis and deduction.

Clements and Batista (1989), argue that the computer language software has proved valuable remark in the teaching of geometry. He also states that it helps students to move from empirical to logical thinking, encourages students to make and test conjectures, facilitate precision and exactness in geometric thinking, encourage the development of autonomy in learning. Students use computers to create geometric shapes, change them, move them around, and combine them into new shapes. In doing so they reach higher levels of geometrical thinking and move on to high achievement in Mathematics.

The mathematics activity room facilitates our students to learn geometrical concepts more logically. Students can get visual ideas from the available resources here. Activities done in the mathematics activity room motivate the students to learn by themselves. Learning by doing also raises the learner's level of recall and retention of mathematical content in long-term memory (Kluwe, 1990).

However, in many Srilankan schools allocation of mathematics rooms has not been properly done and even the schools with mathematics activity rooms are not maintained with needed facilities. Since it's very difficult to conduct the classes with overcrowded students this system does not reach its expected level. Failure to use mathematics activity rooms or less usage of mathematics activity rooms reflects the poor performance of students in geometry.

2.5.3 Students' related factors

A high quality mathematics education will help the students to develop the skills that are essential not only in science and technology but also in everyday life and the workplace. Learning geometrical concepts and gaining conceptual knowledge will lead them to a satisfactory life in the world (Duval ,1998)

Increasing mathematics achievement at all levels of student learning is more important than any other. In Sri Lanka, 50-60 percent of the students consistently fail in mathematics at the GCE O/L examination. According to a 2013 report by the Ministry of Education, about 40 percent of students fail in GCE O-Levels and become drop outs. Although there has been an improvement in the GCE O/L mathematics pass rates in the recent past, the mean score has consistently been low up to 2015 (32 ± 2). There are various factors contributing to the poor performance in mathematics at the GCE O/L examination. (MOE,2013)

Addressing the difficulties in learning geometry, Healy and Hoyles (1998) explained that geometry instruction is often more complex than that of numerical operations or elementary algebra. Failure to master or even understand the geometrical concepts force the students to score low marks in mathematics subject. Several studies have noted the difficulties that students face in learning geometry concepts such as proof (Senk, 1989; Usiskin 1982), area (Mistretta, 2000; Strutchens, 2001), measurement (Strutchens, 2001) and reasoning and relational understanding (Mistretta, 2000).

If students understand and reach the levels step by step, their achievement level should be raised. Battista (1999) found that as students attempt to move from two-dimensional coordinate systems to displays of three dimensions, they first represent the real objects as poorly defined arrays and then with more experience, begin to organize the objects into a cohesive array of units. Battista (1999) further stated that as students developed an understanding of two- and three-dimensional space, the visual and rotational components of spatial sense developed at varying

rates. Students who were more advanced in basic geometric knowledge progressed more rapidly in spatial and reasoning skills than their peers.

Gaining conceptual knowledge in geometry is an important component of learning mathematics since it allows students to analyse and interpret the world they live in as well as equip them with tools they can apply in other areas of mathematics. Many studies depict that most of the students struggle a lot in learning geometry. Numerous causes are identified by scholars and researchers as the factors affecting the development in understanding the geometrical concepts. They mentioned some of the student-related factors such as motivation, enjoyment, value and fearlessness of mathematics, geometry language, visualization abilities, and students' prior knowledge, students' attitude towards geometry and study habits of the students.

Therefore from many inhibiting factors which affect the student achievement of geometry students' prior knowledge, students' interest and study habits cover most of the causes for the poor performance in geometry. Thus this study focuses the attention only on these prime student factors.

2.5.3.1 Students' Prior knowledge

Prior knowledge has a significant influence on learning. One of the factors which affect students' learning in geometry is their existing knowledge before the instruction. Students' prior knowledge will help students to recall and connect new information to their prior understanding.

Prior knowledge of the students in geometry is one of the basic factors identified by the researchers which affect the achievement in geometry. According to Abdusakkir (2010), basically, geometry has a greater opportunity to understand by the students compared with other branches of mathematics. This is because the ideas of geometry are already known by the students early before they enter school, for example, line, area and space. As new learning is constructed on prior knowledge, Geometry is often taught step by step. Students have to understand the concepts in the first level to work on the next level. Hiele theory postulates five levels of geometric thinking which are labelled visualization, analysis, abstraction, formal deduction and rigor. Each level uses its own language and symbols. Students pass through the levels "step by step". This hierarchical order helps them to achieve better understanding and results.

In a recent study, it became clear that students' prior knowledge may impact the effectiveness of interventions designed to promote flexibility (Rittle-Johnson, Star, & Durkin, 2008).

According to Noraini Idris (2006), the lack of understanding in learning geometry often causes discouragement among the students, which invariably will lead to poor performance in geometry. For better understanding, prior knowledge guides the students to learn more readily by forming a link between what they are learning with what they already know.

The report of NEC (2009) said that Students passing from one grade to another without gaining the knowledge and skills that are essential to cope with the work in the higher grade is another contributing factor for the poor performance in mathematics.

Prior knowledge can be a good foundation for building new knowledge, but some researches show that student learning is influenced by the nature of prior knowledge. Prior knowledge can either help or hinder learning. Prior knowledge helps in learning when it is activated. However, Prior knowledge needs to be sufficient, appropriate and, above all, correct. Therefore perfect prior knowledge of the students in geometry will enhance the learning ability and it will help to improve the performance of geometry.

2.5.3.2 Students' interest

Different students are with different abilities. Their talents are varied from one another. Students' interest in understanding geometric concepts is another factor that influences the achievement of the students. Especially in learning geometry students' interest in shape and space plays a vital role as Geometry is the study of shape and space (Güven&Kosa, 2008).

A recent change in mathematics education is that giving the answer is not enough. Students are accepting the expectation that they must offer appropriate support for their answers (Maher, 2002). Maher also states, "Students must be able to convince themselves first and then each other".

Perkins (1996) conducted studies that indicate humans are basically poor geometers. As geometry is abstract, students face difficulty to think abstractly and understand

verbal symbols and still experience difficulties. NorainiIdris (2006) highlighted that spatial visualization has been linked with geometric achievement because geometry is visual in nature. Geometry requires visualizing abilities but many students fail to visualize three-dimensional objects in a two-dimensional perspective as they don't pay much interest in geometry.

According to (Bako,2003; Kouba, 1997) high school geometry students have difficulty in transitioning from two-dimensional to three-dimensional geometry with particular difficulty with volume and surface-area concepts. The three-dimensional context requires students to use formulas to compute surface areas and volumes of figures such as cylinders, prisms, pyramids, and cones.

According to Kariadinata (2012) and Pradika (2012), students' difficulty in learning geometry is caused by a lack of interest. As per Permatasari (2018), the lack of interest of students is caused by internal factors which are having trouble concentrating and lacking the ability to express their ideas. He further says that the lack of interest of students can also be caused by external factors like their residential environments and formal education.

A study by Rukangu (2000) on pupils' development of spatial ability in Mathematics found that 67%of the students did not enjoy learning spatial concepts because they are confusing, abstractly demanding a lot of thinking and difficultly to understand.

Creativity in geometric conceptualization encourages learners to think through problems and apply knowledge and logical reasoning. It develops skills that are necessary for the development of mathematical proof (Senk, 1989). Therefore students interest to think and learn, especially their spatial ability promote the students to achieve geometrical knowledge and concepts more effectively. Hence students' interest is one of the major factors which affect the achievement level of geometry.

2.5.3.3 Students' Study Habits

Geometry is important to success in mathematics. To succeed in mathematics students have to follow certain study habits according to their ability and nature. Choices have to be given to the students to choose the appropriate study habit.

The learning of Mathematics is often viewed as an isolated, individualistic or competitive matter, one sits alone and struggles to understand the material or solve the assignment problems. This process can often be lonely and frustrating. (Davidson Neil, 1990).

Study habits are those activities necessary to organize and complete schoolwork tasks and to prepare for and take tests. (Robbins, 2002). Students can either study individually or in a group. Some may choose the discussion method and others may choose memorizing method. Experts rendered some basic facts about student study habits.

According to Davidson (1994), a Small group provides a forum in which students ask question, discuss ideas, demonstrate to others, learn to listen to others and offer constructive criticism and summarize their discoveries in writing. Through his observation, Davidson noted that small group cooperative learning could avoid the frustration on mathematics.

Rukangu's (2000) statement support the idea of Davidson. Rukangu observed that one of the students' study habits was a discussion between students and students, between teachers and students and between students and parents. According to him this method of teaching enhances psychomotor skills, helps students to discern mathematical relationships in objects and concretizes mathematical concepts.

Unless the students make use of their time effectively and efficiently, the effort they put to understand the concepts will be a failure. Their lack of attention to concepts tends to promote memorization rather than deep understanding as a learning strategy. Students faced Difficulties in learning mathematics because they only memorize concepts in learning mathematics and they are less able to use these concepts if they encounter problems (Trianto, 2007; Hastari, 2018). Students fail to get a clear picture when memorizing the concepts. Thus they are forced to be in trouble when they solve geometrical problems. According to Mondo (2001), games promote learning of Mathematics. Charles and Lynwood (1990) attest that geometry has the characteristic of both game and a puzzle; an intellectual game to be played under accepted rules and

a puzzle to challenge the ingenuity of the student. They argue that these two characteristics in effect constitute the spirit of demonstrative geometry and when students can be brought to approach the subject in this spirit the problem of boredom ceases to be a problem at all.

Since students are at a different level, they should be given the freedom to choose their study habits according to their levels. Different study habits like discussion method, group study, learning by doing, engaging in games, etc facilitate the students to grasp the geometrical conceptual knowledge at a different level. In this study, it is observed that Jaffna students still have a problem in selecting relevant study habits. As study habits affect the students to perform well in geometry, it is considered as a factor that influences the performance of students in geometry.

2.6 Development of geometric conceptual knowledge and skills

Mathematics is a subject that affects all aspects of human life at different levels. It is the realization of the vast applications of mathematics that made to posit that a disciplined and ordered pattern of life can only be achieved through the culture of mathematics. (Eraikhuemen, 2003). However, students' achievement in this important subject has not been encouraging at a secondary level of education in Sri Lanka. Especially in geometry, the poor performance is evident in all parts of Sri Lanka including the Jaffna district.

Geometry can be defined as a science to use graphic representations to find solutions to spatial problems (Pare, 1997). These spatial problems require the ability to use spatial visualization to mentally manipulate and interpret visual information in problem solving situations (Wiley, 1995). The NCTM standards describe geometry as a way to provide students with the ability to visualize and work with spatial relationships and estimation.

If a student wishes to reach a higher order thinking in geometry he must pass through the levels of Van Hiele. It has also shown that materials and methodology can be designed to match levels and to promote growth through the levels (Burger and Shaughnessy, 1986).

Although so many factors affect the developmental stages, students can achieve the target level with the help of proper guidance and instruction. Van Hiele believed that

cognitive progress in geometry can be accelerated by instruction. The progress from one level to the next one is more dependent upon instruction.

The Van Hiele level theory provides a clear picture and a remedy for student difficulty with higher-order cognitive processes required for success in secondary school geometry. Van Hiele's model of geometric thought and the phases of learning deliver a means for identifying a student's level of geometric maturity and suggest ways to help students to progress through the levels.

In geometry learning, students first learn to recognize whole shapes and then to analyze the relevant properties of shapes. Later they can see relationships between shapes and make simple deductions. In this process of learning students have to engage actively in developing mathematical knowledge by exploring, discussing, describing, and demonstrating.

Therefore, by analyzing the influence of factors on achieving the levels of van Hiele and the impact of factors on the development of geometrical knowledge and skills students can get better achievement in geometry. It is important to make sure children are using quality geometry resources in and outside the classroom and make use of effective teaching strategies and their ability by identifying factors that influence the development of geometric concepts and reach the target level.

2.7 Summary

The findings of the literature concluded that though there are so many factors contributing to the performance of learners in geometry at the secondary level of education, some of the domain factors influence a lot in each and every step of the knowledge development in geometry. In this study, three domain factors are chosen under three divisions in relation to teachers, students and school.

According to the scholars, learning geometry for students is not easy, the foremost cause for the poor performance of students in geometry is teachers' method of teaching and content knowledge. As instruction plays a prominent role in students' performance, the availability and usage of instructional materials also affect much in student development. The performance of learners in geometry is also affected by themselves. Factors related to the students such as interest, study habits and prior knowledge cover a wide range of performance rates. Therefore factors like teachers' content knowledge, teaching strategies, learning theory, curriculum, textbooks, availability of mathematics activity room, students' interest, study habits and prior knowledge are selected as important factors which lead to the poor performance of students in geometry.

This study also focuses its attention on Van Hiele model of thinking and how this model support the teachers as well as instructors to facilitate the teaching and learning process and promote them to achieve each level of geometric thinking such as Recognition (L0), Analysis (L1), Order (L2), Deduction (L3) and Rigor (L4).

On the whole, this study focuses on many literatures to find out their reviews on geometrical concepts and the factors which affect the performance of geometry in detail. Moreover, the impact of each domain factor on every level of Van Hiele geometric thinking was also analysed with the help of literature findings. It was used to establish the influence of domain factors on students' development of geometric knowledge and skills.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This study aimed to analyze how the domain factors influence learning geometrical concepts in senior secondary students. The domain factors, which were chosen from the literature of scholars are given focus in the study. As the purpose of this study was to analyze the impact of domain factors on the development of geometrical conceptual knowledge and skills, research design and research methods were chosen according to progress and the need of the research.

“Methods and methodologies of the research represent the procedures and operations that will be followed in carrying out the investigations in the study” (Cohen and Manion, 1992). Furthermore, Bell (1992) suggests that these research methods need to be selected based on whether the methods will be used to collect data that is required to produce a complete picture of reliable and valid research.

The mixed method was used in this study. Here both quantitative and qualitative approaches had been used to investigate the impact of the factors in a more effective way.

This chapter focuses on the approaches and techniques used to collect data relevant to the research study. In addition, it describes and analyses these methods, throwing some light on their limitations and resources.

3.2 Research Design

A research design is “a blueprint for conducting a study with maximum control over factors that may interfere with the validity of the findings”(Burns and Grove, 2003). Research design is the scheme outline or plan that is used to generate answers to research problems. It is an arrangement of conditions and analysis of data in a manner that aims to combine relevance and research purpose (Orodho, 2003).

By focusing the attention on relevance and research purpose, the descriptive survey research design was utilized in the study. In descriptive research, a lot of data was

collected on variables and measured to arrive at causal relationships. A lot of questionnaires and interviews could be used in descriptive research as this research could be done with both quantitative and qualitative research. Moreover, this study emphasizes the relationship between the domain factors and the development of geometrical concepts. The relationship was analysed mainly by the multiple linear regression analysis technique. Hence the descriptive survey research design is more appropriate for the purpose of the study, it was used in the study.

3.3 Research Methods

The research method is a set of guidelines and instruments to be followed in addressing the research problem (Creswell, 2009). Research methodology is the path through which researchers need to conduct their research. It shows the path through which these researchers formulate their problem and objective and present their results from the data obtained during the study period. The research methods summarize the model, which was used in the study. This study adopted the mixed methods design. A mixed method is the type of research in which the researcher combines elements of qualitative and quantitative research approaches for the purposes of breadth and depth of understanding (Creswell, 2010). Both Quantitative and Qualitative methods were used in the study.

3.4 Conceptual frame work

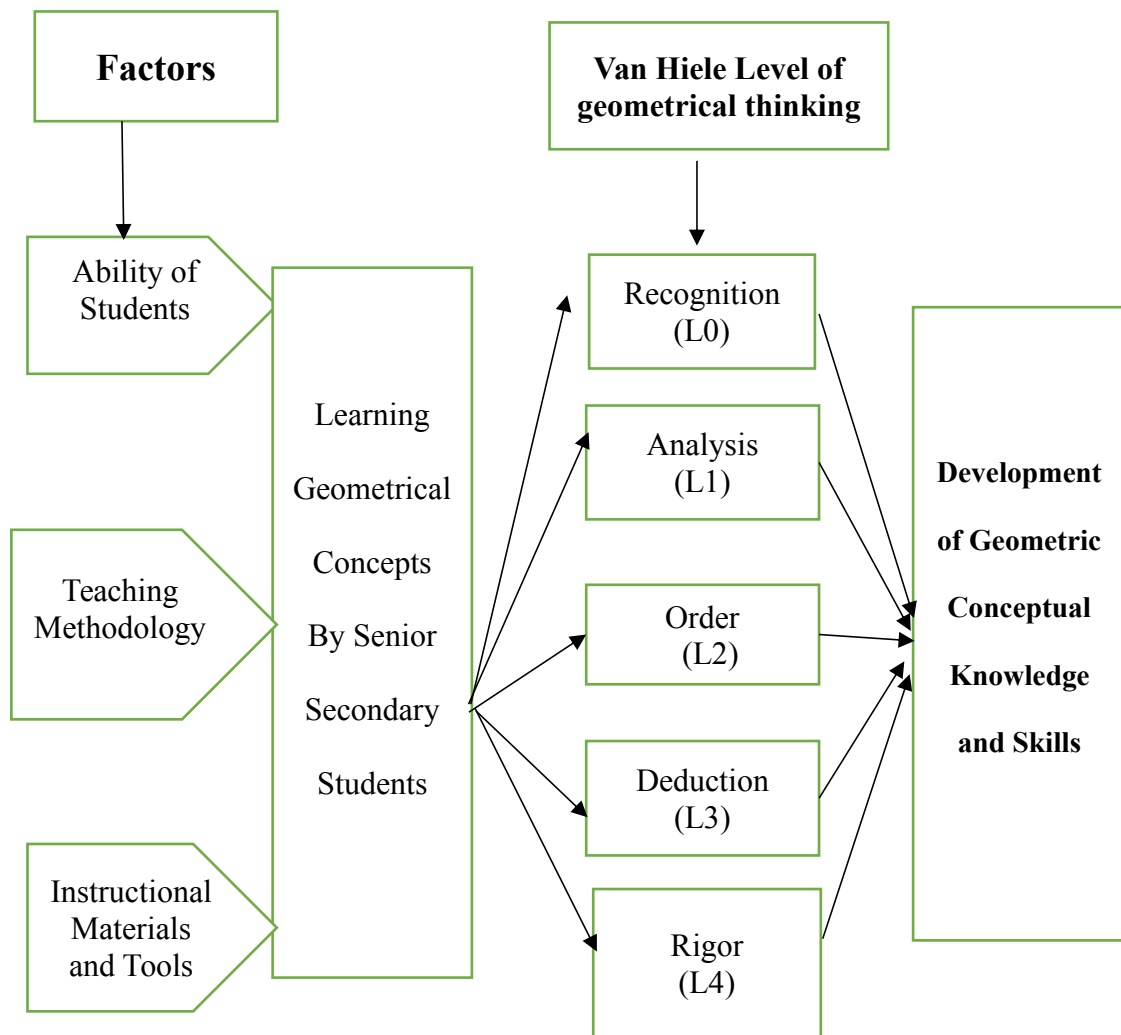


Fig 3.1: Conceptual frame work

For the development of geometric conceptual knowledge and skills, a learner has to achieve the levels of Van Hiele step by step. Van Hiele levels consist of Recognition (L0), Analysis (L1), Order (L2), Deduction (L3) and Rigor (L4). In the Sri Lankan mathematics curriculum, Van Hiele levels are in a hierarchical order in the senior secondary level. If the students are given proper guidance to pass the prescribed levels step by step through their learning, they will reach the target level by developing geometrical knowledge and skills. In this study, how the development of geometric conceptual knowledge and skills is influenced by the above mentioned factors was analyzed.

3.5 Theoretical frame work

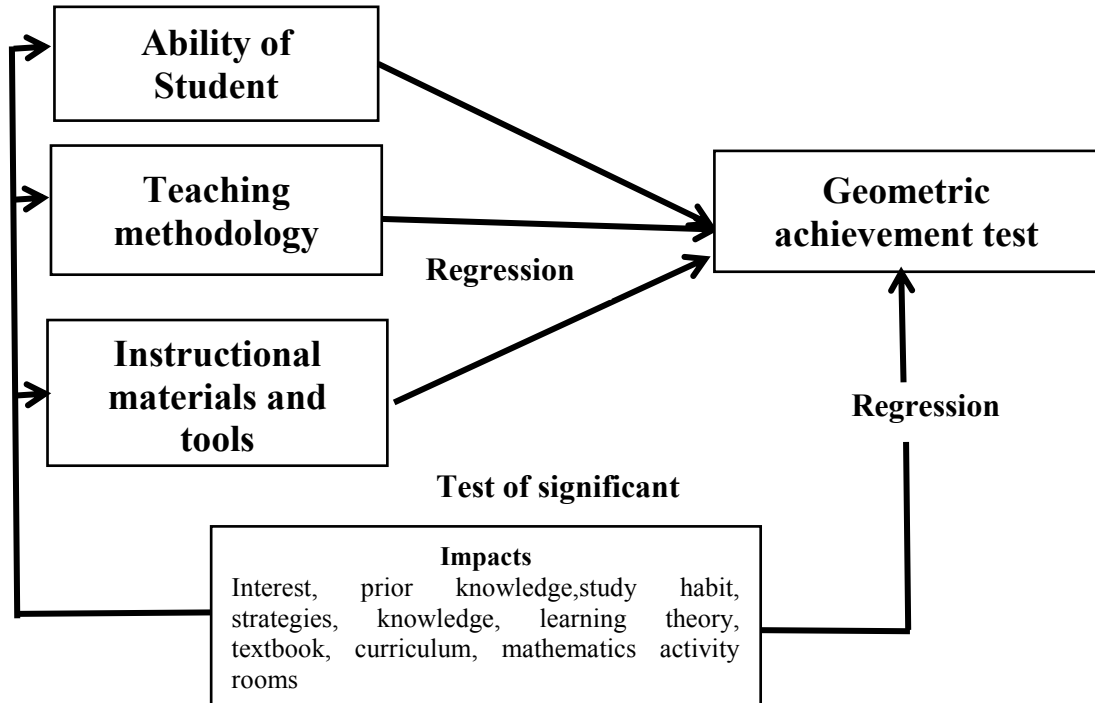


Fig 3.2: Theoretical frame work

Relationship between the three domain factors such as ability of student , teaching methodology and instructional materials and tools with geometric achievement test was analysed by using regression theory. Through entering theory the overall impact of the domain factors and components of the domain factors was analysed. And how the domain factors and components of the domain factors impact was analysed through stepwise theory.

3.6 Research Questions

Research question: What is the impact of domain factors on development of geometric conceptual knowledge and skills?

This study targets to analyze the impact of factors on the development of geometric conceptual knowledge and skills. If the solution is found for the research question the target will be fulfilled. However, it's not possible to include all the factors and analyze their impacts. Due to the practical convenience, three domain factors were chosen to analyze the impact on the development of geometric conceptual knowledge and skills. As the selected domain factors are 'ability of student', 'teaching methodology' and 'instructional materials and tools' the following subsidiary research questions were formed to fulfil the purpose of the study.

Subsidiary research questions

1. What is the impact of ability of students on the development of geometric conceptual knowledge and skills?
2. What is the impact of teaching methodology on the development of geometric conceptual knowledge and skills?
3. What is the impact of instructional materials and tools on the development of geometric conceptual knowledge and skills?

3.7 Hypotheses

The following hypotheses were chosen in order to meet the research objectives.

- $H_{O,1}$: There is no impact of the domain factors on the development of geometric conceptual knowledge and skills.
- $H_{O,2}$: There is no impact of ability of student on the development of geometric conceptual knowledge and skills.
- $H_{O,3}$: There is no impact of teaching methodology on the development of geometric conceptual knowledge and skills.
- $H_{O,4}$: There is no impact of instructional materials and tools on the development of geometric conceptual knowledge and skills.

The alternative hypotheses are:

The following hypotheses were chosen in order to meet the research objectives.

- $H_{A,1}$: There is a significant difference between the impact of the domain factors on the development of geometric conceptual knowledge and skills.
- $H_{A,2}$: There is a significant difference between the impact of ability of students on the development of geometric conceptual knowledge and skills
- $H_{A,3}$: There is a significant difference between the impact of teaching methodology on the development of geometric conceptual knowledge and skills.
- $H_{A,4}$: There is a significant difference between the impact of instructional materials and tools on the development of geometric conceptual knowledge and skills.

3.8 Sampling

To analyze the impact of factors on learning geometrical concepts 50 teachers and 379 students from 22 senior secondary schools were selected from all 5 zones of the Jaffna district. Based on the study purpose the schools which hold at least 50% of pass rate in mathematics exam in G.C.E O/L was chosen for the study. To obtain rich data from the teachers, teachers were selected based on having teaching experience to conduct the class for grade 10 and 11 students and having paper marking experience.

3.8.1 Sampling Techniques

As utilized in qualitative and mixed methods research, in order to select the students with some specific conditions purposive sampling was applied when selecting samples. In this study, 379 students were selected from grades 10 and 11. The researcher utilized a purposive sampling technique to select informants based on their particular needs.

To analyse the impact of the factors on the development of geometric conceptual knowledge and skills, a question paper that includes Van Hiele's all five levels of problems was distributed to the students. This question paper contained the problems in each level in equal portions. Unless students attempt and give the answers to the questions, it's not possible to analyze the impact of the factors. The statistics of the department of education shows that the performance of secondary level students in geometry is not sufficient enough. As 22% of questions in G.C.E O/L mathematics paper contains geometry portion, the sample was selected from the schools which shows at least 50% of pass percentage in mathematics. Since the students from these schools are capable enough to answer the questions in the geometry achievement test, the purposive sampling technique was used to select the schools.

3.8.2 Sample size

According to the need of this study, the researcher has chosen 22 schools from the Jaffna district. Among the 22, 06 were from the Jaffna zone, 06 were from the Valikamam zone, 04 were from the Thenmaradchi zone, 04 were from the Vadamaradchi zone and 02 were from the island zone. Schools were selected from all the categories.

Table 3.1: Sample of the schools

	1AB		1C		Type2		
zone	Total	sample	total	sample	total	sample	total sample
jaffna	12	02	12	02	41	02	06
Valikamam	13	02	14	02	38	02	06
Vadamarachi	09	01	09	01	31	02	04
Thenmarachi	05	01	06	01	24	01	03
Island	04	01	07	01	15	01	03
	33	07	48	07	139	02	22

From each school, students from grades 10 and 11 were chosen. 379 students were engaged in the study including 200 students from grade 11 and 179 students from grade 10.

Table 3.2: Sample of students

Zone	Grade 10	Sample	Grade 11	Sample	Total	Sample
Jaffna	2562	53	2180	56	4742	109
Valikamam	2999	52	2367	54	5366	106
Vadamarachi	2699	35	1832	40	4532	75
Thenmarachi	843	26	708	30	1551	56
Island	757	13	717	20	1474	33
Total	9860	179	7804	200	25341	379

3.9 Research Instruments

The following instruments were used throughout the study.

- ❖ Teacher questionnaire
- ❖ Student questionnaire
- ❖ Interview:
 - i) Researcher with mathematics teachers
 - ii) Researcher with students
 - iii) Researcher with principals
 - iv) Researcher with subject coordinators
- ❖ Document analysis
- ❖ Observation
- ❖ Geometry Achievement Tests

3.10 Questionnaire

3.10.1 Questionnaire For Students

The questionnaire was used as the main data collection tool for collecting data from students. The structure and contents of the questionnaires were carefully and meticulously organized to ensure the reliability and validity of the data for this study. The following points were considered when preparing and finalizing the questionnaires.

1. Question items were prepared considering the research techniques used.
2. The question items were arranged in such a way as to achieve the objectives of the study and to answer the research questions.
3. Preliminary testing was carried out to ensure the correctness and reliability of the questionnaire and the structure and content of the questionnaires were evaluated.

3.10.1.1 Development stages of the questionnaire for students

The questionnaire for students was developed in 4 stages as follows

Phase I: Literature reviews based on the objectives of the study and the research questions were used to analyze the relevant contexts such as family background and attitudes towards the subject matter, their interest in geometry, prior knowledge on geometry, study habits and the difficulties they face when applying geometry and the teaching methods.

Phase II: In the second phase of the questionnaire preparation, the question items were organized in an appropriate order and arranged into parts. The first part consists of general information about the students, the second part includes student-related factors the third part includes teacher-related factors and the fourth part is about instructional materials and tools.

Phase III: The questionnaire was organized in such a way and the advice of the supervisor was obtained and necessary corrections were made accordingly.

Phase IV: The pilot test was conducted to test the structure and content of the questionnaire and the questionnaire was distributed among the students.

3.10.1.2 Preliminary Testing of Questionnaires for Students

The pilot test was conducted in 4 schools from the Jaffna zone. 30 students who were not included in the student sample were randomly selected and were given questionnaires and necessary instructions to answer them. Completed questionnaires were then collected after being given 30 minutes to answer. Are relevant and expected answers to the questions available? Are the questions in the questionnaire correctly and clearly understood by the students? Is the language used in the questionnaire compatible? Such matters were taken into consideration and necessary changes were made and then the questionnaire was finalized.

As a precautionary measure, the researcher decided to give students 40 minutes to answer the questionnaire as they found it difficult to do so in 30 minutes and took into consideration the occasional difficulties faced by the students regarding the language and made necessary corrections in them.

3.10.1.3 Content of the Student Questionnaire

Questionnaire preparation was carried out to ensure the reliability and validity of the data. The objectives of the study, the research questions, and the continuity between the areas were considered and the questions were organized.

The questionnaire contains the question items that deal with student interest in learning geometry, study habits, prior knowledge, teaching strategies, teachers' knowledge, learning theory, textbook, curriculum and mathematics activity room.

The student questionnaire was based on a 5- point Likert scale having the following options; Strongly Agree = S.A, Agree = A, Undecided= U, Disagree = D Strongly Disagree = S.D.

3.10.1.4 Distribution and administration of student questionnaire

A systematic timetable was set up to provide questionnaires and collect answered questionnaires. Prior permission was obtained from the principals regarding the distribution of questionnaires to the students and the day was determined and the distribution of the relevant questionnaires was carried out.

The researcher directly involved in the data collection process and was accompanied by a written document containing instructions for completing the questionnaire and the researcher's assurance that privacy would be maintained. This included the need to respond reliably, the interpretation of the method of answering, and the assurance that the confidentiality of the information provided would be maintained in terms of research ethics. The questionnaires were then given to the students in the already determined manner and allowed to answer them and the completed questionnaires were immediately collected by the examiner.

3.10.2 Questionnaire for teachers

The questionnaire was used as the primary tool for obtaining information from teachers. The following points were taken into consideration while preparing the questionnaires for the teachers.

1. The objectives of the study were well defined and the structure of the questionnaire was organized accordingly.
2. The question items for this questionnaire were prepared considering the research questions used in this study.
3. The question items were organized in such a way as to achieve the objectives of the study and to provide answers to the research questions.
4. Preliminary tests were conducted to determine the suitability of the questionnaires and the suitability of the question items were evaluated.

3.10.2.1 Development stages of the teachers' questionnaire

The questionnaire for teachers was developed in 4 phases.

Phase I: Literature reviews based on the objectives of the study and the research questions were used to analyze the relevant contexts. Teachers' questionnaire was designed to look at what approaches the teachers use in teaching geometry, the difficulties they face when teaching geometry, their professional background and teacher education, vocational qualification and the students' attitude towards learning geometry.

Phase II: In the second phase of the questionnaire preparation, the question items were organized in an appropriate order and arranged into parts. Part I Questionnaire items were organized for obtaining general information about teachers. Part II Questionnaire items were organized for obtaining other needed information.

Phase III: The questionnaire was organized in such a way that it could be distributed and the advice of the supervisor was obtained and necessary corrections were made accordingly.

Phase IV: The pilot test was conducted to test the structure and content of the questionnaire.

3.10.2.2 Pilot testing of teacher questionnaire

The pilot test was conducted in 4 schools in the Jaffna zone. Questionnaires were given to 05 teachers and all the answered questionnaires were collected and the expected answers were obtained. Are the questions correctly and clearly understood by the teachers? Is the language used in the questionnaire appropriate for the teachers to answer? Such matters were taken into consideration and appropriate changes were made and then the questionnaire was finalized.

3.10.2.3 Content of the Teacher questionnaire

Questionnaire preparation was carried out to ensure the reliability and validity of the data. The objectives of the study, the research questions, and the continuity between the areas were considered and the questions were organized.

the questionnaire contains personal information of the teachers, information about the students in the class, study habits of the students in the class, how the teachers deal with difficulties of the students, Assessment methods, teaching methodology, curriculum and geometry.

The questionnaires were in sections A to F. Section A contains questions on the demographic characteristics of the respondents while sections B to F contains questions based on the research objectives. The teacher questionnaire was based on a 5- point Likert scale having the following options; Strongly Agree = S.A, Agree = A, Undecided= U, Disagree = D Strongly Disagree = S.D.

3.10.2.4 Distribution and administration of teacher questionnaire

As already predicted, data was obtained from teachers on the same day as the student questionnaire and focus group discussions were done. The teachers were given the purpose of the study, the content of the questionnaire and instructions for completing the questionnaire and the researcher explained the doubts and ambiguities raised by the teachers.

3.11 Focus group discussion

A Focus group is a technique used for discussing the issue as a group, absorbing the opinions of others and presenting their own opinions based on them (Anderson, 1990). Through these opinions, results can be logically discussed and the data obtained through focus group discussion can be found to be highly organizationally competent (Morgan, 1988; Krueger, 1994). This technique was used to obtain characteristic information and further information needed for the most important reason for the study.

In this study, 2 teachers focus group discussions and 2 students focus group discussions were conducted. Each teachers group had four mathematics teachers and each student group had six students. They presented their views and opinions on the issue and exchanged views on possible steps to be taken to avoid the issues and the opinions were documented by the researcher.

3.12 Interview

Interviews were done with different types of groups (Mathematics ADES, subject coordinators, mathematics teachers, and students) to explore their views, experiences, beliefs and motivations.

All interview guides and questions were developed, tested, and then refined based on what one learns from asking the questions.

According to Cohen (2000), an interview is “an interchange of views between two or more people on a topic of mutual interest”. It sees the centrality of human interaction for knowledge production and emphasizes the social context of research data. In addition, “interviews enable participants to discuss their interpretations of the world in which they live and to express how they regard situations from their own point of view”(McMillan and Schumacher, 2001). It could be argued that an interview is a two-way conversation initiated by the interviewer for the specific purpose of obtaining research-relevant information. This information focuses on content specified by the research aims of systematic description, prediction or explanation. It involves the gathering of data through direct verbal interaction between individuals. The real benefit of an interview is that the researcher is face-to-face with the respondents so that any misunderstandings can be cleared up immediately (Whites, 2000). Respondents can question what they do not understand and the researcher can re-word or re-order the questions if something unexpected happens.

Since the interview schedule is semi-structured during the interviews, the researcher explored how teachers engage students in geometric thinking and the interview process was used to explore the teaching strategies employed by the selected mathematics teachers in geometry learning and teaching against the background of the Van Hiele levels. Furthermore, this process was continued with principals, subject coordinators and teachers regarding how the teachers utilized the resources like activity room and teaching aids also the qualifications and the content knowledge of the teachers.

Interviews were conducted on the specified dates with prior permission and information was obtained.

3.13 Observation

The second data collection technique was observation. (Whites, 2000) describes observation as “an accurate record of what people do and say in real-life situations”. Observation data allows the researcher to gather live data from live situations (Cohen, 2000).

In this study, a strategy of non-participant observation was employed. Non-participant observation is a “technique where the researcher does not become completely involved in the situation which is being researched”. This was the case in the current study.

The observation technique was used to analyse the availability and usage of instructional resources like electronic tools, mathematics activity room, curriculum, textbooks, and student-related variables like their study habits and the ability, in particular how these factors affect learning geometrical concepts.

3.14 Document analysis

Document analysis and records are “useful sources of information, readily available, stable and easily accessible” (Lincoln and Guba, 1985). This method of collecting information also serves the purpose of triangulation.

The following documents were analysed in the study: The geometry syllabus, textbooks, lesson plans, schemes of work, lesson preparations, tasks or activities on geometry and the grade 10, 11 mathematics curriculum. Through documentation how the teachers use the syllabus in terms of their curriculum was analyzed.

Finally, the researcher located and recorded citations to books and other documents which contained useful information and views on the area of the study.

3.15 Mathematics Achievement test

Standardized tests play an important role in providing information concerning student ability, achievement and growth. The geometry achievement test was the major data collection technique used in the study. This test was conducted in grades 10 and 11. The following points were considered when preparing the geometry achievement exam paper.

1. The question items were formed in such a way as to achieve the objectives of the study and to answer the research questions.
2. Preliminary testing was carried out to ensure the correctness and reliability of the geometry achievement test and the structure and content of the exam paper were evaluated.

3.15.1 Content of the mathematics achievement test

For this exam, a question paper was formed including all five levels of geometrical thinking. Altogether twenty five questions were included in the paper, questions were divided into five sections and under each section, five questions were included on each level of Van Hiele geometrical thinking. All the questions were formed based on Usiskin's geometrical test items. But with slight changes to suit the Sri Lankan syllabus.

Table 3.3: Van Hiele levels

van hiele level
Level 1 : Visualization
Level 2 : Analysis
Level 3 : Informal Deduction
Level 4 : Formal Deduction
Level 5 : Rigour

3.15.2 Pilot Study of the Mathematics Achievement test

A pilot study of this study was undertaken on 30 students of the sample population which was not included in the final research. This pilot study was conducted by giving geometry achievement test papers. The pilot study aimed to give the researcher an insight into whether the intended questions were to be given to students and provide the desired data required to answer the research questions. All the answered exam papers were collected and the expected answers were obtained. Are the questions correctly and clearly understood by the students? Is the language used in the exam paper appropriate for the students to answer? Such matters were taken into consideration and appropriate changes were made and then the geometry achievement exam paper was finalized.

3.15.3 Administration of Mathematics Achievement test

As already decided, the achievement test was conducted on the same day as the student questionnaire, teacher questionnaire and focus group discussions were done. The exam papers were then given to the students in the already determined manner and allowed to answer them and the completed exam papers were immediately collected by the examiner.

3.16 Data Collection Procedures

In this study, quantitative and qualitative data collection methods were used. The data required for this study were collected from the selected schools in the Jaffna district. The researcher selected 5 schools from each zone. As the researcher wanted to select the schools with a 50% pass rate the schools were selected to suit accordingly.

Standard data collection procedures were used throughout the study. First, the approval for permission to conduct the study was obtained from the Northern provincial director of education. After that, a request letter was submitted to the principals of the selected schools along with the approval letter.

Since there are five educational zones in the Jaffna district, school visiting was proceeded as one zone per week from 25th of January 2021 to 25th of February 2021. In each zone, the researcher visited the selected schools one by one. The principals guided the researcher towards mathematics subject coordinators and in all the schools, coordinators were very helpful and supportive for the progress of the study.

In the schools, geometry achievement test was conducted to the selected students and questionnaires were given to the same set of students and the selected teachers. 379 students and 50 teachers from all the schools were selected for the study. Then the students were informed verbally that they are going to answer the questionnaire and have a test on geometry to check their achievement level in geometry. Then the students' questionnaires were collected from them and their geometry achievement test marks were carefully analyzed throughout the study.

Teacher questionnaires were distributed to the selected teachers from each school and focus groups discussion were done with selected teachers and students focus groups, Interviews were done with mathematics A.D.E, Mathematics I.S.A, subject coordinators, mathematics teachers, and students.

The data obtained from observation and documentary were also used in the study to analyse the influence of the domain factors on learning geometrical concepts

3.17 Data Analysis Techniques

After the data collection, the researcher turned to the task of analyzing. The following analysing methods had been used in this study.

Quantitative data were analysed by using;

- Descriptive analysis
- Liner Multiple Regression

Qualitative data were analysed by using

- Thematic analysis

3.18 Descriptive analysis

First, descriptive statistics analysis was used for data analysis. quantitative data was analysed to obtain frequencies, percentages and averages. The analysis required a number of closely related operations such as the establishment of categories, the application of those categories to raw data through coding, tabulation and drawing statistical inferences. The researcher classified raw data into purposefully and usable categories.

Data collected on both questionnaires were used to answer the question of whether the domain factors ‘ability of students’, ‘teaching methodology’ and ‘instructional materials and tools’ enhance the understanding of the geometry concepts, and to what extent they do so.

3.19 Liner Multiple Regression

This study mainly emphasizes the relationship between the domain factors and the development of geometrical concepts. “What is the impact of domain factors on the development of geometric conceptual knowledge and skills?”. This is the main research question of the study. To find out the answer to this question, the relationship had been analysed by using the multiple regression analysis technique.

Linear Multiple Regression

The multiple linear regression procedure commenced with screening data for verification of suitability (Francis, 2001)

Multiple linear regression is used to estimate the relationship between two or more independent variables and one dependent variable. Multiple linear regression can be used to know:

1. How strong the relationship is between two or more independent variables and one dependent variable.
2. The value of the dependent variable at a certain value of the independent variables.

The basis of multiple linear regression is to assess whether one continuous dependent variable can be predicted from a set of independent (or predictor) variables. In other words, how much variance in a continuous dependent variable is explained by a set of predictors. Certain regression selection approaches are helpful in testing predictors, thereby increasing the efficiency of analysis.

In the study, how the relationship among the three domain factors and geometric achievement test is being fit in the model had been analysed.

This analysis was done by using two main regression theories. The first one is the standard method of entry theory and the second one is the Selection theory. Under selection theory, the stepwise selection method had been used.

3.19.1 Standard method of entry theory

The standard method of entry is simultaneous (the enter method); all independent variables are entered into the equation at the same time. This is an appropriate analysis when dealing with a small set of predictors and when the researcher does not know which independent variables will create the best prediction equation. Each predictor is assessed as though it were entered after all the other independent variables were entered and assessed by what it offers to the prediction of the dependent variable that is different from the predictions offered by the other variables entered into the model.

3.19.2 Stepwise Selection theory

Selection, on the other hand, allows for the construction of an optimal regression equation along with an investigation into specific predictor variables. Selection aims to reduce the set of predictor variables to those that are necessary and account for nearly as much of the variance as is accounted for by the total set. In essence, selection helps to determine the level of importance of each predictor variable. It also assists in assessing the effects once the other predictor variables are statistically eliminated. The circumstances of the study, along with the nature of the research questions guide the selection of predictor variables.

Stepwise selection is considered a variation of the two methods: Forward selection, Backward elimination. Stepwise selection involves analysis at each step to determine the contribution of the predictor variable entered previously in the equation.

Stepwise linear regression is a method of regressing multiple variables while simultaneously removing those that aren't important. Stepwise regression essentially does multiple regression a number of times, each time removing the weakest correlated variable. At the end, you are left with the variables that explain the distribution best. The only requirements are that the data is normally distributed (or rather, that the residuals are), and that there is no correlation between the independent variables.

3.19.3 Fit in Liner Multiple Regression

How the above mentioned theories fit in the regression model is displayed in detail below.

Step 1

To examine how the predictors combine to influence the dependent variable, standard regression analysis was executed on the SPSS program. In this method, all selected variables are entered into the model at once.

The multiple Regression coefficient (R) is an important statistic in the regression analysis. It is the square root of the coefficient of determination or the correlation squared (R^2), which is the total proportion of variation of the dependent variable explained by dependent variables.

How all three domain factors (AS, TM, IM) fit with the geometric achievement test (GAT) in the model in overall had been analysed.

Similarly, How a domain factor (AS) fit with the geometric achievement test (GAT) in the model in overall had been analysed.

How a domain factor (TM) fit with the geometric achievement test (GAT) in the model in overall had been analysed.

How a domain factor (IM) fit with the geometric achievement test (GAT) in the model in overall had been analysed using the enter method

Step 2

The Stepwise models are composed of different sets of variables. These models are the combinations of variables that best explain the dependent variable. To identify the most influential variable the following steps had been used

First variables should be Entered / Removed. From this how the variables used to build the models is formed.

The next should be the "Model Summary" which gives details of the overall correlation between the variables left in the models and the dependent variable.

The final step should also be forming a Coefficients box, showing the linear regression equation coefficients for the various model variables. The "B" values are the coefficients for each variable, that is, they are the value which the variable's data should be multiplied by in the final linear equation we might use to predict long term

problems. The "Constant" is the intercept equivalent in the equation (i.e. the equation would be $y = \text{constant} + (v1 \times \text{coeff1}) + (v2 \times \text{coeff2}) + \dots$). The Significance (Sig.) figures should be 0.05 or below to be significant at 95 percent. A value of .000 means the figure is too small for a three decimal place representation.

In step 2, variables that are not important have been removed and the most important variables which fit in the model are listed.

Among three domain factors (AS, TM, IM), which most influential factors fit with the geometric achievement test (GAT) in the model had been analysed.

Like that how each domain factor fit with the geometric achievement test (GAT) in the model had been analysed separately.

Finally by using the above theories whether the domain factors fit with the geometric achievement test (GAT) in the model had been analysed. From that the most influential factors were listed.

3.19.4 Equation of Liner Multiple Regression

Multiple regression analysis was conducted to ascertain the relationship between the study variables.

The regression model to be tested is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

Where: - Y = Level of Achievement/performance in Geometry test

Explained Variations of the Model = $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$

X1 = Teaching Methodology

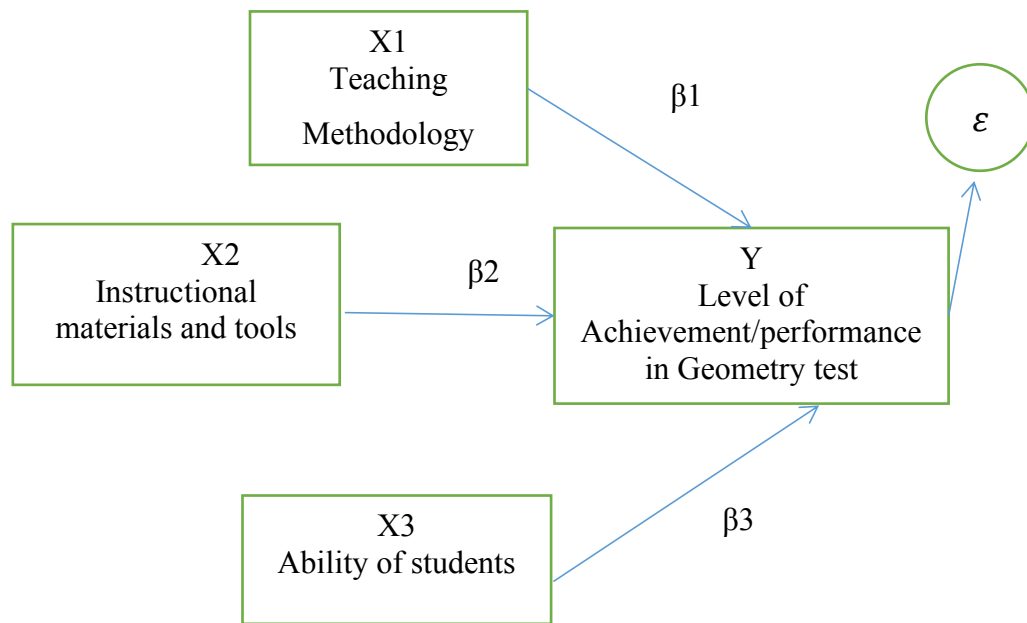
X2 = Instructional materials and tools

X3 = Ability of students

ε = Unexplained Variation i.e. error term, it represents all the factors that affect the dependent variable but are not included in the model either because they are not known or difficult to measure.

β_0 = Constant. It defines the level of achievement in Geometry without the inclusion of predictor Variables.

$\beta_1, \beta_2, \beta_3$ = Regression Co-efficient. Define the amount by which Y is changed for every unit change of predictor variables. The significance of each of the co-efficient was tested at 95 Percent level of confidence to explain the variables.



$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

Multiple regression

Fig 3.3: Multiple regression model

3.20 Thematic analysis

Thematic analysis is one of the most popular qualitative analysis techniques. It is a method for identifying, analyzing, and interpreting patterns of meaning ('themes') within qualitative data.

Table 3.4: Phases and stages of theme development in thematic analysis

Phases	Stages
Initialization	Reading transcriptions and highlighting meaning units; Coding and looking for abstractions in participants' accounts; Writing reflective notes.
Construction	Construction Classifying; Comparing; Labeling; Translating & transliterating; Defining & describing.
Rectification	Rectification Immersion and distancing; Relating themes to established knowledge; Stabilizing.
Finalization	Finalization Developing the story line

Themes were formed from the interview transcripts. The development of the themes was analysed through the above mentioned phases and stages.

3.21 Variables

A number of variables were used in this study to analyse the process of multiple regression.

3.21.1 Variable of Student ability

Table 3.5: Variable of Student ability

Ability of student (AS)	Interest in learning geometry	ASI1	Look forward to do geometry
		ASI2	Study geometry with confident
		ASI3	Never get tired of doing exercises in geometry
	Study habits	ASS1	Study geometry exam oriently
		ASS2	Study geometry through practice
		ASS3	Use technology to study geometry
	Prior knowledge	ASP1	Have a clear knowledge of previous geometry lessons
		ASP2	Can do previous grades geometrical sums now
		ASP3	Need prior knowledge of geometry to learn lessons further

3.21.2 Variable of Teaching Methodology

Table 3.6: Variable of Teaching Methodology

Teaching Methodology (TM)	Teaching Strategies		
		TMTS1	Teaches geometry sum from simple to difficult
		TMTS2	Teacher uses appropriate teaching aids
		TMTS3	Teacher uses new technology in teaching geometry
	Teachers knowledge	TMTK1	Teacher has lot of geometrical knowledge
		TMTK2	Teacher applies his geometrical knowledge appropriately in teaching
		TMTK3	Teacher is always ready to clear doubts at the time
	Learning theory	TMLT1	Teacher uses techniques to motivate learning geometry
		TMLT2	Teacher creates appropriate learning environment
		TMLT3	Teacher gives opportunity to expose real talents in geometry

3.21.3 Variable of Instructional materials and tools

Table3.7: Instructional materials and tools

Instructional materials and tools (IM)	Text book		
		IMT1	Textbook is an important tool in learning geometry
		IMT2	Text book contains enough Geometrical sums to answer the questions in exam
	Curriculum	IMT3	Textbook has vivid explanations to illustrate the geometry lesson for better understanding
		IMC1	Textbook has well designed Geometry lessons to help students in self learning
		IMC2	Geometry lessons move from easy to difficult step by step
	Mathematics activity room	IMC3	Suitable evaluation and assessment is done to give necessary remedial measures.
		IMM1	Use mathematics activity room in learning geometry
		IMM2	Mathematics activity rooms has suitable resources to learn geometry
		IMM3	Feel very comfortable when I do geometry in mathematics activity room

3.22 Ethical consideration

As research ethics is known as the application of moral rules and professional codes of conduct to the collection, analysis, reporting and publication of information about research subjects the investigator handled consent appropriately, both in planning and executing the experiment.

Since this study often involved a great deal of cooperation and coordination among students and teachers in different schools, ethical standards promote the values that were essential to collaborative work, such as trust, accountability, mutual respect and fairness were taken into consideration.

Consent involves the procedure by which an individual may choose whether or not to participate in a study. The participants of the study were informed through a consent letter and a detailed description of the research was given verbally. Both teachers and students gave their consent voluntarily. Therefore the researcher ensured that participants have a complete understanding of the purpose and methods to be used in the study, the risks involved, and the demands placed upon them as a participant.

Ethics is the study of moral principles and values which govern the way an individual or group conducts its activities (Churchill, 1995). As determined earlier the research data of the study would be kept confidential and not used for any other purpose. The anonymity of research participants would be protected.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

In this chapter, the process of analyzing how the domain factors influence the development of geometrical knowledge and concepts is presented. It was then used to address the relationship among these factors. In addition to that, the current status of geometry was analysed as the key step in the progress.

This chapter describes the analyzing techniques that the researcher used to gather information that specifically answer research questions and meet the objectives of the study. The collected raw data were displayed, described, and analyzed, which was originally appeared in the responses to the items in the instruments. From the analysis, research findings were crystallized out.

The data were presented in two separate divisions as biographical information and factors that influence the students' geometrical knowledge development.

4.2 Biographical Information

Biographical information like the type of school, teachers' highest academic qualification, teachers' professional qualification, teachers' experience, participation in geometry seminars, gender of the students and age group of teachers were gathered for the fundamental analysis of the study.

4.2.1 Type of School

According to the need of this study, the researcher has chosen 22 schools from the Jaffna district. Among them Jaffna zone, Valikamam zone, Thenmaradchi, Vadamaradchi and Island zone schools were selected from all the categories. From each zone, 1AB schools, 1C schools and type ii schools were included in the sample.

4.2.2 Gender of the Students

While the participation rate of male students in this study was 55%, the female students were 45% of the sample. The above proportions were maintained to generalize the findings for male and female students.

4.2.3 Teachers' Age group

The findings show that 40% of the teachers who participated in the study were between the age of 30 and 40. Teachers between 40 to 50 years old were 30% and those who were below 30 years were 10% of the total sample while 20% of the teachers were above 50 years.

These findings portray that most of the teachers who participated in the study were above the age of 30. This indicates that teachers who participated in this study had enough experience to understand the factors beneath the poor performance of students in geometry.

Table 4.1: Teachers' Age group

Age group	Numbers	Percentage
Below 25	-	-
25-29 -	05	10%
30-39	20	40%
40-49	15	30%
50-60	10	20%

4.2.4 Teachers' Highest Academic Qualification

In the study, the highest academic qualification of the teachers was also focused on. The findings reveal that almost all the teachers have updated their academic qualifications at least by having a diploma in the subject. This chart shows that the majority of the teachers (64%) were with Bachelors degree and the teachers who got masters degree were 10% whereas the teachers who had diploma qualification were 30%. These findings imply that majority of the teachers who teach mathematics in the Jaffna district were qualified enough to teach mathematics, especially geometry. As such the students in the study region were gifted and were given knowledge by qualified teachers.

Table 4.2: Teachers' Academic qualification

Academic Qualification	Number	Percentage
G.C.E(A/L)	03	06%
Bachelors degree	27	54%
Diploma holders	15	30%
Masters degree	05	10%

4.2.5 Teachers' Professional Qualification

Table 4.3: Teachers' Professional qualification

Professional Qualification	Number	Percentage
Trained teacher	05	10%
NDT	12	24%
PGDE	24	48%
B.Ed	05	05%
M.Ed	02	04%

This table shows all teachers were well trained in teaching mathematics by having any type of professional qualification. Among them, 52% of the teachers have been professionally qualified beyond their first degree. 24% of the teachers were the college of education diploma holders and the rest of them were trained from teacher training colleges.

So, geometrical knowledge was imparted to the students by professionally as well as academically qualified teachers.

4.2.6 Teachers' Experience

The study further tried to find out the experience of the teachers in the Jaffna district. The results show while 84% of them were with 10 years experience or more than that 16% of the teachers have been teaching for 6 to 10 years.

Table 4.4: Teachers' Experience

Teachers' Experience	Numbers	Percentage
Below 5 years	-	-
6-10	08	16%
11-15	21	42%
16-20	11	22%
Above 20	10	20%

This table shows that most of the teachers were with more than ten years of mathematics teaching experience.

4.2.7 Participation in geometry seminars

The finding depicts that teachers in Jaffna attend geometry Seminars frequently. The study also seeks to find how much the teachers were interested in attending geometry seminars for their professional development. The result reveals that 70% of the teachers haven't attended any seminars in the past 5 years. Only 30% have attended geometry seminars and geometry seminars are considered as an essential step where teachers can discuss their work with colleagues and exchange their knowledge to improve their teaching.

Table 4.5: Participation in geometry seminars

Participated in geometry seminars in last 5 years	No.	percentage
Didn't participate in any seminars	35	70%
1-2	08	16%
3-4	07	14%
5-6	-	-
7-8	-	-

Objective- 1

4.3 To identify the current status of students' performance in geometry

For the study 'The domain factors influencing in learning geometrical concepts of senior secondary schools', the current status of students' performance in geometry has to be analysed in detail. The current status plays a prominent role in analysing the factors influencing learning geometrical concepts. As it was not possible to analyse the current status of students' performance in geometry only with the marks obtained in the diagnostic test, the data were gathered from literature, achievement test done by the researcher, teacher questionnaire, documentary and interview. The findings of gathered data were analyzed one by one to achieve the first objective - the current status of students' performance in geometry.

4.3.1 Geometry achievement test

This exam was conducted by the researcher to identify the current status of students' performance in geometry in Jaffna district schools. This exam was held in 22 schools including all types of schools in five zones of the Jaffna district. From the selected schools, 379 students were chosen and the geometry achievement test was conducted. For this exam, a question paper was formed including all five levels of geometrical thinking. Altogether twenty five questions were included in the paper, questions were divided into five sections and under each section, five questions were included on each level of Van Hiele geometrical thinking.

The marks obtained by the student were calculated in the following way:

Table 4.6: marking scheme

Van Hiele level	scheme	marks
Level 1 : Visualization	1x5	05
Level 2 : Analysis	2x5	10
Level 3 : Informal Deduction	3x5	15
Level 4 : Formal Deduction	4x5	20
Level 5 : Rigour	5x5	25
		75

The marks for each level of Van Hiele geometrical thinking was given through the marking scheme and they were added together to get the total marks.

Table 4.7: Percentage of achievement of Van Hiele level

Van Hiele level	percentage
Level 1 : Visualization	90%
Level 2 : Analysis	70%
Level 3 : Informal Deduction	55%
Level 4 : Formal Deduction	37%
Level 5 : Rigour	05%

Questions from each level were analysed separately and the percentage of each level of achievement was calculated to identify the students' performance in each and every level.

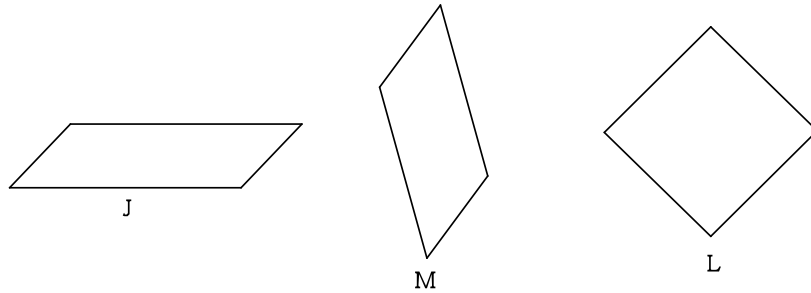
4.3.1.1 Level 1: Visualization

Questions included in this level are about recognizing the figures according to their appearance. In Sri Lanka, students are given knowledge to identify geometrical figures at the junior level and they are capable enough in this level. As even students from the junior level have the capacity to identify geometrical figures, answering the first level questions by senior secondary students was not tough.

However, it is clear that students from the Jaffna district easily achieved the first level of Van Hiele geometrical thinking.

For the following question, all the students identified the triangles, parallelograms and squares. But 10% of the students could not give the correct answer for the final question of level 1

Which of these are parallelograms?



- (A) J only
- (B) L only
- (C) J and M only
- (D) None of these are parallelograms
- (E) All are parallelograms

Among 10% of the students, most of them gave the answer as (C). It reveals that they don't have knowledge of 'square is also a parallelogram'. So even in level-1 also a few students have struggled to identify the figure.

4.3.1.2 Level 2: Analysis

Students understand the properties of shapes in isolation. The second level contained 5 questions. 70% of the students got more than or equal to 4 correct answers.

Table 4.8: Students' performance in level 2

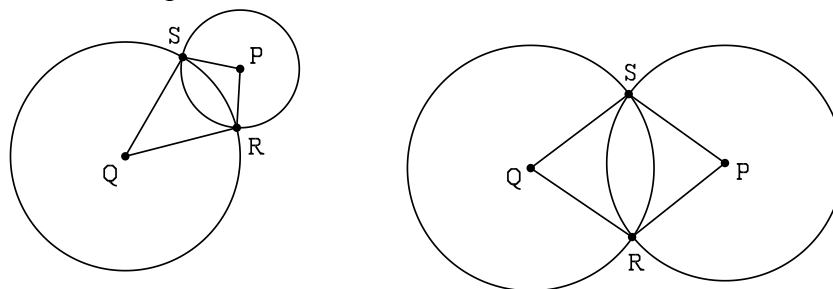
No. of questions	No. of student got the correct answer
1	32
2	102
3	112
4	85
5	48

Questions from 6 to 9 directly deals with the properties of figures emerge.

Where as the 10th question deals with abstract concepts.

(Q-10) Two circles with centers P and Q intersect at R and S to form a 4-sided figure PRQS.

Here are two examples.



Which of (A) – (D) is not always true?

- (A) PRQS will always have two pairs of sides of equal length.
- (B) PRQS will have at least two angles of equal measure.
- (C) PQ and RS will be perpendicular.
- (D) Angles P and Q will have the same measure.
- (F) All of (A) – (D) are true.

Students were struggled to answer the above question. As this sum has some abstract concepts they found it difficult to understand. They were unable to give the correct answer with the knowledge of the properties of quadrilateral only. If the student wants to answer this question, they have to critically analyse the sum. Because this quadrilateral is formed by intersecting two circles. If they related the properties of the circle to find the properties of quadrilateral only they could give the correct answer. But unfortunately, most of the students did not have a clear figure about this.

“If any two circles intersect, a line which joins the intersect points of two circles is perpendicular to the line which joins the centre of the circle”. This is one of the important theorems in the previous curriculum. If the students know this theorem, they could have answered the question very easily.

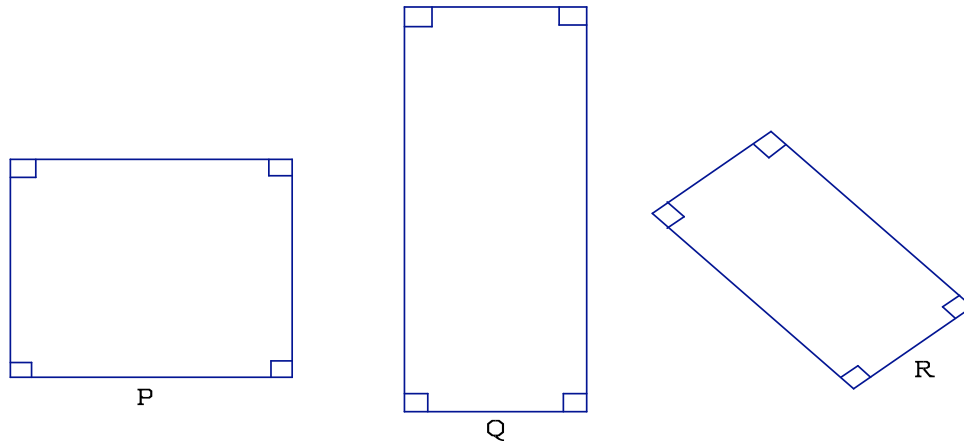
4.3.1.3 Level 3:-Order or informal deduction:

At this level, students can distinguish between necessary and sufficient conditions for a concept. They identify a hierarchical order of connection between groups of different shapes according to their properties and definitions. Definitions and axioms are meaningful for the learners but ordering levels have not been understood by them. For example; they can say ‘ Each square is a rectangle’ but they cannot organize

necessary lists in order to prove this observation. However, at this level, they are incapable of proving claims related to the properties of the geometric shapes.

For example in the question below:

Which of these can be called rectangles?



Which is correct?

- (A) All can.
- (B) Q only
- (C) R only
- (D) P and Q only
- (E) Q and R only

To this question, 70% of the students were capable enough to give the correct answer. Therefore, at this level most of the learners understand the relationship of connection between a rectangle and a square – ‘every square is a rectangle but not every rectangle is a square.

What do all rectangles have that some parallelograms do not have?

- (A) Opposite sides equal.
- (B) Diagonals equal.
- (C) Opposite sides parallel.
- (D) Opposite angles equal.
- (E) None of (A) - (D).

For this question, only 30% of the students gave the correct answer. As the highest number of learners were still unable to prove that the diagonals of a rectangle are the same, they struggled a lot at this level.

4.3.1.4 Level 4:- Formal deduction

At this level, learners understand the roles of basic concepts, axioms, definitions, theorems and proofs and their interrelations. They can use assumptions in order to prove theorems and understand the meaning of necessary and sufficient conditions. At this level, learners are able to provide reasons and arguments for the various levels of the proving process. Moreover, they comprehend the importance of discussing the proofs, the deduction from the particular to the general and even the need for a proof of any kind (Patkin & Levenberg, 2010).

As a result, this level captures the most valuable role in geometry. At this level, students practise to,

- Think and argue deductively
- Understand the relationship between definitions, theorems and axioms
- Construct a formal geometric proof

In the geometry achievement exam paper the questions for level 4 were included in order to analyse the above concepts among Jaffna students.

Table 4.9: Percentage of achievement in level 4

Question no.	Percentage
16	30%
17	25%
18	15%
19	18%
20	10%

The results indicate that Jaffna students were incapable to think and argue deductively, understanding the relationship between definitions, theorems and axioms and constructing the formal geometric proof.

The question given below also supports this.

Which of the following statements always applies to an isosceles triangle but not to all the triangles?

- a. The mid line is parallel to one of the three sides of the triangle.
- b. The mid line is equal to half of one of the three sides of the triangle.

- c. The mid line creates a triangle that is similar to the original one.
- d. The mid-line creates an isosceles trapezium.
- e. The mid line creates a trapezium.

In the above sum, students should have the knowledge of mid point theorem with the knowledge of how this proof constructs in isosceles too. Students with only mid point theorem knowledge too struggled since they were with the lack of knowledge in how to apply the theorem in different situations.

The results of other sums also revealed that students faced difficulties in applying other theorems too. So, it is obvious that unless students know the theorems with their application, deduction and conversion, they cannot reach this level.

4.3.1.5 Level 5:- Rigor

At this level, students understand the necessity for rigor and are able to make an abstract deduction. At this level, non-Euclidian geometry can be understood. It is the highest level of geometrical thinking. In the Sri Lankan syllabus questions from level 5 are included in some portions of geometry. If a student got 3 out of 5 (3/5), it is considered that he/she reached level 5. But in the study, only 18 students from a total sample (379) got three or more than 3. It indicates that our students are not with adequate knowledge to understand the concepts of this level.

4.3.1.6 Results of Geometry achievement test

Table 4.10: Mean value of Van Hiele level achievement test

Van hiele level achievement test	mean
Level 1 : Visualization	4.77
Level 2 : Analysis	9.01
Level 3 : Informal Deduction	10.16
Level 4 : Formal Deduction	8.88
Level 5 : Rigor	4.60
Total	37.06

Purposive sampling was used in this study as the purpose of the study was to analyse the impact of domain factors on the development of geometric conceptual knowledge and skills. Therefore students were selected from the schools which produced a minimum 50 pass percentage. If the capable students attempt this exam only, the impact of domain factors on the development of geometric conceptual knowledge and skill can be analysed. Although this exam was conducted to the students who do mathematics well, they got only 37.06 as the mean value of the exam. Even though it seems that 37.06 mean value was an average score, most of the students scored marks by answering the questions from the first three levels. But for senior secondary students, questions from levels 4 and 5 including proof type are important to develop their knowledge. At this stage, they struggled a lot to give the correct answers for the 4th and 5th level of questions. Therefore the analysis of geometry achievement test reveals that the current status of geometry is poor in senior secondary students.

4.3.2 Geometry diagnostic test

The project was implemented in the Jaffna district in Northern Province in 2019 to promote knowledge in geometry and increase the mathematics pass level. For this purpose Geometry diagnostic test was conducted by the Northern province educational department. This project was implemented for grades 9,10 and 11. The marks of the exam were collected from 40 randomly selected schools in all five educational zones of the Jaffna district. From the schools 1020 grade 11 students with different levels of mathematics were chosen to analyse their achievement level.

This table shows the students' achievement test marks zonal wise. 1AB schools' mean values were greater than the other types of schools. On the other hand, 1C schools and type ii schools show poor performance in geometry. Comparatively schools from Jaffna and Vadamarachi leads forward than other zones. The Island zone remained in the last position in the Jaffna district.

Table 4.11: Marks in zonal wise

zones	1AB	1C	Type ii
	Mean	mean	mean
Jaffna	38.2	34.5	32.3
Valikamam	35.4	31.5	30.5
Thenmarachi	32.3	28.0	27.7
Vadamarachi	36.4	33.3	31.6
Island	30.2	25.2	24.9

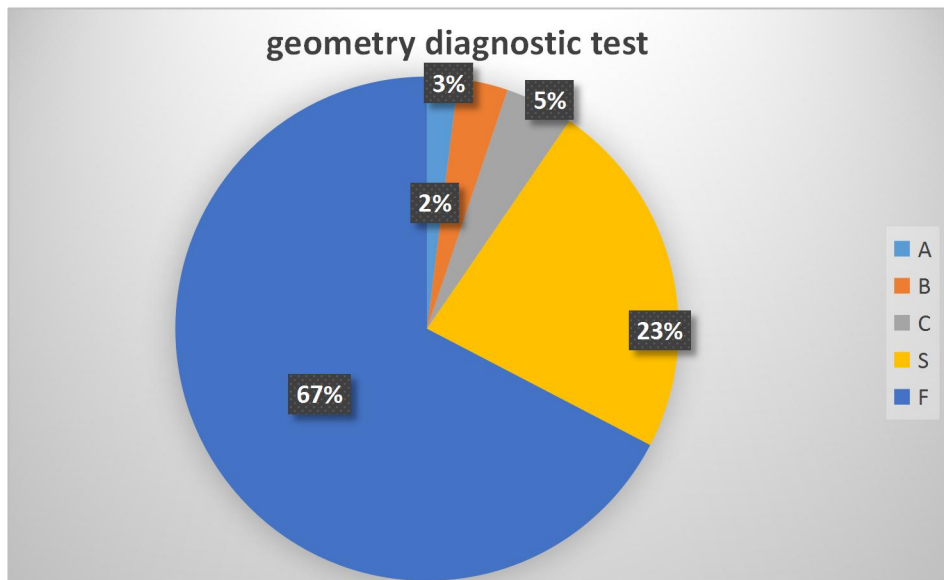


Fig 4.1 geometry diagnostic test

The marks obtained from the exam shows that 67% of the students got fail marks (below 35) and only 10% of the students scored above average.

As most of the students show low performance in geometry, it indicates that the students' performance of geometry in the Jaffna district is currently poor.

4.3.3 Teacher questionnaire

The current status of students' performance in geometry was analysed through teacher questionnaire by considering the items below.

The Questionnaire which was given to 50 teachers revealed:

Item-12

According to your experience, how would you describe the level of geometry of the students in your class?

Check one circle only.

Table 4.12: questionnaire item 12

	No.	percentage
Most are good	02	04%
Most are above average	12	24%
Most are average	09	18%
Most are below average	27	54%

As per item 12

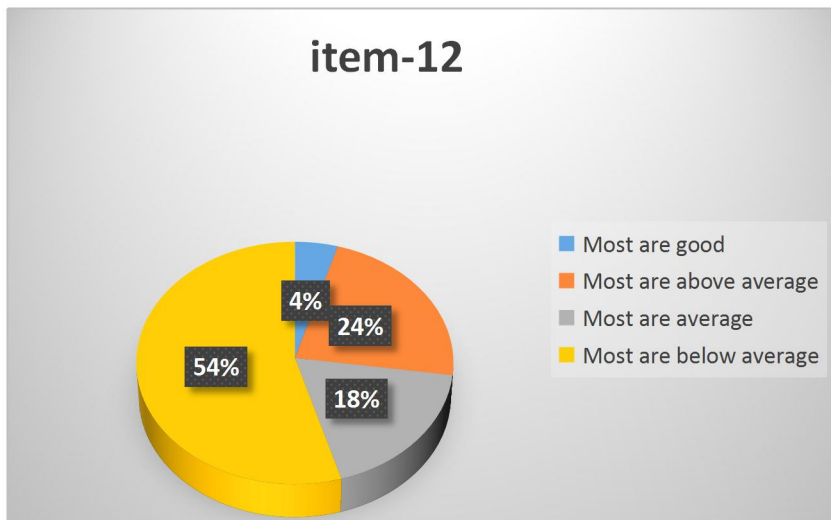


Fig 4.2: questionnaire item 12

54% of teachers said that most of the students are below average. Only 4% of teachers said that students are good.

Item- 13

How would you describe the level of geometry of the students in your previous class?

Table 4.13: questionnaire item 13

	No.	percentage
Most are good	03	06%
Most are above average	05	10%
Most are average	11	22%
Most are below average	31	62%

Item 13

62% of teachers said that most of the students are below average. Only 6% of teachers said that students are good in their previous class.

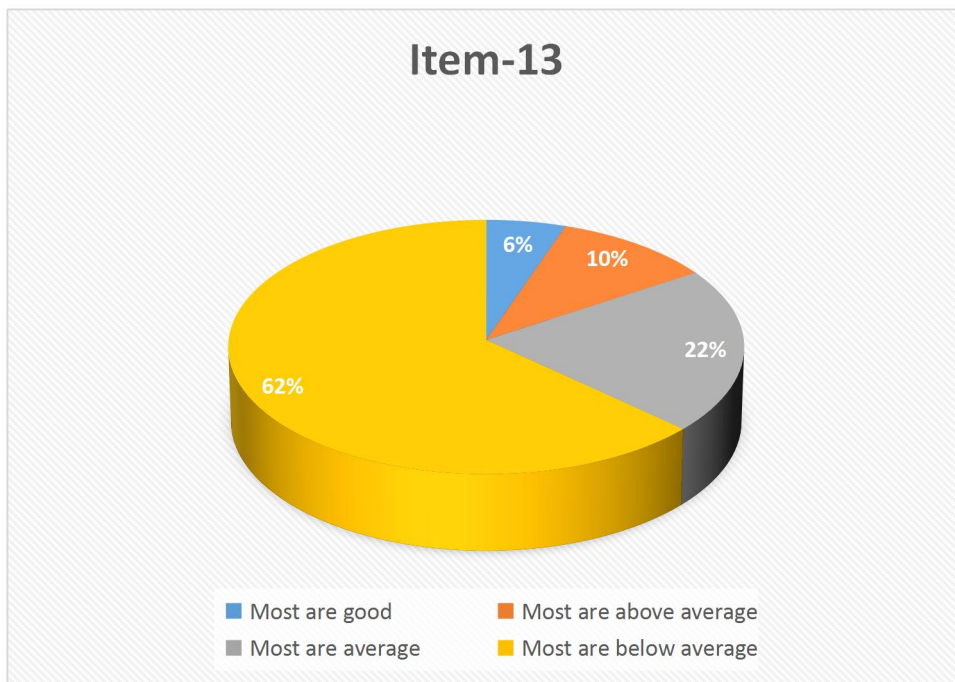


Fig 4.3: questionnaire item 13

Item-15

Table 4.14: questionnaire item 15

Statement	Yes	No
Do students feel difficulty in learning geometry?	36	14
Do students choose geometry questions for the exam curiously?	06	44

Nearly 74% of teachers agreed that students feel difficulty in understanding geometry and 12 % of teachers said that students choose geometry questions for the exam curiously.

Do your students struggle in geometry?

Table 4.15: questionnaire item 15

Always or almost always	54%
Often	32%
Sometimes	12%
Never	2%

As per 86% of teachers' opinions, students always or often struggled in geometry when doing geometrical sums. Whereas only 14% of teachers conveyed that students sometimes or never struggled in geometry. Therefore, according to the teachers' opinion, most of the students struggled in geometry.

On the whole, the findings of the teacher questionnaire revealed that the performance of students in geometry is currently poor.

In addition to the above facts, analysis of the GCE O/L mathematics papers carried out by the Research and Development branch of the Department of Examination also supports this statement. It revealed that in the years 2014 and 2015 the performance of students in geometry was extremely poor. Moreover, in 2014, questions on geometry were selected by less than 45 percentage of the students and among them over 75 percentage of the students scored less than 25 out of a total of 100. Similar trends were seen in 2015 too.

This too indicates that the performance of students in geometry is poor.

4.3.4 Discussion

Analysis of the opinions from teachers focus group interview, students focus group interview, interviews with provincial and zonal mathematics directors, Additional directors of education(A.D.Es) and in-service advisers(ISA) and teacher questionnaire was put together and interview transcripts were formed. From the interview transcripts, thematic analysis was also done to answer the above research question.

The provincial director regrets to express that at the provincial level Northern Province holds the 8th rank in mathematics results. Further, they said that our students are very poor in some portions in mathematics. Especially in geometry sections, they fail to score good marks and they ignore geometry questions in exams. As it is possible to get a distinction pass without attempting geometry proof type sums even the clever student too omit them.

Two A.D.Es of the Jaffna district were interviewed and the opinions were collected from them. According to mathematics achievement level, our zone is currently poor. Most of the students have a mind set with geometry is a difficult subject and they are scared to pay attention on geometry and this leads them to neglect the subject. Moreover, students show low performance in geometry than other sections in mathematics. For them, other sections in mathematics seem to be easy than geometry. ADEs also said that lack of prior knowledge in geometry is the reason behind this. Students fail to understand the basic geometrical concepts at the junior secondary level. Without understanding the geometrical concepts at the junior secondary level students are unable to grasp the concepts at the secondary level.

The interview was also done with in-service advisors to collect data for the current status of students' performance in geometry. According to them, the overall performance of students in geometry is poor in Jaffna schools. Among 1AB, 1C and type II schools, type II schools show low performance in mathematics, especially in geometry. They also add that mathematics teachers also fail to pay attention in the geometry portion.

From teachers' focus group interviews, it was noted that less interest and poor prior knowledge of the students are the biggest challenge and responsibility for the teachers to motivate them and create interest in geometry. In spite of all these mathematics teachers try to motivate the students by explaining the importance of geometry in day to day life and the necessity of geometry in skill development.

Students were interviewed. One of the students focus group replied that when they compared geometry with other sections, geometry is very difficult for them. The other two focus groups agreed with the above statement. Another students focus group said that they could do geometry sums except proof type questions. However, the common opinion gathered from them is 'geometry is a difficult section when compare to other sections in mathematics'

So, the interview transcript also revealed that the current status of students' performance in geometry is poor.

The analysis done through various types of instruments such as achievement test, diagnostic test, questionnaire, documentary and interview revealed almost similar findings. The overall analysis indicates that the current status of students' performance in geometry is poor.

4.4 Linear multiple regression

Multiple linear regression had been utilized to estimate the relationship between domain factors (independent variables) and the development of geometrical knowledge and skill (one dependent variable). Through this estimation, the major research question was answered.

This analysis was done through the two main regression analysis methods. At first, how the selected domain factors impact the development of geometrical knowledge and skill is analysed overall. The second analysis was done by stepwise regression method to find out which factors are the most influential factors among all three domain factors.

4.4.1 Analysis with standard (Enter) regression method

In this method, all selected variables are entered into the model at once.

Table 4.16: Summary statistics of the model

Model	R	R Square	Adjusted R Square	Std.Error of the Estimate
1	.891	.794	.779	4.591

According to Table 4.16, R squared indicates the collective variation of all three independent variables on the dependent variable. According to this, 79.4% of GAT is explained by three variables: AS, TM and IM. The adjusted R-squared value was 0.779. It adjusts statistics based on the number of independent variables in the model. This is the desired property of goodness-of-fit statistics. The difference between R squared and adjusted R squared is not large. This is also a good indicator of the strength of the prediction. In other words, it is a measure of the absolute size of the perdition error. If the standard error is too large R will not be significant.

Objective- 2

To analyse the impact of the domain factors on the development of geometric conceptual knowledge and skills.

For the study ‘The domain factors influencing in learning geometrical concepts of senior secondary schools’ the findings of gathered data were analyzed one by one to achieve the second objective.

All variables were entered into the SPSS program for analysing through standard regression analysis. Results are shown in Table 4.17:

Table 4.17: Model Summary

Model	R	R Square	Adjusted R Square	Std.Error of the Estimate	R Square Change	FChange	Sig.F Change
1	.891	.794	.779	4.591	.794	50.104	.000

a. Predictors: (Constant), AS, TM, IM

b. Dependent Variable: GAT

The model reveals that the R (multiple correlation coefficient) value was 0.891. It measures the degree of the relationship between GAT and the predicted value (AS, TM, IM).

Since the p-value of this model is 0.00, the fitted model having independent variables: AS, TM, IM and the dependent variable GAT is significant.

ANOVA

The ANOVA was used to test the following hypotheses:

- $H_{0,1}$: There is no impact of the domain factors on the development of geometric conceptual knowledge and skills.
- $H_{1,1}$: There is an impact of the domain factors on the development of geometric conceptual knowledge and skills.

Table 4.18: The ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	28512.043	27	1056.002	50.104	.000
Residual	7376.677	350	21.076		
Total	35888.720	377			

a. Predictors: (Constant), AS,TM,IM

b. Dependent Variable:GAT

Since the $p\text{-value}=0.000$ (<0.005), overall the fitted model is significant. Therefore, there is an impact of the domain factors on the development of geometric conceptual knowledge and skills. As these results are highly significant the following research question was answered.

Research question: Is there any impact of domain factors on the development of geometric conceptual knowledge and skills?

Therefore, it was shown that ability of students, teaching methodology, instruction materials and tools influence the development of geometric conceptual knowledge and skills.

Next, the detailed analysis was carried out to express the relationship between the dependent variable GAT and the predictors AS, TM, and IM by applying the stepwise regression.

4.4.2 Results of Stepwise regression analysis

The stepwise regression method was used to find out the most significant factors from the chosen domain factors. Although, it was shown that all three domain factors impact the development of geometric conceptual knowledge and skills through entering method to analyse which domain factor impacts a lot and which domain factor impacts less stepwise regression analysis method was utilized.

The most significant factors were displayed in table 419.

Table 4.19: Variables Entered/Removed

Variables Entered	Variables Removed	Method
ASI1		Stepwise (Criteria: Probability-of-F-to-enter $\leq$.050, Probability-of-F-to-remove $\geq$.100).
ASSH1		Stepwise (Criteria: Probability-of-F-to-enter $\leq$.050, Probability-of-F-to-remove $\geq$.100).
TMTS2		Stepwise (Criteria: Probability-of-F-to-enter $\leq$.050, Probability-of-F-to-remove $\geq$.100).
TMLT3		Stepwise (Criteria: Probability-of-F-to-enter $\leq$.050, Probability-of-F-to-remove $\geq$.100).
TMTK2		Stepwise (Criteria: Probability-of-F-to-enter $\leq$.050, Probability-of-F-to-remove $\geq$.100).

a. Dependent Variable: GAT

Table 4.20: Model summary

R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	Sig. F Change
.883	.780	.777	4.607	.002	4.069	.044

a Predictors: (Constant), ASI1, ASH1,TMTS2,TMLT3,TMTK2

b Dependent Variable: GAT

According to Table 4.20, 77.7% of the total variation is explained by the factors ASI1, ASH1,TMTS2,TMLT3, and TMTK2.

Table 4.21: The ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	27993.810	5	5598.762	264.487	.000
Residual	7895.794	373	21.168		
Total	35889.604	378			

a. Predictors: (Constant), ASI1, ASSH1,TMTS2,TMLT3,TMTK2

b. Dependent Variable:GAT

Since the $p\text{-value}=0.000$ (<0.005), overall the fitted model is significant. Therefore, there is an impact of the domain factors on the development of geometric conceptual knowledge and skills. On the other hand Instructional materials and tools do not influence the development of geometric conceptual knowledge and skills. From the two influential factors (TM, AS), teaching methodology impacts more than the ability of students.

Table 4.22: Coefficients related to regression analysis

	Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	sig
(Constant)	7.458	.875		8.524	.000
ASI1	5.832	.384	.588	15.196	.000
ASSH1	1.676	.300	.200	5.587	.000
TMTS2	.880	.215	.117	4.083	.000
TMLT3	.594	.207	.081	2.866	.005
TMTK2	.436	.216	.058	2.017	.044

Dependent Variable: GAT

Table 4.22, describes the summary of the fitted model having dependent variable GAT and the predictors ASI1, ASH1, TMTS2, TMLT3, and TMTK2.

According to Table 4.22, $p\text{-values}$ of all the selected predictors are less than 0.05, the fitted model is significant. The fitted model is written as

$$\text{GAT} = 7.458 + 5.832 \text{ ASI1} + 1.676 \text{ ASSH1} + 0.880 \text{ TMTS2} + 0.594 \text{ TMLT3} + 0.436 \text{ TMTK2}$$

The interpretation of the coefficients of the above-fitted model is as follows.

- If the ASI1 is increased by one unit, the average marks of GAT is increased by 5.832, while keeping the other predictors constant.
- If the ASSH1 is increased by one unit, the average marks of GAT is increased by 1.676, while keeping the other predictors constant.

- If the TMTS2 is increased by one unit, the average marks of GAT is increased by 0.880, while keeping the other predictors constant.
- If the TMLT3 is increased by one unit, the average marks of GAT is increased by 0.594, while keeping the other predictors constant.
- If the TMTK2 is increased by one unit, the average marks of GAT is increased by 0.436, while keeping the other predictors constant.

Based on the above results the following research question was answered.

Research question: What is the impact of domain factors on the development of geometric conceptual knowledge and skills?

Diagnostic Checking

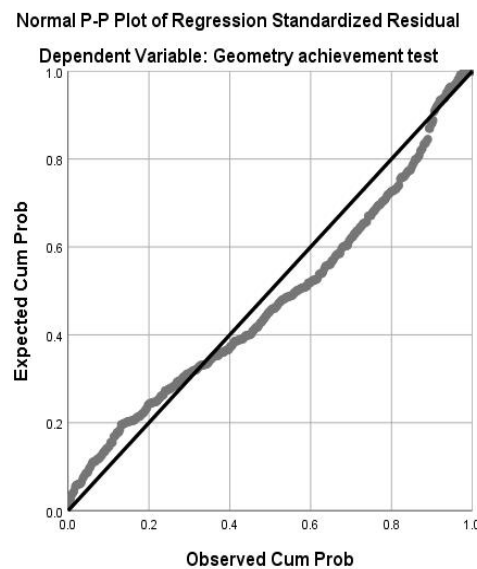


Fig 4.4: Normal Probability plot

Almost all the points are near to the line. The normality assumption of the residuals is satisfied.

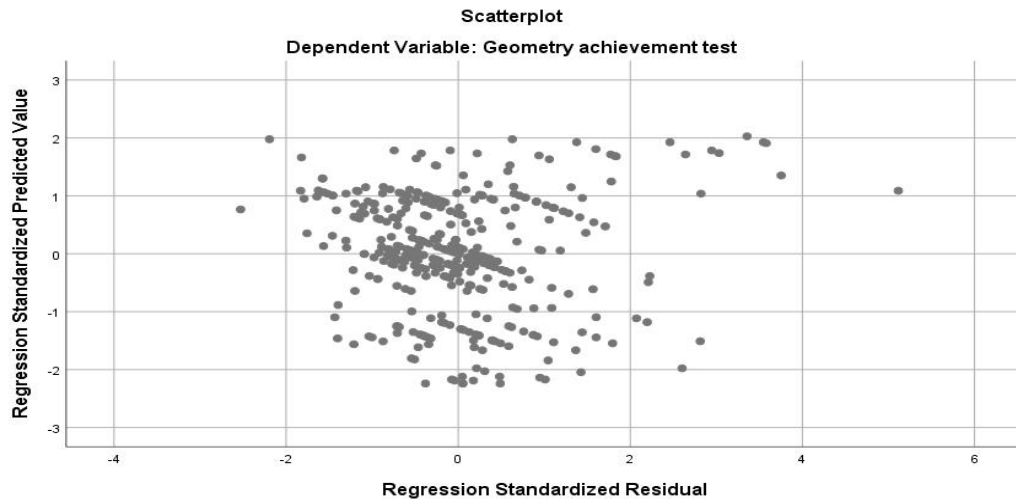


Fig 4.5: Residuals Vs Predicted Values Plot

Since the points are distributed randomly in graph 4.5, one may say that the errors are uncorrelated.

4.4.3 Discussion

Apart from regression analysis, to reach the answer for the above research question interview was done with Provincial mathematics ADE, zonal ADEs, ISAs, teachers focus group and students focus group. Provincial mathematics ADE said that they had a discussion with the panel of zonal ADEs and ISAs to find out the factors for the poor performance of students in geometry. Since most of the reasons given by the above officials are related to teachers and students, a focus group interview was done with students and teachers.

According to the Provincial director, so many factors influence the poor performance of geometry in students. Teachers are the main characters who make changes in the students' performance. Therefore teacher related factors are the most important factors for the low performance of students. Teachers fail to pay attention in adopting suitable methods according to students' ability. Still, they are using old strategies like the lecture method and explanation method.

In service advisors directly deal with students and teachers as well. Their opinions are mostly associated with classroom observations, monitoring teaching learning process

and evaluation of the students' textbooks. As per the opinions from them, both students as well as teachers do not pay much attention on geometry. Though the weight given to geometry becomes less when the curriculum changes, our students struggled in geometry even in the prevailing curriculum. So, we have to take necessary actions to follow the present curriculum to upgrade student achievement.

According to students focus group interviews, most of the teachers love the chalk and talk method. They teach geometry without using any teaching aids and without concerning students' needs. But teachers blame students. According to teachers, lack of interest in geometry and the ability of the students are the major reasons for the poor performance of students in geometry.

From the above interviews, it was revealed that ability of students, teaching methodology, instructional materials and tools impact the development of geometric conceptual knowledge and skills. It was also revealed that teaching methodology and students' ability are more important than the instructional materials and tools which impact the development of geometric conceptual knowledge and skills.

The results of regression analysis also revealed that all three domain factors: ability of students, teaching methodology and instructional materials and tools influence the development of geometric conceptual knowledge and skills. It was also shown that the teaching methodology and ability of students influence a lot than the instructional materials and tools but when teaching methodology is compared with ability of students, teaching methodology impacts more.

Therefore the overall analysis revealed that all the domain factors impact the development of geometric conceptual knowledge and skills. At the same time, ability of students and teaching methodology influence a lot than the instructional materials and tools.

Objective- 3

4.5 To analyse the impact of the ability of students on the development of geometric conceptual knowledge and skills

4.5.1 Results of entering method

Table 4.23: Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	Sig. F Change
1	.879	.772	.766	4.711	.772	138.678	.000

Dependent Variable: GAT

According to this, R Square is 77.2%. As R Square is 77.2% this model shows goodness-of-fit statistics. These results show the relationship between the independent variable (AS) and dependent variable (GAT) are significant at p value= 0.000.

ANOVA

The ANOVA was used to test the following hypothesis:

- $H_{0,1}$: There is no impact of the ability of students on the development of geometric conceptual knowledge and skills.
- $H_{1,1}$: There is an impact of the ability of students on the development of geometric conceptual knowledge and skills.

Table 4.24: The ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	27700.098	9	3077.789	138.678	.000
Residual	8189.506	369	22.194		
Total	35889.604	378			

a. Predictors: (Constant), ASI, ASSH, ASPK

b. Dependent Variable: GAT

Since the $p=0.000$ ($p<0.005$), In overall the fitted model is significant. Therefore, there is an impact of the ability of students on the development of geometric conceptual knowledge and skills. As these results are highly significant the following research question was answered.

Research question: Is there any impact of the ability of students on the development of geometric conceptual knowledge and skills?

Therefore, it was shown that the ability of students influence the development of geometric conceptual knowledge and skills.

Next, the detailed analysis is carried out to express the relationship between the dependent variable GAT and the predictor AS, by applying the stepwise regression.

4.5.2 Stepwise Regression

Though the ability of students is significant in the entering method, under the ability of students which variables influence much had been analysed using the stepwise method.

Table 4.25: Variables Entered/Removed

Variables Entered	Variables Removed	Method
ASI1	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
ASSH1	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
ASPK3	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).

a. Dependent Variable: Geometry achievement test

Table 4.26: Model summary

R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	Sig. F Change
.877	.769	.767	4.706	.009	15.099	.000

a Predictors: (Constant), ASI1, ASH1, ASPK3

b Dependent Variable: GAT

According to Table 4.26, 76.7% of the total variation is explained by the factors ASI1, ASSH1, ASPK3

Table 4.27: The ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	27584.632	3	9194.877	415.182	.000 ^b
Residual	8304.972	375	22.147		
Total	35889.604	378			

a. Predictors: (Constant), ASI1, ASSH1, ASPK3

b. Dependent Variable: GAT

Since the p-value=0.000 (<0.005), overall the fitted model is significant. Therefore, there is an impact of the ability of students on the development of geometric conceptual knowledge and skills. On the other hand, all three variables of the ability of students named students interest(ASI), study habits(ASSH) and prior knowledge(ASPK) influence the dependant variable (GAT).

Table 4.28: Coefficients related to regression analysis

	Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	sig
(Constant)	8.654	.849		10.192	.000
ASI1	6.223	.373	.628	16.682	.000
ASSH1	1.907	.300	.228	6.352	.000
ASPK3	.887	.228	.117	3.886	.000

a. Dependent Variable: GAT

Table 4.28, describes the summary of the fitted model having dependent variable GAT and the predictors ASI1, ASSH1, ASPK3

According to Table 4.28, p-values of all the selected predictors are less than 0.05, the fitted model is significant. The fitted model is written as

$$\text{GAT} = 8.654 + 6.223 \text{ ASI1} + 1.907 \text{ ASSH1} + 0.887 \text{ ASPK3}$$

The interpretation of the coefficients of the above-fitted model is as follows.

- If the ASI1 is increased by one unit, the average marks of GAT is increased by 6.223, while keeping the other predictors constant.
- If the ASSH1 is increased by one unit, the average marks of GAT is increased by 1.907, while keeping the other predictors constant.
- Increasing by 1 unit can increase the predicted performance of GAT by units; If the ASPK3 is increased by one unit, the average marks of GAT is increased by 0.887 while keeping the other predictors constant.

Based on the above results the following research question was answered.

Research question: What is the impact of ability of students on the development of geometric conceptual knowledge and skills?

Diagnostic Checking

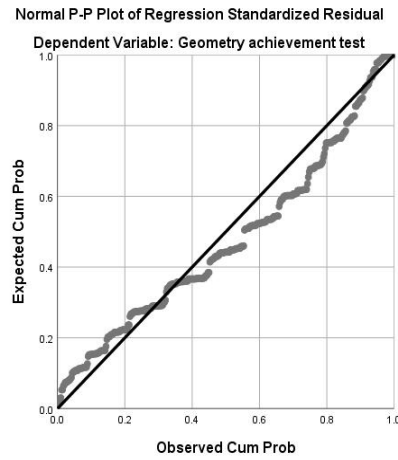


Fig 4.6: Normal Probability plot

Almost all the points are near to the line. The normality assumption of the residuals is satisfied.

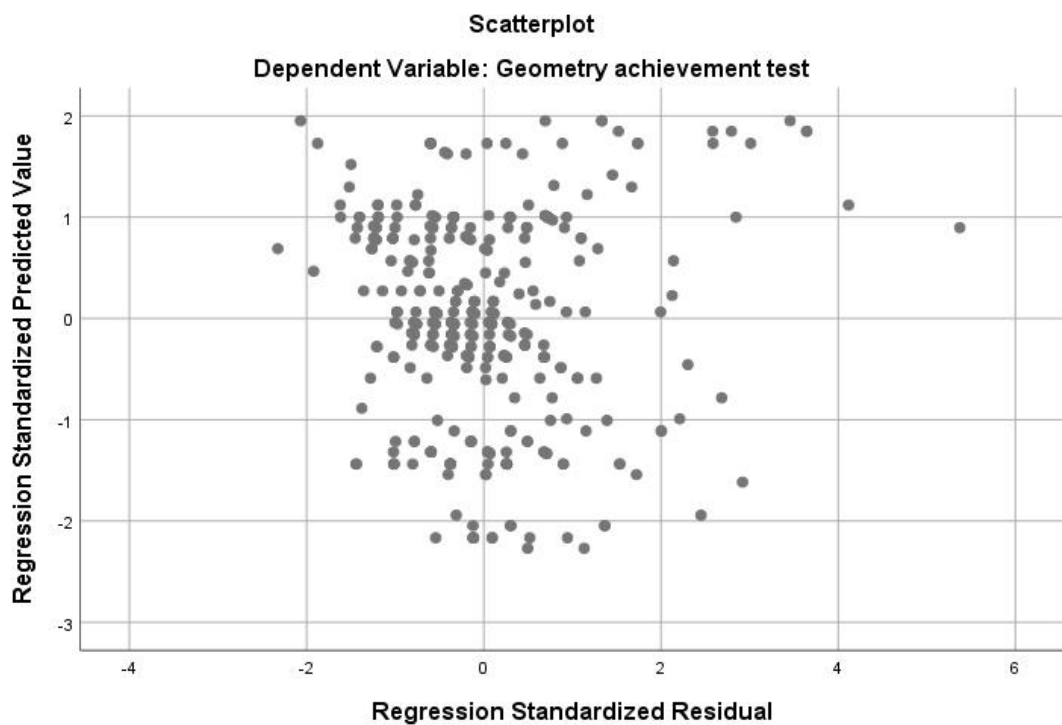


Fig 4.7: Residuals Vs Predicted Values Plot

Since the points are distributed randomly in graph 4.7, one may say that the errors are uncorrelated.

4.5.3 Discussion

In addition to the regression analysis, the opinions gathered from interviews done with provincial and zonal mathematics A.D.Es, in-service advisers and teachers and students focus group were analysed in detail to find out how each component of the ability of student (student interest, study habits, prior knowledge) impact the development of geometric conceptual knowledge and skills.

4.5.3.1 Student interest

According to ADEs and ISAs, students focus their attention on the exam only. They give priority to either passing the exam or scoring good marks. Teachers' opinions indicated that students neglect proof type questions and tend to pass the exam without attempting any geometry questions. This leads our students to pay less interest in learning geometry. On the whole, less interest in learning geometry makes our students struggle a lot and show low performance in geometry.

Opinions collected from student interview too reveal the same. As per their view, they learn geometry only for exam purposes. As per students' view, when teachers teach geometry they try to understand enough. But often they miss that. They couldn't understand geometry when the teacher explains that. All three students groups gave the same reply as they don't like geometry. For example one of the students from the focus group said that when they see proof type questions in geometry they got scared and this made them dislike geometry. Although they know that hard work helps the students to develop geometrical knowledge, they just want to use a chance to pass the exam by learning easier sections than geometry. Another student said that only proof type questions need hard work.

4.5.3.2 Study habits

The first student group expressed that doing a lot of activities and getting enough practice can help them to do geometry sums. But they don't follow this and often omit geometry questions. Another focus group said that they try to get help from friends. but they too are in the same position. At home, no one is there to help them in geometry. So they try to understand the geometry sections from tuition classes. But there also teachers do the same like schools. The third group said that though they get chances to learn geometry through the internet and new technology they don't pay interest in that and they do not search for any other source to learn geometry. As a result, they couldn't understand the geometry section successfully.

4.5.3.3 Prior knowledge

Interview transcripts of teachers revealed that prior knowledge is necessary to understand geometry. Students who fail to learn geometry in previous classes face problems in understanding geometry in the present class and struggle a lot.

The student focus group's reply too supports this. According to them, if students learn geometry in previous grades only they are able to understand geometry in grade 11. they expressed that as they don't have a clear understanding in grade 10 and 11 they couldn't learn geometry in grade 11. All the focus groups students agreed that prior knowledge is important to learn geometry in grade 11.

The analysis of the interview transcript revealed that the components of the ability of students such as students' interest, study habits and prior knowledge influence the development of geometry. It is evident in regression analysis also. In regression analysis, it is shown that if students' interest, study habits and prior knowledge are developed geometrical knowledge and skills will be developed.

Therefore the overall analysis revealed that all the components of the ability of student: students' interest, study habits and prior knowledge impact the development of geometric conceptual knowledge and skills.

Objective- 4

4.6 To analyse the impact of the teaching methodology on the development of geometric conceptual knowledge and skills

4.6.1 Results of entering method

Table 4.29: Model Summary

R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	Sig. F Change
.720	.519	.507	6.851	.519	44.064	.000

Dependent Variable: GAT

According to this, R Square is 51.9%. As R Square is 51.9% this model shows goodness-of-fit statics. These results show the relationship between the independent variable (TM) and dependent variable (GAT) are significant at p value= 0.000.

ANOVA

The ANOVA was used to test the hypothesis:

- $H_{0,1}$: There is no impact of the teaching methodology on the development of geometric conceptual knowledge and skills.
- $H_{1,1}$: There is an impact of the teaching methodology on the development of geometric conceptual knowledge and skills.

Table 4.30: The ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	18614.953	9	2068.328	44.064	.000
Residual	17273.766	368	46.940		
Total	35888.720	377			

a. Predictors: (Constant), TMTS,TMTK,TMLT

b. Dependent Variable:GAT

Since the $p=0.000$ ($p<0.005$), In overall the fitted model is significant. Therefore, there is an impact of the teaching methodology on the development of geometric conceptual knowledge and skills. As these results are highly significant the following research question was answered.

Research question: Is there any impact of teaching methodology on the development of geometric conceptual knowledge and skills?

Therefore it was shown that Teaching methods influence the development of geometric conceptual knowledge and skills.

Next, the detailed analysis is carried out to express the relationship between the dependent variable GAT and the predictors TM by applying the stepwise regression.

4.6.2 Stepwise method

Though Teaching methodology is significant in entering method, under teaching methodology which variables influence much had been analysed in a stepwise method.

Table 4.31: Variables Entered/Removed

Variables Entered	Variables Removed	Method
TMTS1	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
TMTK3	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
TMLT3	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
TMTS2	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
TMTK2	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).

a. Dependent Variable: Geometry achievement test

Table 4.32: Model summary

R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	Sig. F Change
.718	.516	.509	6.833	.007	5.220	.023

Dependent Variable: GAT

According to Table 4.32, 50.9% of the total variation is explained by the factors TMTS1,TMTK3,TMLT3,TMTS2,TMTK2.

Table4.33: The ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	18462.061	5	3692.412	79.028	.000 ^b
Residual	17427.543	373	46.723		
Total	35889.604	378			

a. Predictors: (Constant),TMTS1,TMTK3,TMLT3,TMTS2,TMTK2

b. Dependent Variable:GAT

Since the p-value=0.000 (<0.005), overall the fitted model is significant. Therefore, there is an impact of the teaching methodology on the development of geometric conceptual knowledge and skills. On the other hand, all variables of teaching methodology named teaching strategies(TMTS), teachers' knowledge(TMTK) and learning theory(TMLT) influence the dependent variable (GAT).

Among them, teaching strategies(TMTS) and teachers' knowledge(TMTK) give more impact than the learning theory(TMLT).

Table 4.34: Coefficients related to regression analysis

	Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	sig
(Constant)	13.443	1.237		10.864	.000
TMTS1	1.717	.473	.220	3.631	.000
TMTK3	1.517	.429	.200	3.535	.000
TMLT3	2.178	.287	.296	7.601	.000
TMTS2	1.176	.461	.157	2.548	.011
TMTK2	.996	.436	.132	2.285	.023

Dependent Variable: GAT

Table 4.34, describes the summary of the fitted model having dependent variable GAT and the predictors TMTS1,TMTK3,TMLT3,TMTS2,TMTK2

According to Table 5, p-values of all the selected predictors are less than 0.05, the fitted model is significant. The fitted model is written as

$$\text{GAT} = 13.44 + 1.717 \text{ TMTS1} + 1.517 \text{ TMTK3} + 2.178 \text{ TMLT3} + 1.176 \text{ TMTS2} + 0.996 \text{ TMTK2}$$

The interpretation of the coefficients of the above-fitted model is as follows.

- If the TMTS1 is increased by one unit, the average marks of GAT is increased by 1.717, while keeping the other predictors constant.
- If the TMTK3 is increased by one unit, the average marks of GAT is increased by 1.517, while keeping the other predictors constant.
- If the TMLT3 is increased by one unit, the average marks of GAT is increased by 2.178, while keeping the other predictors constant.
- If the TMTS2 is increased by one unit, the average marks of GAT is increased by 1.176 while keeping the other predictors constant.
- If the TMTK2 is increased by one unit, the average marks of GAT is increased by 0.996 while keeping the other predictors constant.

Based on the above results the following research question was answered.

Research question: What is the impact of teaching methodology on the development of geometric conceptual knowledge and skills?

Diagnostic Checking

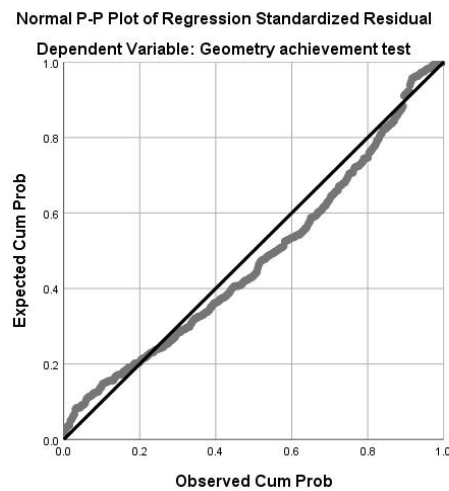


Fig 4.8: Normal Probability plot

Almost all the points are near to the line. The normality assumption of the residuals is satisfied.

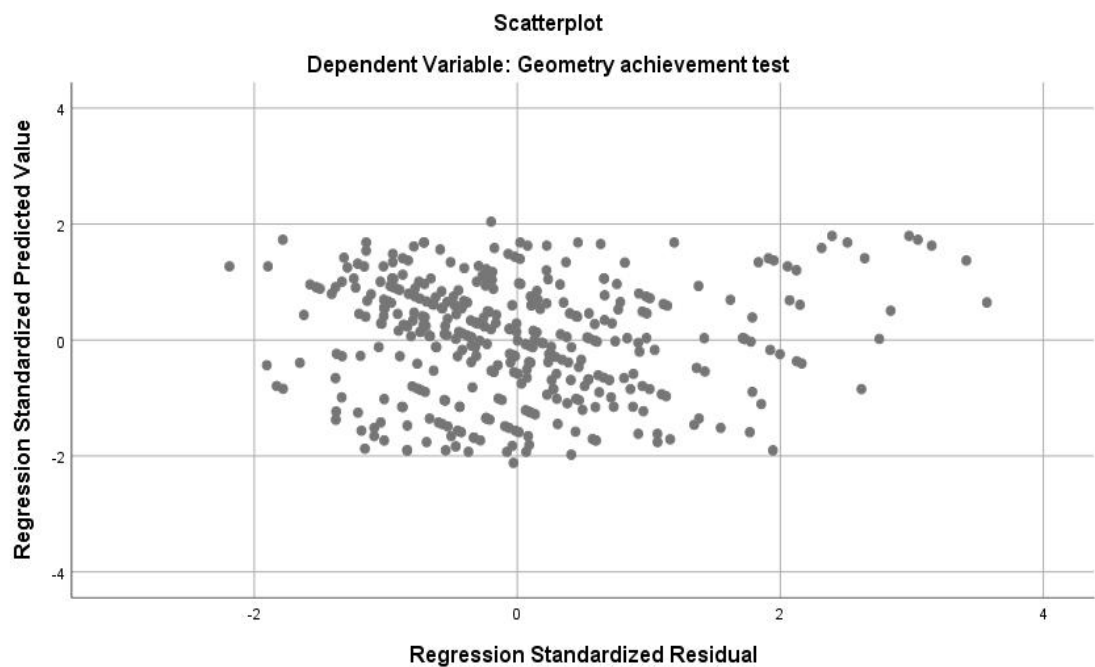


Fig 4.9: Residuals Vs Predicted Values Plot

Since the points are distributed randomly in graph 4.9, one may say that the errors are uncorrelated.

4.6.3 Discussion

In addition to the regression analysis, the opinions gathered from interviews done with the provincial and zonal mathematics A.D.Es, in-service advisers and teachers and students focus group were analysed in detail to find out how each component of teaching methodology (teaching strategies, teachers' knowledge, learning theory) impact the development of geometric conceptual knowledge and skills.

4.6.3.1 Teaching strategies

The provincial director said that only a few teachers love to use different methods of teaching when teaching geometry whereas a vast number of teachers don't utilize modern techniques and new strategies to create an interesting learning atmosphere. Even the schools which have the resources like multimedia and smart board too reluctant to use them in their teaching. Following provincial ADEs' opinion, zonal ADEs too agreed that teachers' role is most important in teaching learning process. When geometry is taught mathematics teachers have to organize students in different ways according to the need. But most of the teachers teach geometry as a whole-class activity. At the same time, they feel it is a real challenge to handle the students as a whole class and it is not easy to make them understand geometry in the whole class. But they are not ready to utilize new teaching strategies to create interest in geometry. The analysis of the transcript from ISAs' interview reveals that during their classroom observation teachers often use the traditional methods of teaching. They don't create opportunities for students' centred learning. As class room learning is not interesting, students are not motivated and they are passive in the class room. According to them, our education system is exam oriented and both students as well as parents are keen only on marks. They focus their attention only on exam questions and teach mathematics for exam purposes only.

Opinions collected from teachers interview depicts that though some teachers agreed that creating same-ability groups or creating mixed-ability groups is a better way to teach geometry rather than whole class activity, in reality within the allocated time it's not possible to form groups and make students understand geometry. They also pointed out they focus their attention mainly on public exams. To teach exam oriented, they often choose the lecture method of teaching. And they are forced to select restricted methods as their classroom situation is not suitable for activity-based

methods or any other suitable methods. Only on some occasions, teachers utilize some other methods according to the need of the lessons.

As per the details gathered from students focus groups, teachers usually teach geometry by merely explaining them on the blackboard. They try to explain by writing and drawing figures on the blackboard. Sometimes they divide the students into groups and do activities. One of the groups said that working in groups gives a lot of enjoyment and it helps them to understand geometry in a better way. They further said teachers don't use any teaching aids while teaching geometry.

Another group responded that they learned geometry on smart board only twice. Learning geometry on smart board gave them a different experience and enthusiasm for them. While learning lessons on surface area and volumes of solids, they could get the real picture from the 3D images.

4.6.3.2 Teachers' subject knowledge

According to provincial(ADE), all the teachers are at least with minimum professional qualifications. But even some teachers skip geometry proof. They hesitate to spend much time explaining geometry questions. And this indicates their poor strength and subject knowledge of the subject.

In ISAs and ADEs view, teachers are efficient and capable enough to clear the doubts of the students in geometry. But they don't give much importance to geometry by telling 'you are struggling in geometry. So do the sums which are easy for you. No need to do all the sums'. They teach geometry proofs on the surface level and they do not explain them in deep. Their intention lies in completing the syllabus on time. ISAs and ADEs report that apart from all these, some of the teachers wish to attend geometrical seminars and workshops to upgrade their geometrical knowledge.

In students' point of view they express that when they ask their doubts in geometry, some teachers insist that it's not important for the students at present and students who chose maths for advanced level only need geometry. Students also add that many teachers do the geometrical sums very clearly, except proof type questions. They don't like to pay attention on proof type questions. Sometimes they skip proofs and move on to the next section.

4.6.3.3 Learning Theory

As most of the teachers are professionally qualified, from their innate knowledge they could apply the learned theories in geometry teaching. This helps them to deal with the students more effectively. Opinions gathered from teachers' focus group indicates that only a few teachers allow students to work independently and motivate their students for self learning. These teachers give opportunities to expose students' real talents in geometry. But according to the students' report, teachers don't create an appropriate learning environment to motivate the students and when they teach geometry teachers always give the sums from the mathematics textbook only. According to students' opinions, some teachers don't try to understand the attitudes of students. They ignore the feelings of students.

The analysis of the interview transcript revealed that the components of teaching methodology such as teaching strategies, teachers' knowledge and learning theory influence the development of geometry. It is evident in regression analysis also. In regression analysis, it is shown that teaching strategies and teachers' knowledge impact more than learning theory.

Therefore the overall analysis revealed that all the components of Teaching methodology such as teaching strategies, teachers' knowledge and learning theory impact the development of geometric conceptual knowledge and skills.

Objective- 5

4.7 To analyse the impact of the instructional materials and tools on the development of geometric conceptual knowledge and skills

4.7.1 Analysis of entering method

Table 4.35: Model Summary

R	R Square	Adjusted R Square	Std. Error of the Estimate	RSquare Change	FChange	Sig.F Change
.714	.510	.498	6.902	.510	42.714	.000

a. dependent variable: GAT

According to this , R Square is 51%. As R Square is 51% this model shows goodness-of-fit statics. These results show the relation between independent variable (IM) and dependent variable (GAT) are significant at p value= 0.000.

ANOVA

The ANOVA was used to test the following hypothesis:

- $H_{0,1}$: There is no impact of the instructional materials and tools on the development of geometric conceptual knowledge and skills.
- $H_{1,1}$: There is an impact of the instructional materials and tools on the development of geometric conceptual knowledge and skills.

Table 4.36: The ANOVA

	Sum of Squares	Df	Mean Square	F	Sig.
Regression	18312.213	9	2034.690	42.714	.000
Residual	17577.391	369	47.635		
Total	35889.604	378			

- a. Predictors: (Constant), IMC,IMT,IMM
- b. Dependent Variable:GAT

Since the $p=0.000$ ($p<0.005$), In overall the fitted model is significant. Therefore, there is an impact of the instructional materials and tools on the development of geometric conceptual knowledge and skills. As these results are highly significant the following research question was answered.

Research question: Is there any impact of instructional materials and tools domain factors on the development of geometric conceptual knowledge and skills?

Therefore it was shown that Instructional materials and tools influence the development of geometric conceptual knowledge and skills.

4.7.2 Results of Stepwise method

Though the factor, instructional materials and tools is significant in entering method, which variables of Instructional materials and tools influence much had been analysed in the stepwise method.

Table 4.37: Variables Entered/Removed

Variables Entered	Variables Removed	Method
IMM2	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
IMT2	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
IMC2	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
IMT1	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).
IMM3	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq .050$, Probability-of-F-to-remove $\geq .100$).

a. Dependent Variable: Geometry achievement test

Table 4.38: Model summary

R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	Sig. F Change
.711	.505	.499	6.900	.017	12.702	.000

According to Table 4.38, 49.9% of the total variation is explained by the factors IMM2,IMT2,IMC2,IMT1,IMM3

Table 4.39: The ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	17837.281	5	3567.456	73.711	.000
Residual	18052.323	373	48.398		
Total	35889.604	378			

a. Predictors: (Constant),IMM2,IMT2,IMC2,IMT1,IMM3

b. Dependent Variable:GAT

Since the p-value=0.000 (<0.005), overall the fitted model is significant. Therefore, there is an impact of the instructional materials and tools on the development of geometric conceptual knowledge and skills. On the other hand, all variables of Instructional materials and tools named mathematical activity room(IMM), textbook(IMB) and curriculum(IMC) influence the dependent variable (GAT).

Among them, mathematical activity room(IMM) and textbook(IMB) give more impact than the curriculum(IMC).

Table 4.40: Coefficients related to regression analysis

	Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	sig
(Constant)	13.491	1.261		10.696	.000
IMM2	1.384	.451	.175	3.069	.000
IMT2	1.368	.375	.174	3.650	.000
IMC2	1.763	.366	.221	4.814	.000
IMT1	1.593	.366	.206	4.346	.000
IMM3	1.551	.435	.194	3.564	.000

a. Dependent Variable: GAT

Table 4.40, describes the summary of the fitted model having dependent variable GAT and the predictors IMM2, IMT2, IMC2, IMT1, IMM3

According to Table 5, p-values of all the selected predictors are less than 0.05, the fitted model is significant. The fitted model is written as

$$\text{GAT} = 13.491 + 1.384 \text{ IMM2} + 1.368 \text{ IMT2} + 1.763 \text{ IMC2} + 1.593 \text{ IMT1} + 1.551 \text{ IMM3}$$

The interpretation of the coefficients of the above fitted model is as follows.

- If the IMM2 is increased by one unit, the average marks of GAT is increased by 1.384, while keeping the other predictors constant.
- If the IMT2 is increased by one unit, the average marks of GAT is increased by 1.368, while keeping the other predictors constant.
- If the IMC2 is increased by one unit, the average marks of GAT is increased by 1.763, while keeping the other predictors constant.
- If the IMT1 is increased by one unit, the average marks of GAT is increased by 1.593 while keeping the other predictors constant.
- If the IMM3 is increased by one unit, the average marks of GAT is increased by 1.551 while keeping the other predictors constant.

Based on the above results the following research question was answered.

Research question: What is the impact of instructional materials and tools development of geometric conceptual knowledge and skills?

Diagnostic Checking

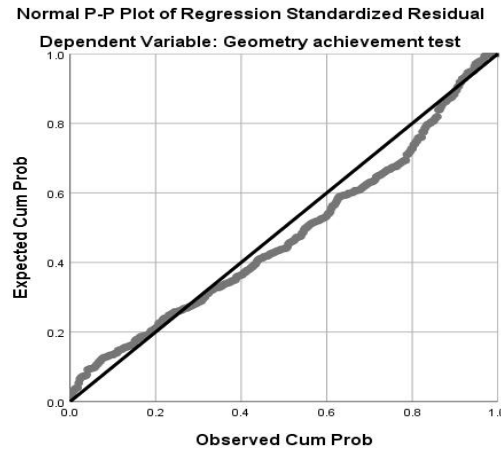


Fig 4.10: Normal Probability plot

Almost all the points are near to the line. The normality assumption of the residuals is satisfied.

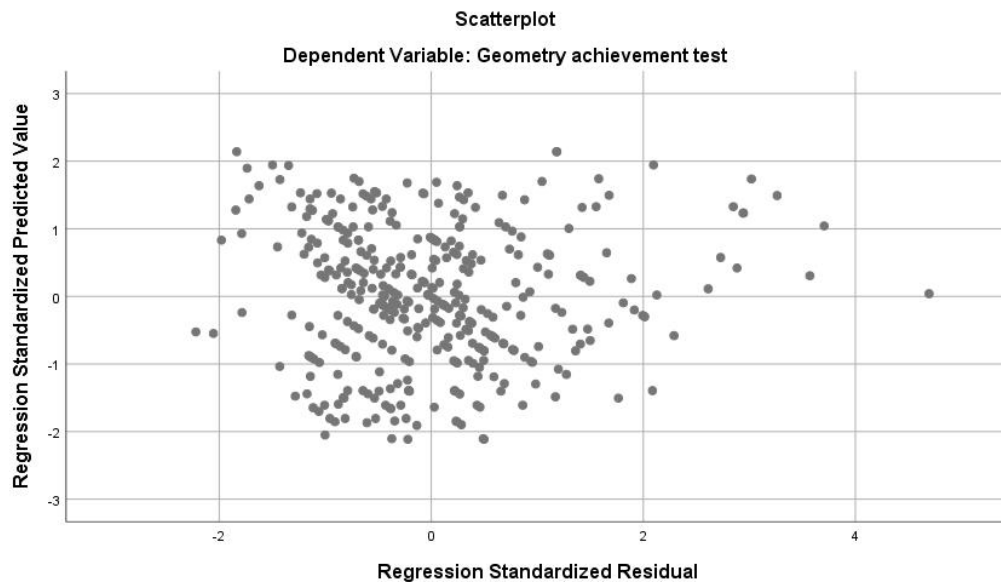


Fig 4.11: Residuals Vs Predicted Values Plot

Since the points are distributed randomly in graph 4.11, one may say that the errors are uncorrelated.

4.7.3 Discussion

With the regression analysis, the opinions gathered from interviews done with provincial and zonal mathematics A.D.Es, in-service advisers and teachers and students focus group were analysed in detail to find out how each component of Instructional materials and tools (mathematical activity room, textbook, curriculum) impact the development of geometric conceptual knowledge and skills.

4.7.3.1 Curriculum

According to the provincial director, to the students who select mathematics stream in the advanced level, our present curriculum causes some struggles to continue their studies at the advanced level. When the curriculum is changed, the weight given to geometry becomes less. Important theorems like the Apollonius theorem are not included in the present curriculum. But there was a contradictory opinion from another ADE. He expressed that the students struggled in geometry even in the prevailing curriculum. So first, we have to take necessary actions to follow the present curriculum to upgrade the student achievement.

4.7.3.2 Text book

Provincial director, ADEs and ISAs' reports reveal that textbooks are the main tool in learning geometry. Since textbooks have enough geometrical sums in a hierarchical order if the teachers do all the sums from textbook they can achieve a satisfactory level in geometry. Both teachers as well as students depend on textbooks to learn geometry.

According to teachers focus group opinion, the mathematics textbook is more important in learning geometry. It is well designed and geometry lessons are included from easy to difficult. Step by step vivid explanations are given in the textbook to illustrate the geometry lesson for better understanding of the students. It helps the students in self-learning too. It is enough to do the geometrical sums in textbook to answer the exam questions and score good marks. One of the teachers focus group pointed out that when they think the gifted students or more average students the weight given to geometry is reduced in a curriculum, it will affect the students' performance in G.C.E A/Ls

Opinions collected from students interviews revealed that teachers use textbooks to explain the lesson and give sums. Students agreed that the textbook is well designed and contained abundant geometrical sums. They also add that textbooks help to get

clear understanding and score good marks in exams. But there was a contradictory opinion from another focus group. According to that teachers explain only some important questions from the textbooks. As some activities like miscellaneous activities are difficult for students many teachers omit these questions. They also expressed that generally, students can do the geometrical sums after the explanation of the teachers. The textbook gives exercises to get self practice. Students do sums and get practice from them. But textbook is not sufficient to learn geometry like other sections in mathematics.

4.7.3.3 Mathematics activity rooms

ISAs revealed that most of the teachers don't utilize these rooms. They use these activity rooms just like class rooms. Teachers have to take much care to get the maximum benefit from mathematics activity rooms.

From the opinions from 1AB school teachers focus group, now- a - days mathematics activity rooms are available in many schools. But in real situations, teachers don't pay much interest to utilize the activity room and the available resources there. Some of the teachers feel very comfortable when teaching geometry in the mathematics activity room. To visualize the real properties and give a clear picture mathematics activity room helps them a lot. It also helps to create students' interest and motivate them. According to these teachers, mathematics activity rooms are the proper tool to maintain an active teaching learning process.

But type II schools don't have activity rooms. So teachers from these schools are highly worried about mathematics activity room and they said that they use the classrooms as the activity room and it causes many problems in reality.

Opinion of student focus group regarding the availability of mathematics activity rooms in the schools revealed that though they have mathematics activity room in their school only one or two mathematics teachers use that to conduct classes. They go there only to take some solids otherwise they don't know much about that.

Some students even don't know about the mathematics activity room. They never go there. One of the student focus groups from 1AB school said that there are no useful instruments except students' creations. They added that though mathematics activity rooms are available in the schools they are not utilized in a proper way to get maximum benefits.

Even though department people stressed the use of teaching aids in geometry teaching most of the teachers are not ready to use them in practice. Although teachers agreed that they are successful with the students' performance in geometry when using appropriate teaching aids in teaching, using the same materials with all students causes some drawbacks as all the students are not at the same level. Teachers focus group further says that however in practical, it is not possible to use different materials with the students at different levels. It requires much time and additional workload too. So our teachers do not use the instructional materials often and fail to get the benefit from them.

The analysis of the interview transcript revealed that the components of Instructional materials and tools such as mathematical activity room, textbook and curriculum influence the development of geometry. According to them, mathematical activity room and textbook are more important than curriculum. It is evident in regression analysis also. In regression analysis, it is shown that mathematical activity room and textbook give more impact than the curriculum.

Therefore the overall analysis revealed that all the components of Instructional materials and tools such as mathematical activity room, textbook and curriculum impact the development of geometric conceptual knowledge and skills.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Introduction

In the study which was done to find out the factors which affect the development of geometrical knowledge and skills of the students, how far the domain factors impact the geometrical knowledge and skills were analysed. This chapter presents major findings, conclusions and recommendations of the study.

5.2 Major findings

The major findings were categorized according to the research questions.

The analysis done through various types of instruments such as achievement test, diagnostic test, questionnaire, documentary and interview revealed almost similar findings. As the overall analysis indicated that the current status of students' performance in geometry is poor, the impact of domain factors on the development of geometric conceptual knowledge and skills were discussed and findings were portrayed.

Research question: What is the impact of domain factors on the development of geometric conceptual knowledge and skills?

From the analysis, it was shown that the domain factors: Student ability, Teaching methodology, Instructional materials and tools influence the development of geometric conceptual knowledge and skills.

The findings also revealed that among the three domain factors, teaching methodology and ability of students influence a lot than the instructional materials and tools. According to the findings, instructional materials and tools do not influence a lot on the development of geometric conceptual knowledge and skills. It was also shown that from the two influential factors, teaching methodology and ability of students, teaching methodology impacts more than the ability of students.

Research question: What is the impact of the ability of students on the development of geometric conceptual knowledge and skills?

It was shown that the ability of students influences the development of geometric conceptual knowledge and skills. It was also revealed that all three variables of the ability of students named students interest, study habits and prior knowledge influence the students' performance in geometry.

Research question: What is the impact of teaching methodology on the development of geometric conceptual knowledge and skills?

It was shown that teaching methodology influences the development of geometric conceptual knowledge and skills. It is also revealed that all variables of teaching methodology named teaching strategies, teachers' knowledge and learning theory influence the students' performance in geometry. Among them, teaching strategies and teachers' knowledge give more impact than the learning theory.

Research question: What is the impact of instructional materials and tools on the development of geometric conceptual knowledge and skills?

It was shown that instructional materials and tools also influence the development of geometric conceptual knowledge and skills. It was also revealed that all variables of instructional materials and tools named mathematical activity room, Textbook and Curriculum influence the students' performance in geometry. Among the three variables, the mathematical activity room and Textbook give more impact than the curriculum.

5.3 Conclusions

Based on the major findings presented above the following conclusions were drawn: Despite the poor performance of geometry at present, the first two levels (basic levels) of Van Hiele's theory is successfully achieved by the students. But the next three levels of Van Hiele's theory including deductive reasoning is comparatively poor in students' performance. This shows that students find difficult and struggle to learn proof type geometry problems.

Important of domain factors on development of geometric conceptual knowledge and skills

As the domain factors Student ability, Teaching methodology and Instructional materials and tools influence the development of geometric conceptual knowledge and skills, great attention should be paid to all of them. They all contribute to the improvement of the geometrical performance of the students. Since the student ability and teaching methodology affect the performance lot, special care should be taken on both.

Important of student ability on development of geometric conceptual knowledge and skills

As there is an impact of all three variables of the ability of students named students interest, study habits and prior knowledge on development of geometric conceptual knowledge and skills, student ability should be taken into consideration much. Lack of willingness to learn geometry and the negative attitude towards geometry influence the performance. Students who have much interest in geometry score better than those who are with less interest. In addition to that, utilizing proper study habits according to the students' needs help them to develop geometrical performance in a better way. Moreover, the greater contribution of prior knowledge is the fundamental need to the successful geometrical performance of the students.

Important of teaching methodology on the development of geometric conceptual knowledge and skills

Since all three variables of teaching methodology such as teaching strategies, teachers' knowledge and learning theory impact the development of geometric conceptual knowledge and skills, it was considered as one of the major factors which affect the

geometrical performance of the students. Inappropriate teaching methodology turns the students into passive participants in learning geometry. Using appropriate teaching methods, teaching aids and new technology in teaching geometry motivates the students and develop interest in learning geometry. This guides them to pay full attention and improve a lot. It was also expressed that teachers should be possessed with good command of geometrical knowledge to apply their geometrical knowledge appropriately in teaching and always be ready to clear doubts at the time. Moreover, teachers have to use techniques to motivate students in learning geometry and create an appropriate learning environment to give opportunities to expose real talents in geometry.

Importance of instructional materials and tools on the development of geometric conceptual knowledge and skills.

The variables of Instructional materials and tools named mathematical activity room, textbook and curriculum influence the development of geometric conceptual knowledge and skills. As such it is clear that instructional materials and tools affect the students' performance in geometry. Inadequate and inappropriate instructional materials and tools are one of the reasons behind the poor performance of students in geometry. The availability and use of mathematical activity room in teaching and learning geometry enhance the geometrical performance of students in a better way. Using textbooks by teachers as well as students produces greater achievement and improvement in teaching and learning geometry than not using them in the classroom and out of the classroom. Therefore, proper usage of mathematical activity room and textbooks must be essential for the success of learning geometry.

5.4 Recommendations

Based on the study findings the study recommends;

The findings of the study revealed that the current status of geometry is poor and the students fail to perform successfully in geometry. Therefore responsibility should be taken to overcome the obstacles which affect the successful performance of the students in geometry.

All the mathematics teachers should be given training with the relevant skills and knowledge. It is important that the teachers have to be trained to have Van Hiele level of geometry thinking determined. They should take care of establishing professional development workshops and seminars to develop teachers' geometrical knowledge. The Ministry of Education should carry out frequent feedback and monitoring to schools and organize courses for teachers through the relevant zonal education departments.

As geometry is mingled with each and every aspect of real life, teachers have to teach geometry in relation with day to day activities. Teachers should try as much as possible to relate geometry lessons to the real life situations in order to reduce the abstract nature of geometric concepts. They should take special care when designing geometry lessons. Effective learning of geometry requires much more care. As geometry is not focused much in G.C.E (O/L), our teachers fail to spend much time on enhancing the knowledge of geometry in lower levels such as in grades 9 and 10. They just ignore the misconceptions on geometry and move on to the next level. Teachers have to think of the strategies they use in teaching/learning geometry. Students' levels of geometry have to be identified by the teachers to take appropriate remedial action and prepare them for learning the next levels of Van Hiele theory. The administration of the schools should encourage the teachers through a school based teacher development programme to work as a team and share ideas and collaborate with their colleagues in order to develop their knowledge, develop teaching materials and try new instructional methods. Identification of the level of Van Hiele level of each student, providing appropriate content knowledge, providing a remedy for developing an appropriate level of thinking among students can make progress gradually.

It was expressed that the students' ability affects the performance of geometry. Students struggle a lot in grasping the knowledge of geometry. Efforts should be

taken to increase interest in geometry and encourage learners to engage in geometry activities. Teachers should help them to study geometry with confidence and look forward to do geometry. It requires a series of learning activities that enable students in order to develop appropriate abstract concepts of geometry. It was also recommended to allocate a separate period for geometry in the timetable.

In addition to this, the Department of Education has to take action to provide schools with the relevant resources in order to promote the performance of geometry in schools. Inappropriate instructional materials in schools influence students to perform less. Therefore it is necessary to ensure that needed instructional materials are available for teachers as well as students.

Geometry has been given little emphasis in the curriculum. The weight given on geometry becomes less when the curriculum changes. But in G.C.E (A/L) geometry is included almost everywhere. Unless the students master the subject within G.C.E (O/L) they won't be able to do the sums in A/L. As the Sri Lankan curriculum is exam oriented students leave out geometrical questions and try to pass the exam. Geometry problems would not affect students unless the effect of leaving problems reduces the students' marks. So importance given to geometry in G.C.E O/L should be increased much in order to meet the challenges in the G.C.E A/L syllabus.

A textbook is an important tool in learning geometry. The textbook has step by step vivid explanations to illustrate the geometry lesson for better understanding and well-designed lessons with enough geometrical sums help students in self learning too. Therefore, teachers should guide the students to get the maximum benefit from the textbooks.

Technology is very important in the teaching and learning of geometry and it improves the performance of learners. It is appropriate to provide such facilities to motivate teaching and learning geometry in schools. Many schools have allocated a separate room with needed equipment and resources for mathematics learning. Using mathematics activity room in learning geometry will motivate the students, encourage them by learning by doing and be engaged in the learning process actively. As a result, modern equipment like computers, overhead projectors, multimedia and even smart boards should be utilized by mathematics teachers to teach geometry by utilizing the mathematics activity room effectively.

5.5 Implications for Further Research

By analyzing the obstacles faced during the study the researcher could suggest some ideas for further research.

This study was restricted only to the Jaffna district. Further research could include other districts of Sri Lanka to analyse the impact of domain factors on geometrical knowledge development in other districts too.

As the purposive sampling technique was utilized in the study, the study focus its attention only on the same pattern of the sample and it was applicable only for the same sampling. Future research could use random sampling techniques to improve the generalizability.

This study provided an analysis of the impact of domain factors on geometrical knowledge development and focused on how the selected domain factors affect students' performance in geometry. Further research could also be done to find out other factors which influence geometrical knowledge development.

5.6 Limitations

Since the objective of the study was to analyse the impact of domain factors on the development of geometrical knowledge and concepts some limitations were affected the process of the study. The purposive sampling technique was used to select the students who can attempt geometry sums even at the lower level. For this purpose, only the schools which obtained 50 pass percentage in mathematics were selected. Therefore, while the study was in progress it affected the generalizability of the results. Moreover, due to the covid 19 pandemic, the schools were not functioning properly. Thus, when selecting students from senior secondary schools syllabus was not covered by the teachers and there was a limitation for selecting the sample on time.

5.7 Summary

Geometrical knowledge and concepts are very essential for the students. Since senior secondary students show poor performance in geometrical knowledge and concepts, it is very important to develop their achievement level in geometry. Hence this study was attempted to find out the factors which affect the performance of students in geometry. From the findings, it was revealed that the domain factors such as Student ability, Teaching methodology and Instructional materials and tools influence the development of geometric conceptual knowledge and skills of the students. In addition to that, all the variables of each domain factors contribute much for the students' performance of geometry. Variables of the ability of students named students interest, study habits and prior knowledge; variables of teaching methodology named teaching strategies, teachers' knowledge and learning theory and the variables of instructional materials and tools named mathematical activity room, textbook and curriculum affect the students' performance in geometry. Therefore it is very important to find out the remedies to give solutions for the problems caused by the above factors by considering Van Hiele theory and make necessary actions to follow the hierarchical levels of Van Heile theory step by step. Improving the achievement level of the students depends on the ministry of education, department of examinations, department of education, principals, educators, teachers and students. Therefore all have to work hard towards successful achievement in geometry. The findings and recommendations of this study will contribute a lot to the improvement of teaching and learning geometry and achieve a satisfactory level in geometry not only in Jaffna but also in all the districts of Sri Lanka.

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APPENDICES

Appendix-1: Permission letter

நவராஜா பிரசாந்தன்,
ஆசிரியர், யா/வேம்படி பெண்கள் உயர்தரப் பாடசாலை.
14.10.2020

மாகாணக் கல்விப் பணிப்பாளர்,
மாகாணக் கல்வி அலுவலகம்,
யாழ்ப்பாணம்.

கனம் மாகாணக் கல்விப் பணிப்பாளர் அவர்களுக்கு,

ஐயா,

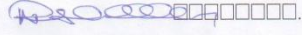
கல்வியியல் ஆய்வுக்கான தரவு சேகரிப்பதற்கான அனுமதி கோரல்.

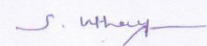
தங்களின் அனுமதியுடன் கொழும்பு பல்கலைக் கழகத்தில் முதுதத்துவமாணி கற்கை நெறியைத் தொடர்கின்றேன். அக்கற்கைநெறிக்காக

□THE DOMAIN FACTORS INFLUENCING IN LEARNING GEOMETRICAL CONCEPT OF SENIOR SECONDARY STUDENTS□

என்னும் தலைப்பில் ஆய்வினை முன்னெடுக்கின்றேன். அவ்வாய்வுக்காக வடமாகாணத்தின் பாடசாலைகளிலும் கல்விப்புலம் சார்ந்தவர்களிடமும் தரவு சேகரிப்பதற்கு அனுமதி தருமாறு தாழ்மையுடன் கேட்டுக்கொள்கின்றேன்.

தரவு சேகரிப்பின்போது கொரோனோ வைரஸ் நோயிலிருந்து பாதுகாப்பதற்கான சுகாதார நடைமுறைகள் மற்றும் பாடசாலைகளின் ஒழுங்கு விதிகளுக்குக் கட்டுப்பட்டு நடந்துகொள்வேன் எனவும் உறுதியளிக்கின்றேன்.

இப்படிக்கு,
தங்களின் உண்மையுள்ள

(ந.பிரசாந்தன், N.I.C.:791653428V)

அனுமதி வழங்கப்படுகின்றது.

S.Uthayakumar
Provincial Director of Education
Provincial Department of Education
Northern Province

Appendix-2 : Geometry Achievement Test**Geometry Achievement Test****(Van Hiele Test)**

Directions:

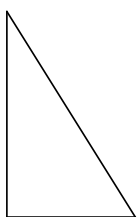
This test contains 25 questions. It is not expected that you know everything on this test. Please answer each question to the best of your ability.

When you are told to begin:

1. Read each question carefully.
2. Decide upon the answer you think is correct. There is only one correct answer to each question. Carefully circle in the correct answer on the answer sheet.
3. If you want to change an answer, please make sure you completely erase the first answer.
4. Use scrap paper for finding the solution to a problem. DO NOT mark in the test booklet.

Van Hiele Geometry

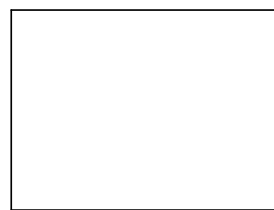
1. Which of these are squares?



K



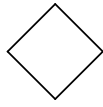
L



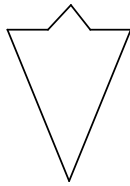
M

- (A) K only
(B) L only
(C) M only
(D) L and M only
(E) All are squares

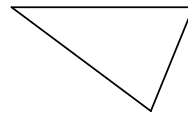
2. Which of these are triangles?



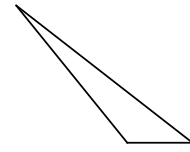
U



V



W



X

(F) None of these are triangles.

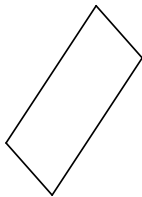
(G) V only

(H) W only

(I) W and X only

(J) V and W only

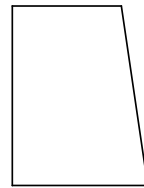
3. Which of these are rectangles?



T



S



U

(A) S only

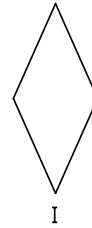
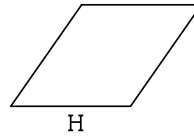
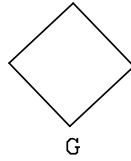
(B) T only

(C) S and T only

(D) S and U only

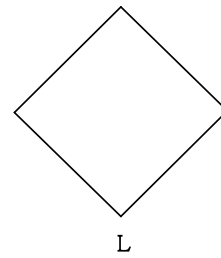
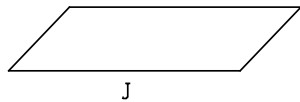
(E) All are rectangles

4. Which of these are squares?



- (A) None of these are squares.
- (B) G only
- (C) F and G only
- (D) G and I only
- (E) All are squares

5. Which of these are parallelograms?

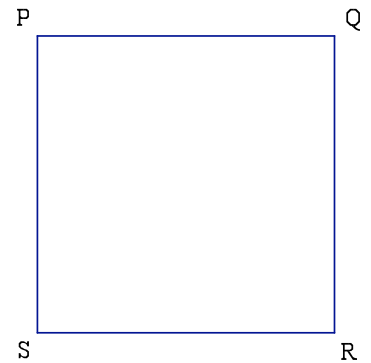


- (A) J only
- (B) L only
- (C) J and M only
- (D) None of these are parallelograms
- (E) All are parallelograms

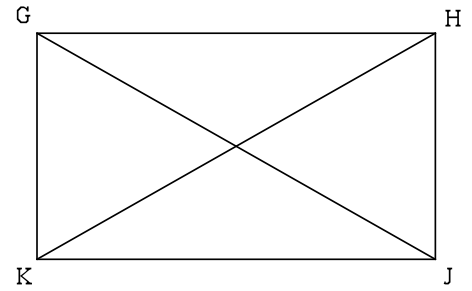
6. PQRS is a square

Which relationship is true in all squares?

- (A) PR and RS have the same length.
- (B) QS and PR are perpendicular.
- (C) PS and QR are perpendicular.
- (D) PS and QS have the same length.
- (E) Angle Q is larger than angle R.

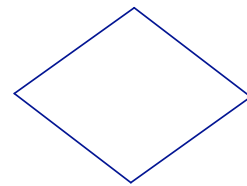
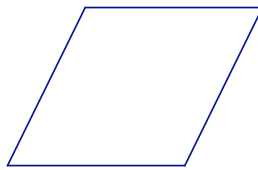
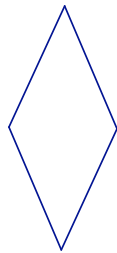


7. In the rectangle GHJK, GJ and HK are the diagonals.



Which of the (A) – (D) is not true in every rectangle?

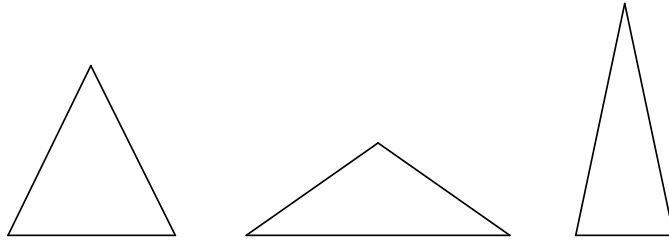
- (A) There are four right angles.
 - (B) There are four sides
 - (C) The diagonals have the same length.
 - (D) The opposite sides have the same length.
 - (E) All of (A) – (D) are true in every rectangle.
8. A rhombus is a four sided figure with all sides of the same length. Here are some examples.



Which of (A) – (D) is not true in every rhombus?

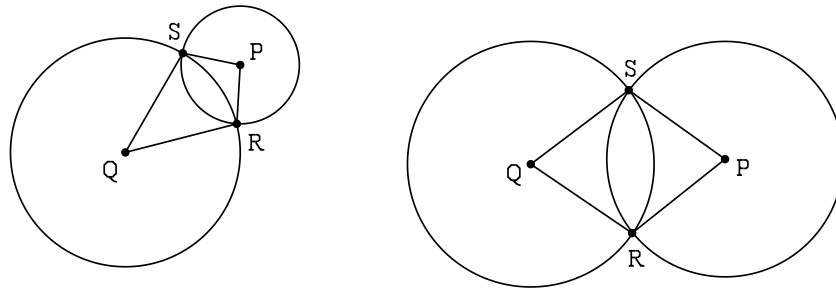
- (A) The two diagonals have the same length.
- (B) Each diagonal bisects two angles of the rhombus.
- (C) The two diagonals are perpendicular.
- (D) The opposite angles have the same measure.
- (E) All of (A) – (D) are true in every rhombus.

9. An isosceles triangle is a triangle with two sides of equal length. Here are three examples.



Which of (A) – (D) is true in every isosceles triangle?

- (A) The three sides must have the same length.
 - (B) One side must have twice the length of another side.
 - (C) There must be at least two angles with the same measure.
 - (D) The three angles must have the same measure.
 - (E) None of (A) – (D) is true in every isosceles triangle.
10. Two circles with centers P and Q intersect at R and S to form a 4-sided figure PRQS.



Here are two examples.

Which of (A) – (D) is not always true?

- (A) PRQS will always have two pairs of sides of equal length.
- (B) PRQS will have at least two angles of equal measure.
- (C) PQ and RS will be perpendicular.
- (D) Angles P and Q will have the same measure.
- (E) All of (A) – (D) are true.

11. Here are two statements.

Statement 1: Figure F is a rectangle.

Statement 2: Figure F is a triangle.

Which is correct?

- (A) If 1 is true, then 2 is true.
- (B) If 1 is false, then 2 is true.
- (C) 1 and 2 cannot both be true.
- (D) 1 and 2 cannot both be false.
- (E) None of (A) - (D) is correct.

12. Here are two statements.

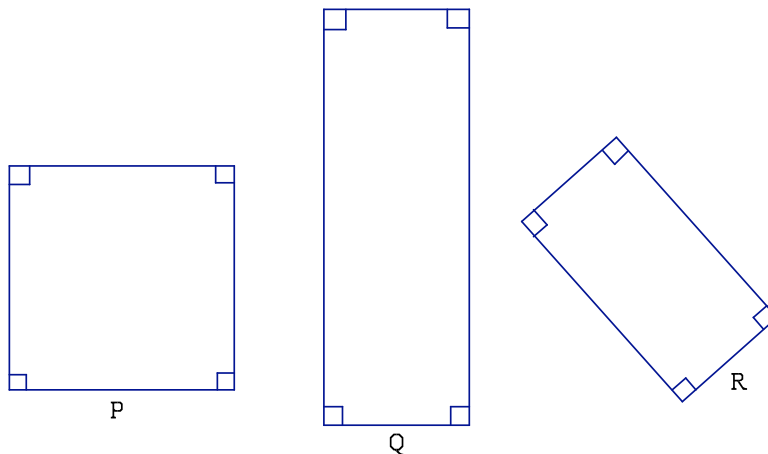
Statement S: $\triangle ABC$ has three sides of the same length.

Statement T: In $\triangle ABC$, angle B and angle C have the same measure.

Which is correct?

- (A) Statements S and T cannot both be true.
- (B) If S is true, then T is true.
- (C) If T is true, then S is true.
- (D) If S is false, then T is false.
- (E) None of (A) - (D) is correct.

13. Which of these can be called rectangles?



Which is correct?

- (A) All can.
- (B) Q only
- (C) R only
- (D) P and Q only
- (E) Q and R only

14. Which is true?

- (A) All properties of rectangles are properties of all squares.
- (B) All properties of squares are properties of all rectangles.
- (C) All properties of rectangles are properties of all parallelograms.
- (D) All properties of squares are properties of all parallelograms.
- (E) None of (A) - (D) is true.

15. What do all rectangles have that some parallelograms do not have?

- (A) Opposite sides equal.
- (B) Diagonals equal.
- (C) Opposite sides parallel.
- (D) Opposite angles equal.
- (E) None of (A) - (D).

16. Here are three properties of a figure.

Property D: It has diagonals of equal length.

Property S: It is a square.

Property R: It is a rectangle.

Which is true?

- (A) D implies S which implies R.
- (B) D implies R which implies S.
- (C) S implies R which implies D.
- (D) R implies D which implies S.
- (E) R implies S which implies D.

17. Here are two statements.

- I: If a figure is a rectangle, then its diagonals bisect each other.
- II: If the diagonals of a figure bisect each other, then the figure is a rectangle.

Which is correct?

- (A) To prove I is true, it is enough to prove II is true.
- (B) To prove II is true, it is enough to prove I is true.
- (C) To prove II is true, it is enough to find one rectangle whose diagonals bisect each other.
- (D) To prove II is false, it is enough to find one non-rectangle whose diagonals bisect each other.
- (E) None of (A) - (D) is correct.

18. All pyramids have in common:

- a. All pyramids have a triangular-shaped base.
- b. The lateral surface is constructed of triangles.
- c. The base is a quadrangle and the lateral surface is comprised of triangles.
- d. All faces of the pyramid are triangles.
- e. Statements a – d are false.

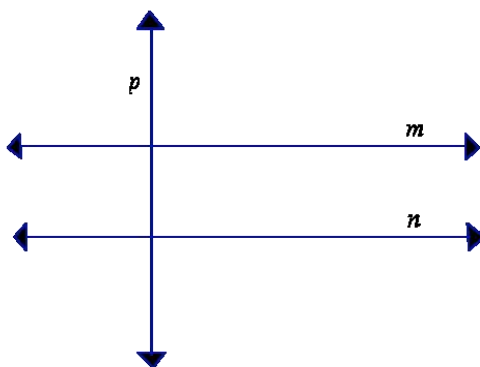
19. Which of the following statements always applies to an isosceles triangle but not to all the triangles?

- a. The mid line is parallel to one of the three sides of the triangle.
- b. The mid line is equal to half of one of the three sides of the triangle.
- c. The mid line creates a triangle which is similar to the original one.
- d. The mid-line creates an isosceles trapezium.
- e. The mid line creates a trapezium.

20. Examine these three sentences.

- (1) Two lines perpendicular to the same line are parallel.
- (2) A line that is perpendicular to one of two parallel lines is perpendicular to each other.
- (3) If two lines are equidistant, then they are parallel.

In the figure below, it is given that lines m and p are perpendicular and lines n and p are perpendicular. Which of the above sentences could be the reason that line m is parallel to line n ?



- (A) (1) only
- (B) (2) only
- (C) (3) only
- (D) Either (1) or (2)
- (E) Either (2) or (3)

21. In F-geometry, one that is different from the one you are used to, there are exactly four points and six lines. Every line contains exactly two points. If the points are P, Q, R, and S, the lines are $\{P,Q\}$, $\{P,R\}$, $\{P,S\}$, $\{Q,R\}$, $\{Q,S\}$, and $\{R,S\}$. Here are how the words “intersect” and “parallel” are used in F-geometry. The lines $\{P,Q\}$ and $\{P,R\}$ intersect at P because $\{P,Q\}$ and $\{P,R\}$ have point P in common. The lines $\{P,Q\}$ and $\{R,S\}$ are parallel because they have no points in common.

From this information, which is correct?

- (A) $\{P,R\}$ and $\{Q,S\}$ intersect.
- (B) $\{P,R\}$ and $\{Q,S\}$ are parallel.

- (C) $\{Q,R\}$ and $\{R,S\}$ are parallel.
- (D) $\{P,S\}$ and $\{Q,R\}$ intersect.
- (E) None of (A) - (D) are correct.

22. To trisect an angle means to divide it into three parts of equal measure. In 1847, P.L. Wantzel proved that, in general, it is impossible to trisect angles using only a compass and an unmarked ruler. From this proof, what can you conclude?

- (A) In general, it is impossible to bisect angles using only a compass and an unmarked ruler.
- (B) In general, it is impossible to trisect angles using only a compass and a marked ruler.
- (C) In general, it is impossible to trisect angles using any drawing instruments.
- (D) It is still possible that in the future someone may find a general way to trisect angles using only a compass and an unmarked ruler.
- (E) No one will ever be able to find a general method for trisecting angles using only a compass and an unmarked ruler.

23. There is a geometry invented by a mathematician J in which the following is true:

The sum of the measures of the angles of a triangle is less than 180° .

Which is correct?

- (A) J made a mistake in measuring the angles of the triangle.
- (B) J made a mistake in logical reasoning.
- (C) J has a wrong idea of what is meant by “true”.
- (D) J started with different assumptions than those in the usual geometry.
- (E) None of (A) - (D) is correct.

24. Two geometry books define the word rectangle in different ways.

Which is true?

- (A) One of the books has an error.
- (B) One of the definitions is wrong. There cannot be two different definitions for a rectangle.
- (C) The rectangles in one of the books must have different properties from those in the other book.
- (D) The rectangles in one of the books must have the same properties as those in the other book.
- (E) The properties of rectangles in the two books might be different.

25. Suppose you proved statements I and II.

I. If p , then q .

II. If s , then not q .

Which statement follows from statements I and II?

- (A) If p , then s .
- (B) If not p , then not q .
- (C) If p or q , then s .
- (D) If s , then not p .
- (E) If not s , then p .

Appendix-3: Questionnaire for students

Questionnaire for students

This questionnaire is given to you for a study related to geometry. This information will be used for research purposes only and will be kept confidential.

Personal information

1 Name of the school.

2. Grade: -

3. Sex: - Male- 1 Female -2

Mark the right one

Strongly Disagree -1 Disagree - 2 Neutral -3 Agree -4 Strongly Agree -5

1	Ability of student	1	2	3	4	5
1.1	Interest in learning geometry					
1.1.1	Look forward to do geometry					
1.1.2	Study geometry with confident					
1.1.3	Never get tired of doing exercises in geometry					

Mark about your opinion of geometry learning

1.2	Study habits	1	2	3	4	5
1.2.1	Study geometry exam oriented					
1.2.2	Study geometry through practice					
1.2.3	Use technology to study geometry					

To respond in relation to learning geometry of your prior knowledge

1.3	Prior knowledge	1	2	3	4	5
1.3.1	Have clear knowledge of previous geometry lessons					
1.3.2	Can do previous grades geometrical sums now					
1.3.3	Need prior knowledge of geometry to learn lessons further					

Remember your maths teacher for the items given below and mark the correct ones

2	Teaching Methodology	1	2	3	4	5
2.1	Teaching Strategies					
2.1.1	Teaches uses appropriate teaching methods					
2.1.2	Teacher uses appropriate teaching aids					
2.1.3	Teacher uses new technology in teaching geometry					
2.2	Teachers knowledge					
2.2.1	Teacher has lot of geometrical knowledge					
2.2.2	Teacher applies his geometrical knowledge appropriately in teaching					
2.2.3	Teacher is always ready to clear doubts at the time					
2.3	Learning theory					
2.2.1	Teacher uses techniques to motivate learning geometry					
2.2.2	Teacher creates appropriate learning environment					
2.2.3	Teacher gives opportunity to expose real talents in geometry					

To respond in relation to Textbook

	Textbook	1	2	3	4	5
3.1.1	Textbook is an important tool in learning geometry					
3.1.2	Textbook contains enough Geometrical sums to answer the questions in exam					
3.1.3	Textbook has vivid explanations to illustrate the geometry lesson for better understanding					

To respond in relation to Mathematics activity room

	Mathematics activity room	1	2	3	4	5
3.3.1	Uses mathematics activity room in learning geometry					
3.3.2	Mathematics activity rooms has suitable resources to learn geometry					
3.3.3	Feel very comfortable when I do geometry in mathematics activity room					

To respond in relation to curriculum

	Curriculum	1	2	3	4	5
3.2.1	Textbook has well designed Geometry lessons to help students in self learning					
3.2.2	Geometry lessons move from easy to difficult step by step					
3.2.3	Suitable evaluation and assessment is done to give necessary remedial measures.					

Remember your maths teacher's instruction for the following and identify the correct ones

Never -1 Some times -2 Neutral -3 Many times -4 Always -5

4.1	Related to the method used to teach geometry	1	2	3	4	5
4.1.1	Teaches only by using the blackboard					
4.1.2	Classifies and teaches students in groups					
4.1.3	Teaches using Smart Board when teaching geometry					

Mark 'Yes' or 'No'

3.2	About mathematics activity room	Yes	No
3.2.1	Does your school have a math mathematics activity room?		
3.2.2	Whether it is found with sufficient resources in operation		

Never -1 Some times -2 Neutral -3 Many times -4 Always -5

	A place where the students need extra help beyond the geometry teaching at school	1	2	3	4	5
	Among peers					
	At home					
	In private Tuition classes					

Appendix-4: Questionnaire for Teachers

Questionnaire for Teacher

This questionnaire is addressed to teachers of mathematics who are asked to supply information about their academic and professional background, classroom resources and the instructional materials and activities used to teach geometry and promote the development of students' skills and strategies.

Section I: Personal Information

1..School Name: _____

2:school type:- 1AB.....1C..... type ii.....

3. How long have you been working as a teacher?

1--5 years 6-10 years 11-20 years... More than 20 years...

4. How old are you?

Under 25 ----☐ 25-29 ----☐ 30-39 ----☐
40-49 ----☐ 50-60 ----☐

5. Are you female or male?

Female ----☐ Male ----☐

6. What is the highest level of formal education you have completed?

G.C.E(A/L)..... B.A..... B.Sc..... B.Com.....others.....

7. professional qualification

Trained teacher..... NDT..... PGDE.....Others

8.Other qualification

M.Ed.....M.phil..... Phd....

9. Have you attend any professional development / workshops or seminars that dealt directly with teaching geometry?

Yes..... No.....

If yes, how long? Check one circle only.

Less than 6 hours ----☐ 6-15 hours ----☐

16-35 hours ----☐ More than 35 hours ----☐

10. Where do you prepare materials for instruction?

Check one circle only.

Only at home ----☐ Mostly at home ----☐ About equally at home and at school ----☐

Mostly at school ----☐ Only at school ----☐

11. How strongly do you agree or disagree with the following statements ...

Please mark one choice in each row.

Strongly Disagree-1 Disagree-2 Neutral 3 Agree -4 Strongly Agree -5

a)	All in all, I am satisfied with my job as a mathematics teacher	1	2	3	4	5
b)	I feel that I am making a significant educational difference in the lives of my students.					
c)	If I try really hard, I can make progress with even the most difficult and unmotivated students in geometry					
d)	I am successful with the students' performance in geometry in my class.					
e)	I usually know how to get through to students.					

Questions 12-13 ask about the students in your class.

12. According to your experience, how would you describe the level of geometry of the students in your class?

Check one circle only.

Most are above average ----☐

Most are average ----☐

Level of geometry varies greatly ----☐

Most are below average ----☐

13. According to your experience, how would you describe the level of prior knowledge of geometry of the students in your class?

Check one circle only.

Most are above average ----☐

Most are average ----☐

Level of geometry varies greatly ----☐

Most are below average ----☐

14. When you teach geometry how often do you organize students in the following ways?

Always or almost always -1 Often -2 Sometimes -3

Never-4

1	I teach geometry as a whole-class activity	1	2	3	4
2	I create same-ability groups	1	2	3	4
3	I create mixed-ability groups	1	2	3	4
4	I use individualized instruction for teaching geometry	1	2	3	4
5	Students work independently	1	2	3	4

Difficulties of students

Questions 15 asks about how you deal with difficulties of the students in your class.

15. What do you usually do if a student begins to fall behind in geometry?

Check one circle for each line.

Yes -1

No-2

a)	Do your students struggle in geometry?	1	2
b)	Do students feel difficulty in Learning geometry?	1	2
c)	Do students understand the portion of geometry in previous grades?	1	2
d)	Do students choose geometry question for exam curiously?	1	2
f)	Do you think geometry is an important section in mathematics?	1	2
g)	Do you give importance to geometry when teaching?	1	2

16. Please mark one choice in each row.

Always or almost always -1 Often -2 Sometimes -3 Never-4

		1	2	3	4
1	Do you teach geometry using appropriate teaching aids ?				
2	When students struggle in geometry, have you used the appropriate methods to suit the students level ?				
3	As it is a difficult section, do you ignore geometry when teaching?				
4	Do you use the mathematics activity room when teaching geometry?				

17. How much do you agree with the following statements?

Mark one choice in each row.

Disagree a little -1 Disagree a lot-2 Neutral -3 Agree a little-4 Agree a lot -5

1	The importance given to the geometry in reforms is decreased step by step.	1	2	3	4	5
2	It is enough to do the geometrical sums in text book to answer the exam questions					
3	When the weight given to geometry is reduced in a curriculum, it will affect the students performance in G.C.E A/Ls					
4	Geometry should be taught in secondary education according to the subjects students choose in A/L					
5	Geometry Students have to be categorized in secondary education in accordance with the subjects they choose in A/L					
6	In Addition to mathematics period, more time is needed to teach					
7	The importance given to the geometry in G.C.E Ordinary level curriculum is reduced.					
8	Now the number of questions on geometry in G.C.E Ordinary level exam paper is reduced than in former curriculum					
9	geometry is everywhere in A/L. but the focus given to the geometry up to O/L is not enough. Therefore, there is a vast gap between O/L and A/L geometry					
10	Mathematics activity room is with needed equipment and resources for learning geometry.					

18. Mark one choice in each row.

Very often -1 Sometimes-2 Rarely-3 Never-4

1	Activity methods	1	2	3	4
2	Discussion methods	1	2	3	4
3	When teach geometry use smart board	1	2	3	4
4	When teach geometry use mathematics activity room	1	2	3	4

19. Mark one choice in each row.

Very satisfied-1 Satisfied-2 Almost satisfied-3 Not satisfied-4

1	Exercises given in the text book for geometry are enough	1	2	3	4
2	The current status of geometry in mathematical curriculum is Geometry exercises are organized in mathematics text book step by step	1	2	3	4

20. Mark one choice in each row.

Highly increased-1 Slightly increased-2 Not changed-3

Highly Reduced - 4 Slightly reduced-5

1	when curriculum was changed the weight given to the proof type questions in G.C.E O/L exam paper is	1	2	3	4
2	when curriculum was changed, the weight given to geometry is	1	2	3	4

Appendix-5: Teachers' Interview schedule

Teachers' Interview schedule

Research site:

Interview date:..... Interview number:.....

Interview starts:..... Interview ends:.....

Interviewer:..... Interviewee:.....

Introduction: The researcher introduces himself and provides the purpose of the interview. Informed consent question: Do you consent freely to participate in this interview?

1. How long you have been a mathematics teacher?
2. Do you enjoy teaching geometry lessons? Please elaborate.
3. When you teach a typical geometry lesson, how do you introduce your lesson?
4. According to your experience, how would you describe the level of geometry of the students in your class?
5. Can you describe what students feel about geometry?
6. How is their interest in geometry learning?
7. Do the Students feel geometry as a difficult subject and it is not easy to learn geometry?
8. What about your students who fail to learn geometry in previous classes?
9. What are the reasons for the poor performance of students in geometry?
10. How do teachers usually teach geometry?
11. What are the teaching strategies used to teach geometry?
12. What are the new technologies you use to create interest in geometry?
13. Do you think the importance given to the geometry in reforms is decreased step by step?
14. How do you feel regarding the importance given to the geometry within G.C.E Ordinary level curriculum?
15. How does the text book help in teaching and learning geometry?

16. What do you think about the geometrical sums and explanations included in the text book?
17. How do you use mathematics activity room? Can you create an interest by using it in a proper way?
18. What are the remedial actions taken to overcome the problems?
19. What are the methods used by the students to learn geometry?
20. What do you think about attending geometrical seminars and work shops to upgrade your geometrical knowledge?
21. How else you can develop your geometrical knowledge?

Appendix-6: Students' Interview schedule

Students' Interview schedule

Research site:

Interview date:..... Interview number:.....

Interview starts:..... Interview ends:.....

Interviewer:..... Interviewee:.....

Introduction: The researcher introduces himself and provides the purpose of the interview. Informed consent question: Do you consent freely to participate in this interview?

1. Do you enjoy geometry learning?
2. Why do you feel difficulty in understanding geometry?
3. Please think about the reasons for lack of understanding in geometry?
4. Why don't you study the portion of geometry in previous grades?
5. Do you choose geometry question for exam curiously?
6. How does your teacher teach geometry?
7. In which way he/she motivate you to learn geometry? Does your teacher use appropriate and interesting teaching methods with appropriate teaching aids?
8. How does your teacher clear your doubts in geometry?
9. How do you learn geometry?
10. How do your teachers use text books to teach geometry?
11. Do your teachers depend only on text book?
12. What do you think about the text books?
 - i. Is geometry lesson move from easy to difficult step by step in the text books?
 - ii. Does the text book contain vivid explanation to illustrate the geometry lesson for better understanding?
 - iii. Does the textbook give opportunity for self learning?
 - iv. Is text book an important tool in learning geometry?

13. How often you use mathematics activity room in learning geometry?
14. Are there adequate and appropriate resources in the mathematics activity room?
15. How do you feel when you do geometry in mathematics activity room?

Appendix-7: ADS ,ISA Interview schedule

ADS ,ISA Interview schedule

Research site:

Interview date:..... Interview number:.....

Interview starts:..... Interview ends:.....

Interviewer:..... Interviewee:.....

Introduction: The researcher introduces himself and provides the purpose of the interview. Informed consent question: Do you consent freely to participate in this interview?

1. What is the current status of students' performance in geometry?
2. What are the reasons for the poor performance of students in geometry?
3. "Teachers are the main characters who make changes in the students performance". What do you think about teachers' knowledge in geometry?
4. How do you feel teachers use different methods of teaching when teaching geometry?
5. When geometry is being taught, how mathematics teachers organize the students?
6. How do teachers create appropriate learning environment to motivate the students?
7. How do the teachers clear the doubts of the students in geometry? do they pay much interest in that?
8. How do you think the importance given to the geometry in reforms?
9. How do you feel regarding the importance given to the geometry within G.C.E Ordinary level curriculum?
10. Do you have any ideas to upgrade their geometrical knowledge?
11. Have you arranged any geometrical seminars?
12. How can we develop the geometrical knowledge of the students? Could you please give any suggestions?